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A Techno-Economic Environmental Approach to Improving the Performance of Photovoltaic, Battery, Grid-Connected, Diesel Hybrid Energy Systems: A Case Study in Kenya

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Abstract

Backup diesel generator (DG) systems continue to be a heavily polluting and costly solution for institutions with unreliable grid connections. These systems slow economic growth and accelerate climate change. Photovoltaic (PV), energy storage (ES), grid connected, DG – Hybrid Energy Systems (HESs) or, PV-HESs, can alleviate overwhelming costs and harmful emissions incurred from traditional back-up DG systems and improve the reliability of power supply. However, from project conception to end of lifetime, PV-HESs face significant barriers of uncertainty and variable operating conditions. The fit-and-forget solution previously applied to backup DG systems should not be adopted for PV-HESs.

To maximize cost and emission reductions, PV-HESs must be adapted to their boundary conditions for example, irradiance, temperature, and demand. These conditions can be defined and monitored using measurement equipment. From this, an opportunity for performance optimization can be established. The method demonstrated in this study is a techno-economic and environmental approach to improving the performance of PV-HESs. The method has been applied to a case study of an existing PV-HES in Kenya. A combination of both analytical and numerical analyses has been conducted. The analytical analysis has been carried out in Microsoft Excel with the intent of being easily repeatable and practical in a business environment. Simulation analysis has been conducted in improved Hybrid Optimization by Genetic Algorithms (iHOGA), which is a commercially available software for simulating HESs.

Using six months of measurement data, the method presented identifies performance inefficiencies and explores corrective interventions. The proposed interventions are evaluated, by simulation analyses, using a set of techno-economic and environment key performance indicators, namely: Net Present Cost (NPC), generator runtime, fuel consumption, total system emissions, and renewable fraction. Five corrective interventions are proposed, and predictions indicate that if these are implemented fuel consumption can be reduced by 70 % and battery lifetime can be extended by 28 %, net present cost can be reduced by 30 % and emissions fall by 42 %. This method has only been applied to a single PV-HES; however, the impact this method could have on sub-Saharan Africa as well as similar regions with unreliable grid connections is found to be significant. In the future, in sub-Saharan Africa alone, over $500 million dollars (USD) and 1.7 billion kgCO_2 emissions could be saved annually if only 25 % of the fuel savings identified in this study were realized. The method proposed here could be improved with additional measurement data and refined simulation models. Furthermore, this method could potentially be fully automated, which could allow it to be implemented more frequently and at lower cost.

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Contents

Abbreviations

Abbreviation	Description
A/C	Air Conditioning
AC	Alternating Current
bi-Inv	bi-directional Inverter
DC	Direct Current
DG	Diesel Generator
ES	Energy Storage
GHI	Global Horizontal Irradiation
HES	Hybrid Energy System
HOMER	Hybrid Optimization of Multiple Energy Resources
iHOGA	improved Hybrid Optimization by Genetic Algorithms
KPI	Key Performance Indicator
K-HES	PV/Energy Storage/Grid/Diesel Generator - Hybrid Energy System in Kenya
NE	North East
NPC	Net Present Cost
O&M	Operation and Maintenance
PV	Photovoltaic
PVSyst	Photovoltaic system simulation software
PoA	Plane of Array
PV-HES	PV/Energy Storage/Grid/Diesel Generator -Hybrid Energy System
RF	Renewable Fraction
SAM	System Advisor Model
SMA-McB	SMA Multicluster Box - 36
SoC	State of Charge
SW	South West
UPS	Uninterrupted Power Supply

Nomenclature

Symbol	Description	Unit
A	Diesel generator fuel curve coefficient	N/A
B	Diesel generator fuel curve coefficient	N/A
E_{Chrg}	Battery charge energy	kWh
E_{DChrg}	Battery discharge energy	kWh
E_{DGChrg}	Diesel generator charge energy	kWh
E_{DG}	Generator energy	kWh
E_{DG-L}	Diesel generator energy to load	kWh
E_{Gr}	Grid energy	kWh
E_{GrChrg}	Grid charge energy	kWh
E_{Gr-L}	Grid energy to load	kWh
E_{L}	Load energy	kWh
E_{PV}	PV energy	kWh
E_{PVChrg}	PV charge energy	kWh
E_{PVCurt}	PV curtailment energy	kWh
E_{PV-L}	PV energy to load	kWh
$E_{\mathrm{PV-NE}}$	Estimated energy production from North-East facing PV arrays	kWh
E_{PVnom}	Nominal PV Energy	kWh
E_{PV-SW}	Estimated energy production from South-West facing PV arrays	kWh
E_{PVth}	Theoretical PV energy	kWh
E_{T}	Total energy production	kWh
GA	Grid availability	Binary
G_{STC}	Irradiance under standard test conditions	W/m^2
$G_{\mathrm{PoA}}(t)$	Simulated irradiance in the plane of array at time t	W/m^2
$G_{\mathrm{PoA-SW}}$	Pyranometer measured irradiance in the SW plane of array	W/m^2
$G_{\mathrm{PoA-NE}}$	Pyranometer measured irradiance in the NE plane of array	W/m^2
$I_{\mathrm{DG\_chrgmax}}$	Maximum DG charging current	A
Inv_{eff}	Inverter efficiency	%
$\mathrm{I1}, \mathrm{I2}, \mathrm{I3}$	3 phase line currents	A
$NRMSE$	Normalized root mean square error	N/A
P_{DG}	Diesel generator power	kW
$\mathrm{P}_{\mathrm{DG}}(\mathrm{t})$	Simulated diesel generator power at time t	kW
$\mathrm{P}_{\mathrm{DG-rated}}$	Rated power of diesel generator	kW
P_{Gr}	Grid power	kW
$P_{\mathrm{Gr/DG}}$	Sum of grid and diesel generator power	kW
$P_{\mathrm{DGcritical}}$	Critical DG power	kW
P_{DGMin}	Minimum DG power	kW (also expressed as a percentage of rated power)
P_{L}	Load power	kW
P_{PV}	PV power	kW
$P_{\mathrm{PV}}(t)$	Simulated PV power at time t	kW
P_{tot}	Total measured power with ABB A44 energy meter	kW
P_{ThrBatt}	Threshold battery power	kW

PV_{CapNE}	Total PV capacity facing north east	kW
PV_{CapSW}	Total PV capacity facing south west	kW
PV_{derate}	General PV production scaling term	N/A
PV_{Pcoeff}	PV module temperature coefficient of power	%/˚C
PV_{TSTC}	PV module temperature under standard test conditions	˚C
SoC_{min}	Minimum state of charge	%
SoC_{stp}	State of charge stop	%
T_{amb}	Ambient temperature	˚C
T_{batt}	Battery temperature	˚C
$T_{\mathrm{batt/R}}$	Battery room temperature	˚C
t_{DG}	DG runtime	h
T_{NOTC}	Nominal operating cell temperature of the PV module	℃
t_{PO}	Power outage length	min
T_{PV}	measured module temperature	℃
$\mathrm{V1}, \mathrm{V2}, \mathrm{V3}$	3 phase line voltages	V
V_{PV}	DC voltage at maximum power point	V
$\varphi 1, \varphi 2, \varphi 3$	3 phase power factor angles	°

1 Introduction

In sub-Saharan Africa alone, nearly 90 kilo-barrels of diesel are consumed every day when backup diesel generators (DG) respond to grid outages. The annual fuel consumption during grid outages costs consumers $5 billion USD, and this consumption is responsible for emitting approximately 18 billion $kgCO_2$ emissions [1]. The International Energy Agency has stated that businesses in sub-Saharan Africa frequently cite inadequate power supply as detrimental to their operations. In fact, estimations show that, on average, 4.9 % of sales in this region are lost due to grid outages. This average value is significantly reduced by areas in southern Africa with relatively stable grid connections, whereas areas in central Africa suffer much higher losses [1]. For business that can afford backup DGs, their operations are restricted by high energy costs. For business that cannot afford backup DGs, their operations can suffer even greater consequences. Backup DGs are a poor choice for consumers, and they are a sub-optimal method for increasing energy access.

Backup DGs are clearly not a long-term sustainable economic and environmental solution for institutions with unreliable grid connections. Renewable Hybrid Energy Systems (HESs) offer an alternative to traditional backup DG systems. This study focuses on a promising configuration of HESs comprised of photovoltaics (PV), energy storage (ES), grid connection, and DGs – hereinafter referred to as PV-HESs. The viability of PV-HESs has been demonstrated with ground breaking projects. In Gaza, for example, several medical facilities have restored access to affordable power [2]. The same technology has also been proven to be feasible in countless off-gird projects, such as in the Hawaiian Islands where fuel consumption and emission reductions between 50-100 % have been achieved [3]. Furthermore, PV costs have been steadily declining and the trend is expected to continue [4]. The cost of ES is falling even more rapidly; McKinsey & Company reported an 80 % decline in the cost of lithium ion batteries between 2010 and 2016 [5]. All considered, the long-term economics of the components required to construct PV-HESs are auspicious.

However, the successful adoption of PV-HESs should not be expected to automatically follow the declining costs of PV and ES technologies. The high complexity of PV-HESs requires a comprehensive design process to mitigate significant uncertainties. Stochastic variables, such as PV energy yields, electrical load profiles, cost of fuel, component degradation profiles, replacement costs, and source/system availabilities, must be carefully considered. The appropriate size and type of system components are contingent on the interpretation of these variables. Similarly, control measures must be adapted appropriately. In some cases, unforeseeable or unpredictable variations to the boundary conditions could be highly compromising or even detrimental to the performance of PV-HESs. Examples of cases where the replacement costs of HESs have surpassed those of the original DG system can be found in [6], [7]. Additionally, when it comes to engineering, procurement, and construction, there are inevitable compromises to the idealized system design. Decisions hinging on communication channels, time-constraints, supply, customer relations, funding and other considerations can all have an impact on the design and performance of a PV-HES. As stated by one of the collaborators on this project:

> "due to the nature of contracting for HES, feasibility, design and pricing is often completed up to two years before the commissioning of a project leading to sub-optimal designs or control strategies being implemented with little contractual mechanisms or incentives for EPC contractors to improve solutions. This can lead to frustrating issues where well-meaning engineers are unable to optimise a system and advance technology in the face of irrational contractual barriers. These issues are often compounded when customers are poorly educated about how to optimise systems on site, especially when they receive conflicting information from EPCs, consultants and on-site teams [8]".

Considering the numerous design challenges and the looming threats of shifting boundary conditions, it is easily concluded that PV-HESs face substantial barriers to achieving optimal economic and environmental performance (hereinafter, performance refers to both economic and environmental performance).

The intent of this study is to provision for the long-term viability of PV-HESs. Accordingly, a method to evaluate and improve system performance has been developed in response to the foreseeable need for performance interventions. A techno-economic approach has been taken to justify interventions. The complex nature of the interaction between the numerous uncertain variables make it difficult to predict and optimize system performance. Short of trial and error, the optimization process requires a comprehensive method for predicting how a system will respond to design and control modifications. Simple analytical methods are not adequate; therefore, simulation models employing robust numerical methods are employed. This study makes use of a simulation software called improved Hybrid Optimization by Genetic Algorithms (iHOGA) to build numerical simulation models.

The method developed in this study is applied to a case study of a PV-HES in Kenya. The system utilizes industry standard equipment and is thought to be representative of similar PV-HESs in the region; although, no data to support this inference is presented.

(Note: all costings in this study are reported in USD)

1.1 Aims and Objectives

This study aims to demonstrate a method for improving and optimizing the performance of PV-HESs. A PV-HES installed in Kenya, henceforth referred to as the K-HES, is used as a case study. The objectives of the study are:

- Develop a tool to interpret measurement data and characterize the performance of the K-HES in terms of power and energy consumption for each of the system components.
- Critically analyze the performance of the K-HES for inefficiencies to formulate and propose corrective interventions.
- Model the existing K-HES in the iHOGA simulation environment and quantify technical, economic, and environmental key performance indicators (KPIs), namely: Net Present Cost (NPC), generator runtime, fuel consumption, total system emissions, and renewable fraction.
- Model each of the proposed corrective interventions in the iHOGA simulation environment and compare the KPIs to the simulation results from the existing K-HES.
- Discuss and evaluate the observed change in the KPIs to determine the technical, economic and environmental implications of the proposed interventions.

Before describing the method, which is found in Section 1.4, it is useful to first discuss previous works and the details of the K-HES configuration.

1.2 Previous work

There is an abundance of literature concerning the techno-economic optimization of HESs. Most of this literature, for example, concerns the design and feasibility of new HESs, as discussed in [9]–[15]. Generally, these studies focus on new locations and/or applications for novel or common configurations of HESs. Conversely, there is very little literature concerning the techno-economic or environmental optimization of existing HESs. In fact, it is difficult to find literature dedicated to this topic. The reasons for this are suspected to be commercial and contractual barriers surrounding existing installations

– there is little incentive for system owners to promote the optimization of their HES within academic literature. Studies concerning existing HESs which have either evaluated system performance, discussed performance improvements, or predicted the impact of design changes, are reviewed in this section.

The performance of a 4 kW grid connected residential wind-PV HES with battery storage is investigated in [16]. Two years of measurement data is evaluated using a time series parameter-based approach. Unmet load, grid availability, power quality, and the impacts of wind and PV intermittency are explored. It is concluded the system has achieved substantial emission reductions and supports the utility with demand side reduction. No suggestions for improving the system performance are justified. In [6], several HESs in Rwanda are evaluated and found to be suffering from accelerated battery degradation. The study develops a method to evaluate battery lifetime in off-grid PV systems. It was found that accelerated battery degradation was due to undersized PV-systems, and changes in the social behavior of the system's users. Simulation results produced with HOMER (Hybrid Optimization of Multiple Energy Resources) and Matlab showed the total operational costs of the HESs investigated could be reduced by increasing the capacity of the PV systems, and limiting overnight power usage. The techno-economics of a PV-diesel HES installed in the Peruvian Amazon are analyzed in [7]. The performance of the system is analyzed with the intent to pass on lessons to institutions in similar regions in the developing world. Despite having adequate solar resource, it is found the system is not economical. The primary cause is identified to be oversizing of the systems components. It is concluded that high capital costs, high operation and maintenance (O&M) costs, and poor efficiencies eroded the systems economics. The authors of [17] investigate the performance of a PV-wind-diesel-battery hybrid energy system. A novel control optimization method is proposed and tested. The method aims to reduce the total operational costs of the system, including all replacement and O&M costs. Measurement results are compared from before and after the implementation of the control method. It is concluded the daily optimization of the control parameters results in up to 7.8 % savings when compared to the yearly optimization of the control parameters. Furthermore, compared to the simplistic but common load-following control method, up to 37.7 % savings are achieved.

These studies offer insight into how various types of HESs perform under different boundary conditions. Different methods are demonstrated for evaluating the performance of various types of HESs. The common strategy is a time-series parameter-based approach, which is to be adopted in this study. The results offer system designers and engineers topics to consider when designing or optimizing similar HESs. However, no studies were identified which address the performance evaluation or optimization of systems with the specific configuration addressed in this study. Furthermore, barring [6] and [17], the impact of proposed improvements are not quantified. Moreover, a comprehensive method to identify performance deficiencies, recommend interventions, and quantify techno-economically and environmentally viable solutions has not been presented. To the knowledge of the author, no such method as proposed herein has been demonstrated for PV-HESs.

1.3 K-HES Overview

The K-HES is located within a degree of latitude from the equator, and it has been in operation since 2017. Prior to its installation, the sites primary energy sources were the Kenyan national grid and two backup DGs. The national grid is unreliable; therefore, backup power generation is essential to ensure the near continuous availability of power. The K-HES was designed to minimize emissions and the cost of energy, more specifically, to minimize diesel fuel consumption. In order to reduce capital costs, the original power system components were not replaced when the system was converted from a backup DG system to a PV-HES. The PV, ES, and control systems were installed parallel to the existing power system. A basic schematic of the K-HES is shown in Figure 1.1, and the capacities of the primary components in the system are summarized in Table 1.1.

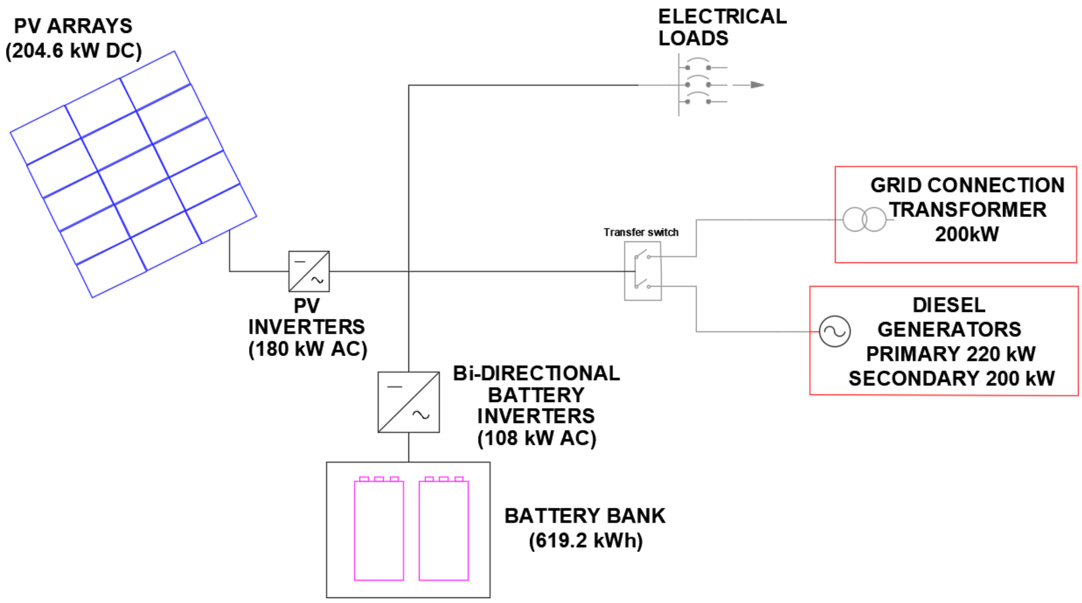

Figure 1.1: Basic schematic of the K-HES including all power sources and inverters.

Table 1.1: Summary of K-HES energy sources and battery inverter capacities.

Component	***Description***
PV system	204.6 kW DC/180 kW AC
DGs	Primary 220 kW (275 kVA)
	Secondary 200 kW (250kVA)
National Grid	200 kW Transformer
Lead Acid Battery Bank	619.2 kWh (309.6 kWh usable capacity)
Bi-directional Battery inverters	108 kW AC Power

1.3.1. PV system

The PV system has a nominal DC power of 204.6 kW, and nominal AC power of 180 kW. There are 26 strings in total, each having 20 or 21 modules, which are distributed on the rooftops of 3 buildings. The layout of the PV system is shown in Figure 1.2, and the array configuration is described in Table 1.2.

Figure 1.2: Layout of the PV system. Left side: administration rooftops. Right Side: workshop rooftops [18].

Table 1.2: PV system array configuration.

Building	***Array***	***Inverter***	***No. Strings***	***Mods per String***	***No. Inverters***	***No. Mods***	***kW DC***	***kW AC***
Admin	A1	SMA STP 20000TL	4	20	1	80	22.0	20
Workshop	W1-W4	SMA STP 20000TL	4	21	1	84	23.1	20
Workshop	W2	SMA STP 20000TL	4	20	1	80	22.0	20
Workshop	W3	SMA STP 20000TL	4	20	1	80	22.0	20
Workshop	W1	SMA STP 25000TL	5	21	2	210	57.8	50
Workshop	W4	SMA STP 25000TL	5	21	2	210	57.8	50
		Totals				**744**	**204.6**	**180**

The orientation of the modules and the primary components of the PV system are summarized in Table 1.3. Approximately half of the PV modules face north-east (NE) at 116° and half face south-west (SW) at -64°. Shading of the PV arrays is negligible due to the latitude and unobstructed horizon.

Table 1.3: PV system orientation and primary components.

Array Slope	*18°*
Orientation	50.5 % of modules face 116°
	49.5 % of modules face -64°
PV Module	Canadian Solar CS6K-275M
PV Inverters	SMA 20000TL
	SMA 25000TL

The primary role of the SMA 20000TL and SMA 25000TL PV inverters is to convert DC power to AC power. The power conversion efficiency is dependent on the DC input voltage and the output power of the inverter. At the nominal operating cell temperature (NOCT), which is defined when the ambient temperature is 20°C, the DC voltage at the maximum power point of the 20 and 21 module strings is 570.0 V and 598.5 V, respectively. From the inverter datasheets the optimal inverter input voltage is 600V [19]. The average daytime ambient temperature on site, according to SolarGIS, is 21°C [20]. Therefore, the inverters are expected to operate near to their defined nominal operating conditions. Additionally, the PV inverters communicate with the control system and are capable of curtailing their output when requested using either frequency-based control or using commands over the communication system.

1.3.2. Diesel Generators

The primary DG is rated for standby conditions at 275 kVA/220 kW and the secondary DG is rated for standby conditions at 250 kVA/200 kW. The two DGs cannot run simultaneously. The secondary DG is run once the primary DG has run for 4 hours. The DGs require a minimum warmup period of 5 minutes and must run for a minimum of 15 minutes after being connected to the load – which is inconvenient when the site does not have battery storage to support the load when the grid fails. The fuel consumption data for standby operating conditions is summarized in Table 1.4 from the primary DG datasheet [21], and the corresponding fuel efficiency curve is shown in Figure 1.3.

From the data collected in this study, it is not possible to determine which of the two DGs is running. Therefore, a simplification is made which assumes the system only operates with the primary DG.

Table 1.4: Primary DG rated 275kVA/220kW fuel consumption under standby operating conditions [21].

Percentage of rated output power, %	***Fuel Consumption, L/h***
50	37.0
75	49.1
100	60.3

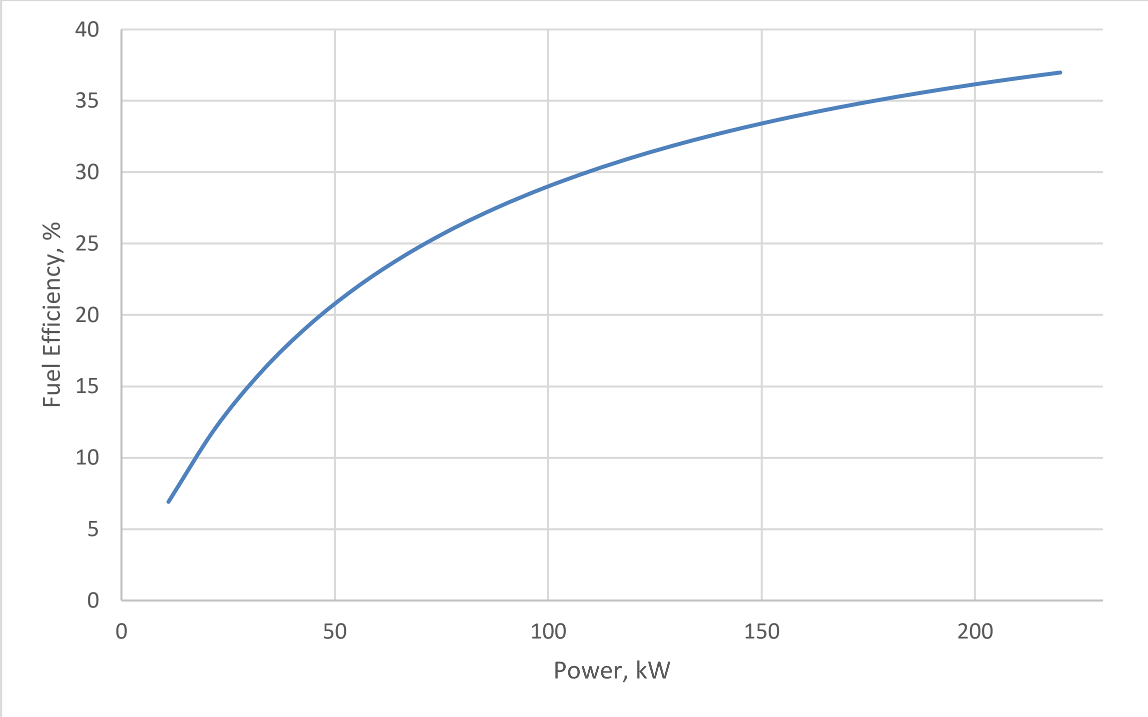

Figure 1.3: Primary DG rated 275kVA/220 kW fuel efficiency curve under standby operating conditions (the curve has been generated assuming fuel consumption as per Table 1.4 and generator efficiency defined as described in [17]).

1.3.3. Battery System

The battery system is housed in a concrete building, which relies on natural ventilation for cooling and for venting of hydrogen gas. The primary components of the battery system are the SMA Sunny Island bi-directional inverters (bi-Inv), the Hoppecke 16OPzS solar.power flooded lead acid batteries, and the SMA Multicluster Box – 36 (SMA-McB). The batteries are connected via the bi-Invs to the AC bus as depicted in Figure 1.1. Figure 1.1 does not include the SMA-McB, which acts as the main AC distribution board and the connection point between the bi-Invs and the remainder of the AC power system. The SMA-McB is also the central communications hub facilitating the control measures discussed in Section 1.3.4. The primary components of the battery system are summarized in Table 1.5.

Table 1.5: Summary of primary battery system components.

Component	***Nominal Size***	***No. Units***	***Total Capacity***
SMA Sunny Island 8.0H	8 kW for 30 min, 6 kW indefinitely	18	144 kW for 30 min, 108 kW indefinitely
SMA Multiclustbox-36	300 kW	1	300 kW
Hoppecke 16OPzS - solar.power 2900	2150Ah @C10, 2V cell x 24, 48V bank	144	619.2 kWh

1.3.4. Control System

The control system is formed by the bi-Invs, SMA-McB, and a bespoke external control network. The SMA-McB measures power to/from each component and the load. Measurement data is passed from the SMA-McB to the master bi-Inv, which then implements control measures via all bi-Invs accordingly. The control measures can be understood by considering scenarios for each of the two operating modes, which determine the energy source that is given priority to the load. In mode #1, active from 2:00-17:59, the priority order is PV, battery, grid, DG. In mode #2, active from 18:00-1:59 (that is, 1:59 the following day), the priority order is PV, grid, DG, battery.

Mode #1

While operating in mode #1, if enough PV power is available, no additional energy sources are called. If excess PV power is available, the batteries are charged with a maximum current of 1200 A DC, and any additional current is curtailed. If there is not enough PV power available, the battery system will make up the difference provided the total load does not exceed 95 kW, and the battery SoC remains above the minimum SoC SoC_{min}. The value of SoC_{min} is time-dependent in order to reduce generator runtime during PV ramp up:

- From 7:00-11:59 $SoC_{\mathrm{min}} = 55\ \%$ and $SoC_{\mathrm{stp}} = 60\ \%$
- From 12:00-6:59 $SoC_{\mathrm{min}} = 60\ \%$ and $SoC_{\mathrm{stp}} = 65\ \%$

Once SoC_{min} is reached, the grid/DG is run until SoC stop SoC_{stp} is reached. Similarly, if the load exceeds 95 kW, the grid/DG is called until the load falls below 85 kW. In both cases, when the grid/DG is called the grid has priority ahead of the DG; the DG is only started if the grid is not available. The external control network is responsible for making the grid available whenever possible. As previously mentioned, the DG requires a 5-minute warm up period. During this warm up period the battery supplies the load.

Mode #2

While in mode #2, PV power supply is generally less than the load due to the timeframe. Therefore, the grid is called, but if it is not available, the DG is called. If the grid and DG are not available, the battery is called as the final option. In mode #2, the battery is typically only utilized to support the load while the DG warms up following a grid outage.

In both control modes, while the grid/DG is running, the battery system operates under droop control; whereby, the battery rate of charge/discharge depends on the system frequency. If the frequency falls below the nominal value (50 Hz), the battery charge rate is reduced, or having the same effect, the battery discharge rate is increased until the frequency returns to the nominal value. Similarly, if the frequency increases above the nominal value, the battery charge rate is increased, or the discharge rate is reduced until the system frequency returns to the nominal value. Droop control also regulates the power factor of the grid and DG to unity by supplying reactive power to the load. Furthermore, the battery system behaves as an uninterrupted power supply (UPS) when a power source is abruptly impeded. However, the battery will not operate under droop control, or behave as a UPS, if its SoC reaches 35 % or below. In this case, the battery system is shutdown to protect the battery from being damaged.

Battery charging occurs according to a set of parameters recommended by the battery manufacturer. The charge parameters are entered in the bi-Inv charge control settings; they are listed in Table 1.6. The charge parameters are for the most part self-explanatory, but descriptions for clarification can be found in the SMA Sunny Island operating manual [22].

Table 1.6: Charge control parameters for Sunny Island inverter.

Charge Parameter	***Value***
Max charging current	200 A
Absorption time boost charge	180 min
Absorption time full charge	8 h
Absorption time equalization charge	8 h
Cycle time full charge	14 d
Cycle time equalization charge	180 d
Charge voltage boost	2.40 V
Charge voltage full	2.5 V
Charge voltage float	2.23 V

The availability of the grid is determined by the external control network as per the voltage and frequency limits in Table 1.7. These constraints are enforced to avoid damaging the sites electronic equipment.

Table 1.7: Voltage and frequency grid disconnection limits.

Setting	***Trip Point***	***Time Delay (s)***
Over Voltage Stage 1	252 V	0.5
Over Voltage Stage 2	253 V	0.2
Under Voltage Stage 1	215	1
Under Voltage Stage 2	200	0.2
Voltage Unbalance	3 %	0
Over Frequency Stage 1	52	90
Over Frequency Stage 2	52.25	0.2
Under Frequency Stage 1	47.5	20
Under Frequency Stage 2	47	0.01

1.4 Method

1.4.1. Data Collection

The proposed method begins with collecting measurement data from a monitoring system comprised of several measurement instruments. Each instrument is connected via a communications network to a central datalogger which uploads data to an online server. The instruments are exterior to the primary components of the system, which in general have been found to reduce measurement uncertainty compared to using data directly from the inverters [23]. Measurements are taken at sub-intervals every minute to produce 15-minute averages that are saved to the datalogger memory. The monitoring system includes the following instruments:

1. 2 x Pyranometer: Kipp and Zonen SMP11
2. 4 x Energy meter: ABB A44
3. 1 x Voltage and frequency meter: Intelipro G59
4. 4 x Temperature sensor: Pt100 thin film element

Data collection started in 2017 and is ongoing. Historical data is stored and accessed through an online web service. Datasets can be downloaded for any duration starting from the first day the monitoring system was brought online up until real-time. The remainder of this section is used to describe each of the measurement instruments and their accuracy.

Pyranometer: Kipp and Zonen SMP11
Irradiation measurements are taken with two SMP11 thermopile pyranometers manufactured by Kipp & Zonen. The pyranometers are secondary standard devices classified according to ISO 9060; the classification is described in detail in [24]. As listed in [23], and discussed further in [24]–[26], the following factors introduce uncertainty to measurements taken with this type of pyranometer:

1. Calibration uncertainty
2. Drift over time
3. Directional response (as a function of the azimuth and zenith angle)
4. Offset originated by the thermal radiation
5. Offset originated by the temperature change
6. Temperature dependency of the sensitivity
7. Non-linearity
8. Spectral response
9. Tilt response
10. Long time drift of the measuring system
11. Error of the analog to digital converter of the measuring unit

It has been shown that with extremely careful work, measurement uncertainties as low as ±2 % can be achieved with this type of thermopile pyranometer; whereas uncertainties of ±5 % are more realistic [24]. The SMP11 pyranometers used in the study are less than a year old, which suggests factors contributing to increasing uncertainty over long periods are relatively small, such as factors #2 and #10 from the list above. However, the service and calibration schedules for the pyranometers in this study are not well documented. Uncertainties arising from the remaining factors are not quantified in this study. Uncertainty of ±5 % is assumed in this study. One pyranometer is oriented in the SW Plane of Array (PoA) and the other is oriented in the NE facing PoA.

Energy Meter: ABB A44
Four separate ABB A44 energy meters are used to measure the following quantities:

1. Electrical Load Meter – measures the total power consumed by the load. The meter is placed directly in front of the main AC distribution panel.
2. Grid Power Meter– measures power imported from the grid (no energy is exported to the grid). The meter is placed at the connection point to the grid.
3. Grid/DG Power Meter – this meter is downstream from the grid connection point and the DG. It measures power imported from the grid, as well as power generated by the DGs.
4. PV Output power Meter – measures power exported to the AC bus from the PV system. The meter is placed following the AC output of the PV inverters.

The location of each meter in the system is shown in Figure 1.4. The ABB A44 energy meters are rated class B (Cl 1) with a measurement accuracy of ±1%. The 3-phase meter measures line voltages $\mathrm{V1, V2, V3}$, line currents $\mathrm{I1, I2, I3}$, and line power factor angles $\varphi 1, \varphi 2, \varphi 3$. The meter internally calculates total power $\mathrm{P_{tot}}$ as per Equation 1.1,

$$P_{tot} = V1(I1)\cos\varphi 1 + V2(I2)\cos\varphi 2 + V3(I3)\cos\varphi 3 \pm 1\% \qquad \textit{Equation 1.1}$$

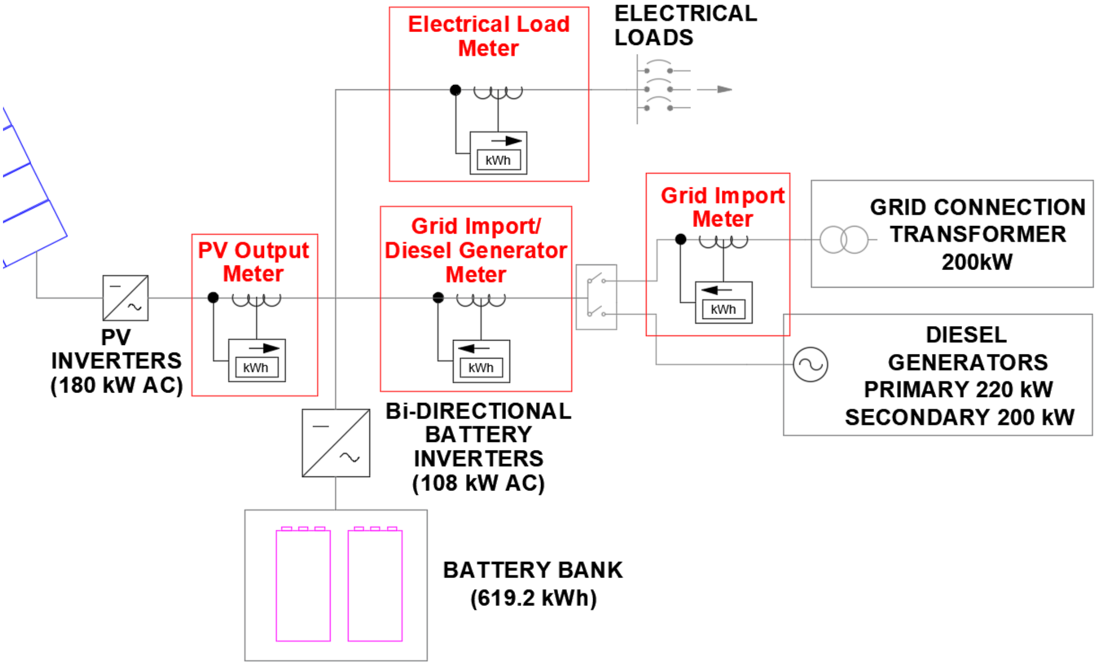

Figure 1.4: Locations of the four ABB A44 energy meters in the K-HES.

Voltage and frequency meter: Intelipro G59

The intelipro G59 is used to measure 3-phase grid voltage and frequency. The voltage accuracy is ±1 % at 20˚C, and between 50Hz - 60Hz. Over the complete frequency and temperature range of the device, -30˚C -80˚C and 30 Hz -70 Hz, the accuracy is ±1.5 %. The G59 measurements are used to enforce the grid disconnection limits as described in Table 1.7. Measurement uncertainty of ±1.5 % is assumed in this study.

Temperature sensors: Pt100 thin film element

The Pt100 element is adhered to the back of a PV module to measure cell temperature. The single module temperature is assumed to be representative of all PV modules. There is also a Pt100 element adhered to a battery in each battery bank to measure cell temperatures; for calculations the average temperature is taken. The outdoor ambient temperature and the battery room temperature are also measured with Pt100 elements. The Pt100 element is classified under IEC 60751 class B. The measurement uncertainty, μ_T, is defined by Equation 1.2,

$$\mu_T = \pm(0.3 + 0.005|T|) \qquad \textit{Equation 1.2}$$

where, $|T|$ is the absolute value of the measured temperature in ˚C. Equation 1.2 is valid for measured temperatures between -196˚C and 500˚C.

Measurement instruments and their associated uncertainties are summarized in Table 1.8.

Table 1.8: Summary of measurement instruments and measurement uncertainties.

Instrument	***Model***	***Measured Variable***	***Measurement Uncertainty***
Pyranometer	Kipp and Zonen SMP11	Irradiance (W/m^2)	±5 %
Energy meter	ABB A44	Power (kW)	±1 %
Voltage and frequency meter	Intelipro G59	Voltage and frequency (V, Hz)	±1.5 %
Module and ambient temperature sensors	Pt100 thin film element	Temperature (°C)	$\pm(0.3 + 0.005\lvert T \rvert)°C$

1.4.2. Performance Evaluation

A bespoke data analysis tool has been developed in Excel to process measurement data and evaluate the performance of the K-HES. Excel was chosen as the platform for this tool because it allows numerous logical operations and mathematical calculations to be efficiently computed. Furthermore, Excel is a widely used and accessible software in the business environment, which makes the proposed method more easily adopted. The parameters listed below are generated with this tool. They have been selected according to the available measurement data and their ability to characterize system performance. When applicable, parameters are calculated for each measurement interval in the dataset. Detailed calculations for each parameter are described in Section 2.

1. PV energy, kWh
2. DG energy, kWh, and average DG power, kW
3. Grid energy, kWh, and average grid power, kW
4. Load energy, kWh, and average load power, kW
5. Total energy production, kWh
6. Battery charge, kWh
7. Battery discharge, kWh
8. PV energy to charge batteries, kWh
9. DG energy to charge batteries, kWh
10. Grid energy to charge batteries, kWh
11. PV energy to the load, kWh
12. DG energy to the load, kWh
13. Grid energy to the load, kWh
14. DG runtime, min
15. Grid availability, yes/no
16. Length of power outage, min
17. PV Inverter efficiency, %
18. Theoretical PV energy, kWh
19. Nominal PV energy, kWh

Analytical methods and logical operations are utilized to interpret the significance of parameters above. In turn, interventions are formulated to improve the performance of the K-HES. iHOGA is then employed to predict the outcome of the proposed interventions through simulation analysis.

1.4.3. Simulation Analysis

iHOGA is commonly known as a state-of-the-art simulation software for sizing and optimizing HESs. In this study it is used to model, simulate, and optimize the performance of the K-HES. Each of the K-HES components is modelled in the simulation environment by a set of characteristic parameters. The load and grid availability profiles are generated from measurement data and subsequently imported into the software. iHOGA uses monthly irradiation data to generate synthetic hourly irradiation values which are then calibrated to the measurement data. The simulation analysis is executed over discrete timesteps of 1 hour. iHOGA is comparable to HES software such as HOMER, SAM, Hybrid 2, RETScreen, INSEL, Hybrids, and HySim. An independent review and comparison of many HES simulation software packages, including iHOGA and HOMER is conducted in [27]. There are several pros and cons to all software packages; iHOGA has been selected as an alternative to other software for several reasons:

- Multi-objective optimization. iHOGA is not limited to single-objective optimization functions. Software using single-objective optimization functions, for example HOMER, are restricted to minimizing either NPC or system emissions. iHOGA, on the other hand, can find solutions that simultaneously minimize NPC and emissions using multi-objective optimization functions.
- Advanced, customizable, multi-year model for estimating the lifetime of lead acid batteries. The Schieffer lead acid battery model is specifically intended for lead acid batteries in HES applications [28]. Critical to this study, is the model's ability to account for capacity degradation and the impacts of temperature. The Schieffer battery model has been shown to be more accurate than the lead acid battery models used by comparable software [29]. The model is discussed further in Section 4.1.5.
- Custom grid availability profile. The grid model allows the user to define availability with a temporal resolution of one hour for one year. The complex behavior of the connection to the grid can thus be included in the simulation analysis by using measurement data to construct the availability profile.
- Compensation for auto discharging of the batteries when the batteries are in standby mode. This feature is not included in similar software packages but is observed in all lead acid battery systems.
- Detailed cost model. Component level costs can be defined, including initial capital, replacement, O&M, inflation (general inflation, as well as additional inflation of energy storage and diesel fuel costs can be defined). Fixed initial and annual costs are also included.
- Customizable bi-Inv model includes: separate limits for charging and discharging currents, time dependent surge power, power dependent inverter efficiency, and rectifier efficiency.
- The DG model accounts for increased ageing and O&M costs due to operation outside of optimal running conditions. This feature is not available in HOMER.
- Detailed account of CO_2 emissions. Emissions generated from fuel consumption and externalities can be accounted for in the model.

iHOGA has two fundamental roles in this study. The first role is to create a base-case model which accurately simulates the performance of the existing K-HES. The purpose of the base-case model is to generate KPIs which can be used as a point of reference. The second role is to model each of the proposed interventions, and thereby generate a new set of KPIs that can be directly compared to the base-case. Variations of the simulated KPIs will be the basis for determining the effectiveness of each proposed intervention.

2 Calculations

The parameters listed in Section 1.4.2 are calculated by the performance evaluation tool developed in Excel. As previously stated, the tool imports measurement data and subsequently performs a series of logical operations and calculations. Each of the operations and calculations is described in this section.

The measurement period began on October 12th, 2017 and ended on April 12th, 2018, spanning 183 days, and 17472 measurement intervals. The measurement resolution is one minute, but values are averaged and saved over 15-minutes intervals. The measured variables are summarized in the first column of Table 2.1. Measurement data has been filtered to exclude non-physical values believed to have occurred from random instrument errors. Large positive and negative values lying outside of the maximum and minimum limits in Table 2.1 are discarded from the dataset. The number of discarded values above/below the limiting values is shown in brackets. Less than 0.03 % of all measurement values have been discarded from the dataset. The impact of discarding these values has not been investigated in this study.

All operations and calculations are conducted assuming:

- All measurement intervals are exactly 15 minutes.
- The measurement instruments are installed according to the manufacturers' recommendations and requirements, and accordingly that they meet the manufacturers' reported uncertainties.
- For power measurements, during intervals when components are started or stopped (i.e. the component does not operate over the entire 15-minute interval), the measured power is artificially lowered. For example, if the grid operates for 10 minutes at 50 kW, and 5 minutes at 0 kW, the average power recorded over the 15-minute interval is 33 kW. Therefore, start and stop values are discarded when calculating minimum power values and average power values. The impact of discarding these values is not considered.
- The temporal resolution captures the characteristic system behavior i.e. all sub-minute fluctuations of the measured values are assumed to average out across the entire dataset.

Additional assumptions are discussed throughout this section as they arise. All calculation uncertainties are propagated according to the error analysis guidelines described in [30], which are briefly summarized here:

if variables x,…,w, with uncertainties δx,…,δw, are used to calculate the value q then there is an uncertainty δq, which depends on the operations performed on variables x,…,w, as follows:

if q is the sum and/or the difference, i.e. $q = x + \cdots + z - (u + \cdots + w)$ then,

$$\delta q \leq \delta x + \cdots + \delta z + \delta u + \cdots \delta w \qquad \textit{Equation 2.1}$$

if q is the product and/or quotient, i.e. $q = \frac{x * \ldots * z}{u * \ldots * w}$ then,

$$\delta q \leq \frac{\delta x}{|x|} + \cdots + \frac{\delta z}{|z|} + \frac{\delta u}{|u|} + \cdots + \frac{\delta w}{|w|} \qquad \textit{Equation 2.2}$$

if q the product and/or quotient of a scaler with an exact known value B, i.e. $q = Bx$, then

$$\delta q = |B|\delta x \quad \text{Equation 2.3}$$

The uncertainty of each measurement from Section 1.4.1 is summarized in Table 2.1.

Table 2.1: Summary of measurement variables and limits for discarding outlying measurement values.

Measured Variables	***Unit***	***Symbol***	***Measurement Uncertainty***	***Maximum Value, (measurements discarded)***	***Minimum Value, (measurements discarded)***
Grid/DG Power	kW	$P_{\mathrm{Gr/G}}$	± 1 %	120 (18)	0 (1)
Grid Power	kW	P_{Gr}	± 1 %	120 (8)	0 (0)
Load Power	kW	P_{L}	± 1 %	120 (1)	0 (0)
PV Power	kW	P_{PV}	± 1 %	250 (3)	0 (0)
Voltage L1	V	V_1	± 1.5 %	300 (3)	0 (0)
Voltage L2	V	V_2	± 1.5 %	300 (0)	0 (0)
Voltage L3	V	V_3	± 1.5 %	300 (0)	0 (0)
Irradiance PoA (SW)	W/m^2	$G_{\mathrm{PoA-SW}}$	± 5 %	1200 (2)	0 (0)
Irradiance PoA (NE)	W/m^2	$G_{\mathrm{PoA-NE}}$	± 5 %	1200 (2)	0 (0)
Module Temperature	°C	T_{PV}	$\pm(0.3 + 0.005\lvert T_{\mathrm{PV}}\rvert)$	100 (0)	0 (0)
Battery Temperature	°C	T_{batt}	$\pm(0.3 + 0.005\lvert T_{\mathrm{batt}}\rvert)$	50 (0)	0 (16)
Battery Room Temperature	°C	$T_{\mathrm{batt/R}}$	$\pm(0.3 + 0.005\lvert T_{\mathrm{batt/R}}\rvert)$	50 (0)	0 (0)
Ambient Temperature	°C	T_{amb}	$\pm(0.3 + 0.005\lvert T_{\mathrm{amb}}\rvert)$	50 (0)	0 (0)

2.1.1. PV Energy

The measured AC power generated by the PV system $P_{PV\,(i)}$ over interval i is used to calculate PV energy $E_{\mathrm{PV\,(i)}}$,

$$\mathrm{E}_{\mathrm{PV\,(i)}} = \mathrm{P}_{\mathrm{PV\,(i)}}\left(\frac{1}{4}\right)\ \mathrm{kWh} \quad \text{Equation 2.4}$$

2.1.2. DG Energy

The DG power $P_{\mathrm{DG\,(i)}}$ over each interval i is not measured directly. The grid/DG power measurement $P_{\mathrm{Gr/DG(i)}}$ is the sum of the grid power $P_{\mathrm{Gr(i)}}$ and the DG power over interval i; therefore, the DG power is simply the difference:

$$P_{\mathrm{DG\,(i)}} = P_{\mathrm{Gr/DG(i)}} - P_{\mathrm{Gr(i)}}\ kW \quad \text{Equation 2.5}$$

The grid and DG cannot operate simultaneously, which means in most cases when calculating $P_{\mathrm{DG(i)}}$ the grid power is zero. However, it is possible to have non-zero values for $P_{\mathrm{DG\ (i)}}$ and $P_{\mathrm{Gr(i)}}$ during intervals when the load is transferred from the grid to the DG, or vice versa. The DG energy $E_{\mathrm{DG\ (i)}}$ over interval i is:

$$\mathrm{E_{DG\ (i)}} = \mathrm{P_{DG\ (i)}}\left(\frac{1}{4}\right)\mathrm{kWh} \qquad \textit{Equation 2.6}$$

2.1.3. Grid Energy

The average grid power $P_{\mathrm{Gr(i)}}$ over interval i is used to calculate grid energy $E_{\mathrm{Gr(i)}}$:

$$\mathrm{E_{Gr\ (i)}} = \mathrm{P_{Gr\ (i)}}\left(\frac{1}{4}\right)\mathrm{kWh} \qquad \textit{Equation 2.7}$$

2.1.4. Total Energy Production

The systems total energy production $E_{\mathrm{T(i)}}$ from all sources over interval i is the sum of the energy produced by all power generators:

$$\mathrm{E_{T\ (i)}} = \mathrm{E_{PV\ (i)}} + \mathrm{E_{DG\ (i)}} + \mathrm{E_{Gr\ (i)}}\ \mathrm{kWh} \qquad \textit{Equation 2.8}$$

2.1.5. Load Energy

The average load power $P_{L(i)}$ over interval i is used to calculate load energy $E_{L(i)}$:

$$E_{\mathrm{L\ (i)}} = P_{\mathrm{L\ (i)}}\left(\frac{1}{4}\right)kWh \qquad \textit{Equation 2.9}$$

2.1.6. Battery Charge and Discharge Energy

There is no energy meter monitoring the battery system; therefore, the charged and discharged energy over each time step are calculated as the difference between the total energy production and consumption. In other words, the charge energy $E_{\mathrm{Chrg(i)}}$ and discharge energy $E_{\mathrm{DChrg(i)}}$ is taken as the difference between the total energy production and the load as follows:

$$\begin{aligned}
&if\ E_{\mathrm{T\ (i)}} - E_{\mathrm{L\ (i)}} > 0 \text{ then,}\\
&\qquad E_{\mathrm{Chrg\ (i)}} = E_{\mathrm{T\ (i)}} - E_{\mathrm{L\ (i)}}\ kWh\\
&else,\\
&\qquad E_{\mathrm{Chrg\ (i)}} = 0\ kWh
\end{aligned} \qquad \textit{Equation 2.10}$$

$$\begin{aligned}
&if\ E_{T\ (i)} - E_{L\ (i)} < 0\ then,\\
&\qquad E_{DChrg\ (i)} = E_{T\ (i)} - E_{L\ (i)}\ kWh\\
&else,\\
&\qquad E_{DChrg\ (i)} = 0\ kWh
\end{aligned} \qquad \textit{Equation 2.11}$$

It has been assumed that electrical losses between the metered points are negligible. The transmission distance between the energy meters and the battery system range between 5m to 300m. The calculation method does not neglect electrical losses, but rather attributes all losses to the battery system.

2.1.7. PV Charge Energy

As per the control philosophy, PV energy is assumed to be directed towards the load before any other source. However, when it comes to charging, the grid and the DG are assumed to supply charging energy ahead of the PV system. In other words, over each interval i, PV energy is only assumed to charge the batteries when the cumulative energy from the grid and DG does not account for the total charging energy. This definition of PV charging ensures no charging energy is attributed to the PV system while $\mathrm{P_{PV\ (i)}} < P_{\mathrm{L(i)}}$.

The PV energy to charge the batteries E_{PVChrg} is calculated as the PV energy minus the load energy, DG charging $\mathrm{E}_{DGChrg\ (i)}$, and grid charging energy $E_{\mathrm{GrChrg(i)}}$. The logic is expressed as follows:

$$if\ E_{\mathrm{PV\ (i)}} > (E_{\mathrm{L\ (i)}} + \mathrm{E}_{\mathrm{DGChrg\ (i)}} + E_{\mathrm{GrChrg(i)}})\ then,$$
$$E_{PVChrg\ (i)} = E_{PV\ (i)} - E_{L\ (i)} - E_{DGChrg\ (i)} - E_{GrChrg(i)}\ kWh$$
$$else,$$
$$E_{PVChrg\ (i)} = 0\ \ kWh$$

Equation 2.12

2.1.8. DG Charge Energy

If the DG and one or both of the other energy sources supply the load during a measurement interval while the battery is charging, there is a mix of energy sources charging the batteries. There is no distinction for identifying the origin of power directed to the batteries versus power to the load. Therefore, the determination is subjective. In such cases, the DG is assumed to charge the batteries. If the total DG energy is greater than or equal to the total charging energy, then all charging energy is attributed to the DG. Otherwise, grid energy and/or PV energy also contributed to charging over the measurement interval, and DG charging energy is simply equal to the total DG energy over the interval. This logic is defined by the following operations and calculations:

$$if\ E_{Chrg\ (i)} > 0\ and\ E_{DG\ (i)} > 0\ then,$$
$$if\ E_{DG\ (i)} > E_{Chrg\ (i)}\ then,$$
$$E_{DGChrg\ (i)} = E_{Chrg\ (i)}\ kWh$$
$$else,$$
$$E_{DGChrg\ (i)} = E_{DG\ (i)}\ kWh$$
$$else,$$
$$E_{\mathrm{DGChrg\ (i)}} = 0\ kWh$$

Equation 2.13

2.1.9. Grid Charge Energy

Following the logic from the previous two sections, grid energy to charge the batteries $E_{\mathrm{GrChrg(i)}}$ is calculated as follows:

$$if\ E_{Chrg\ (i)} > 0\ and\ E_{Gr\ (i)} > 0\ then,$$
$$if\ (E_{Chrg\ (i)} - E_{DGChrg\ (i)}) > E_{Gr\ (i)}\ then,$$
$$E_{GrChrg\ (i)} = E_{Gr\ (i)}\ kWh$$
$$else,$$
$$E_{GrChrg\ (i)} = E_{Chrg\ (i)} - E_{DGChrg\ (i)}\ kWh$$
$$else,$$
$$E_{\mathrm{GrChrg\ (i)}} = 0\ kWh$$

Equation 2.14

2.1.10. PV Energy to the Load

All PV power is delivered directly to the load unless surplus PV power is found to charge the batteries as per Equation 2.12. Therefore, the PV energy to the load $E_{\mathrm{PV-L}}$ is:

$$\mathrm{E}_{\mathrm{PV-L\ (i)}} = \mathrm{E}_{\mathrm{PV\ (i)}} - \mathrm{E}_{\mathrm{PVChrg\ (i)}}\ \mathrm{kWh}$$

Equation 2.15

2.1.11. DG Energy to the Load

All DG power is delivered directly to the load unless surplus DG power is found to charge the batteries as per Equation 2.13. Therefore, the DG energy to the load $E_{\mathrm{DG-L}}$ is:

$$\mathrm{E}_{\mathrm{DG-L\ (i)}} = \mathrm{E}_{\mathrm{DG\ (i)}} - \mathrm{E}_{\mathrm{DGChrg\ (i)}} \qquad \textit{Equation 2.16}$$

2.1.12. Grid Energy to the Load

All grid power is delivered directly to the load unless surplus grid power is found to charge the batteries as per Equation 2.14. Therefore, the grid energy to the load, $E_{\mathrm{Gr-L}}$, is:

$$\mathrm{E}_{\mathrm{Gr-L\ (i)}} = \mathrm{E}_{\mathrm{Gr\ (i)}} - \mathrm{E}_{\mathrm{GrChrg\ (i)}}\ \mathrm{kWh} \qquad \textit{Equation 2.17}$$

2.1.13. DG Runtime

For each measurement interval when the DG energy production is greater than 6 kWh, the DG is assumed to be running for the entire duration of the measurement interval. The total DG runtime at the end of each interval, $t_{\mathrm{DG(i)}}$, is determined by:

$$\begin{aligned} & if\ E_{\mathrm{DG(i)}} > 6\ kWh\ \ then, \\ & \qquad t_{\mathrm{DG(i)}} = t_{\mathrm{DG(i-1)}} + 15\ min \\ & else, \\ & \qquad t_{\mathrm{DG(i)}} = 0\ min \end{aligned} \qquad \textit{Equation 2.18}$$

The 6 kWh limit is defined to average the distribution of DG runtime during intervals when the DG is started or stopped. The average DG power of 49 kW corresponds to 12 kWh of energy production over a 15-minute interval. It is assumed, on average, if the DG runs for more than half the interval it will generate more than 6 kWh. Similarly, if the DG runs for less than half of the interval it will generate less than 6 kWh. Therefore, the calculation error is assumed to be zero during intervals when DG runs for the entire interval and ±7.5 minutes for intervals when the DG starts or stops.

2.1.14. Grid Availability

If the voltage across each phase, V_1, V_2, V_3 is non-zero and balanced, the grid is defined as available. The convention adopted for grid availability GA, is $GA = 1$ if the grid is available, and $GA = 0$ if the grid is unavailable. GA is determined as follows:

$$\begin{aligned} & if\ (V_1 + V_2 + V_3) > 0\ and, \\ & if\ \frac{Max(|V_1 - V_2|, |V_1 - V_3|, |V_2 - V_3|)}{\frac{(V_1 + V_2 + V_3)}{3}} < 0.03\ then, \\ & \qquad GA_{\mathrm{(i)}} = 1 \\ & else, \\ & \qquad GA_{\mathrm{(i)}} = 0 \end{aligned} \qquad \textit{Equation 2.19}$$

Where $Max(|V_1 - V_2|, |V_1 - V_3|, |V_2 - V_3|)$ returns the maximum absolute value of the differences in voltages across the phases.

As will be shown in Section 4.1.4, the grid availability is required in hourly format to be integrated with the iHOGA simulation software. If the average GA over each hour is greater than the 0.5 then GA is said to equal 1, otherwise $GA = 0$.

Note: the method for calculating GA assumes that all grid outages are identified by voltage imbalance. Manual inspection of the cause for grid outages confirms this is a reasonable assumption, but a quantitative analysis is not conducted.

2.1.15. Length of Power Outages

The length of each power outage t_{PO} must be rounded to intervals of 15-minutes, and determined based on the grid availability in the current and previous intervals by the following logic:

$$
\begin{aligned}
&if\ GA_{(\mathrm{i})} = 0\ then, \\
&\qquad t_{\mathrm{PO(i)}} = t_{\mathrm{PO(i-1)}} + 15\ min \\
&else, \\
&\quad t_{PO(i)} = 0\ min
\end{aligned}
$$
Equation 2.20

2.1.16. Theoretical PV Energy

The PV power is measured; however, to estimate PV curtailment – which occurs when the maximum available PV power cannot be distributed to the load or to charge the batteries – the theoretical PV energy E_{PVth} must be calculated for comparison. E_{PVth} is calculated over each 15-mintute interval as follows:

$$E_{\mathrm{PV-NE(i)}} = 0.25 \cdot PV_{\mathrm{CapNE}} \cdot PV_{\mathrm{derate}} \cdot \eta_{\mathrm{Inv(i)}} \left(\frac{G_{\mathrm{PoA-NE(i)}}}{G_{\mathrm{STC}}}\right)\left(1 + PV_{\mathrm{Pcoeff}} \cdot \left(T_{\mathrm{PV(i)}} - PV_{\mathrm{TSTC}}\right)\right)\ kWh$$
Equation 2.21

Where, $E_{\mathrm{PV-NE(i)}}$ is the estimated energy production from the NE facing PV arrays over interval i.

$$E_{PV-SW(i)} = 0.25 \cdot PV_{CapSW} \cdot PV_{derate} \cdot \eta_{\mathrm{Inv(i)}} \left(\frac{G_{\mathrm{PoA}-SW(i)}}{G_{STC}}\right)\left(1 + PV_{Pcoeff} \cdot \left(T_{\mathrm{PV(i)}} - PV_{\mathrm{TSTC}}\right)\right)\ kWh$$
Equation 2.22

Where, $E_{\mathrm{PV-SW(i)}}$ is the total estimated energy production from the SW facing PV arrays over interval i.

Therefore, the total theoretical PV energy is:

$$E_{\mathrm{PVth(i)}} = E_{\mathrm{PV-NE(i)}} + E_{\mathrm{PV-SW(i)}}\ kWh$$
Equation 2.23

Constants

- $PV_{\mathrm{CapNE}} = 103.4\ \mathrm{kW\ DC}$: total PV capacity facing NE (from Table 1.3).
- $PV_{\mathrm{CapSW}} = 101.2\ \mathrm{kW\ DC}$: total PV capacity facing SW (from Table 1.3).
- $PV_{\mathrm{TSTC}} = 25\ ^{\circ}C$: PV module temperature under standard test conditions.
- $G_{STC} = 1000\,\frac{W}{m^2}$: irradiance under standard test conditions.
- $PV_{\mathrm{Pcoeff}} = 0.41\,\frac{\%}{^{\circ}C}$: PV module temperature coefficient of power.

Note: when calculating uncertainty, all constants are assumed to have an error of ±1 %.

Measurement Input Data

- $G_{\mathrm{PoA-NE}}$: pyranometer irradiance in the NE plane of array (PoA)
- $G_{\mathrm{PoA-SW}}$: pyranometer irradiance in the SW plane of array (PoA)
- T_{PV}: measured module temperature.

The use of PoA irradiation data measured with pyranometers (without modification of the data) has been justified by National Renewable Energy Laboratories (NREL). They have shown simulation models using PoA irradiation data consistently returns results with lower error when compared against traditional methods for calculating the PoA irradiance, such as transposing measured horizontal irradiation data into the PoA [31].

Inverter Efficiency, η_{Inv}
Inverter efficiency is calculated by linearly interpolating to the measured PV power on the inverter efficiency curve found in the inverter datasheet [19]. It has been assumed that:

1. When PV power is curtailed inverter efficiency is only affected to a small degree.
2. PV curtailment only occurs while the PV system is operating at powers higher than the load.

The inverter efficiency is relatively constant above 15 % of the total inverter power (15 % × 180 kW AC = 27 kW), where efficiency only ranges from 98.0 % to 98.4 %. Thus, when PV power is above 27 kW the inverter efficiency varies by ±0.4 %. It will be shown in Section 3 that the average load is well above 40 kW. PV curtailment is assumed to only occur when the PV power is greater than the load. Additionally, the string voltage (which fluctuates primarily depending on module temperature) impacts the inverter efficiency. To account for errors introduced by PV curtailment and string voltage, the uncertainty when calculating inverter efficiency is estimated to be ±2 %.

PV Derating, PV_{derate}
There are several factors that introduce losses to PV systems, such as: soiling, incident angle modifier, shading, power transmission, irradiance level, module quality, light induced degradation, module mismatch, and auxiliary power consumption. Estimating each of these factors is a difficult task, which can introduce significant uncertainties when calculating PV energy production. An alternative option is to use a general scaling term, PV_{derate}, to account for all losses. PV_{derate} can be estimated using measurement data; however, PV curtailment has a significant impact on the value of PV_{derate} by inflating apparent losses. Therefore, measurement data during periods of PV curtailment must not be used when estimating PV_{derate}.

The daily PV generation profile is used to identify periods of PV curtailment. An initial guess for PV_{derate} is required to generate an approximate generation profile, $PV_{\text{derate}} = 0.85$ is taken. From Equation 2.23, PV power is calculated and compared against the measured PV power to identify periods without PV curtailment. Twenty-one days were identified showing zero or very minimal PV curtailment, the criteria being a match in the shape of the PV production profiles. An example from January 10th, 2018 is shown in Figure 2.1. In comparison, most days throughout the measurement period show significant PV curtailment, as shown in Figure 2.2.

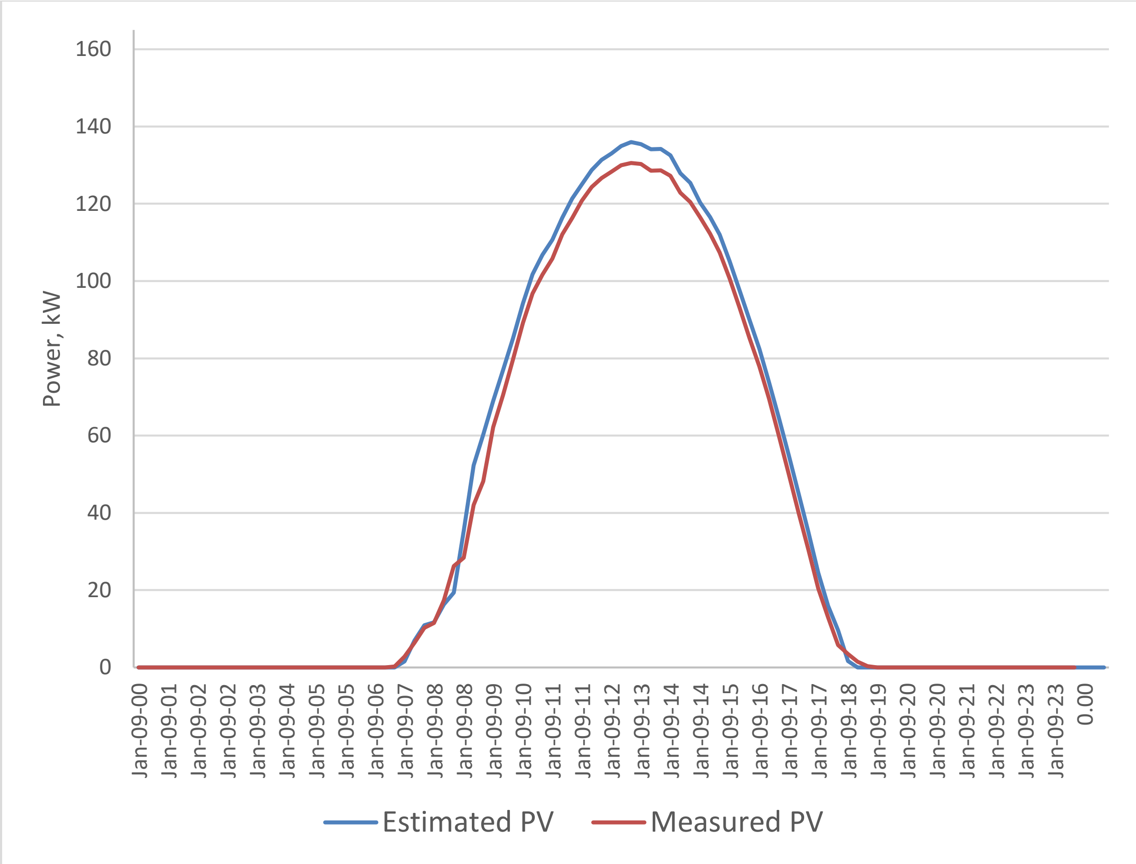

Figure 2.1: January 10th, 2018 estimated PV power compared to measured PV power where PV_{derate} *is assumed to be 0.85.*

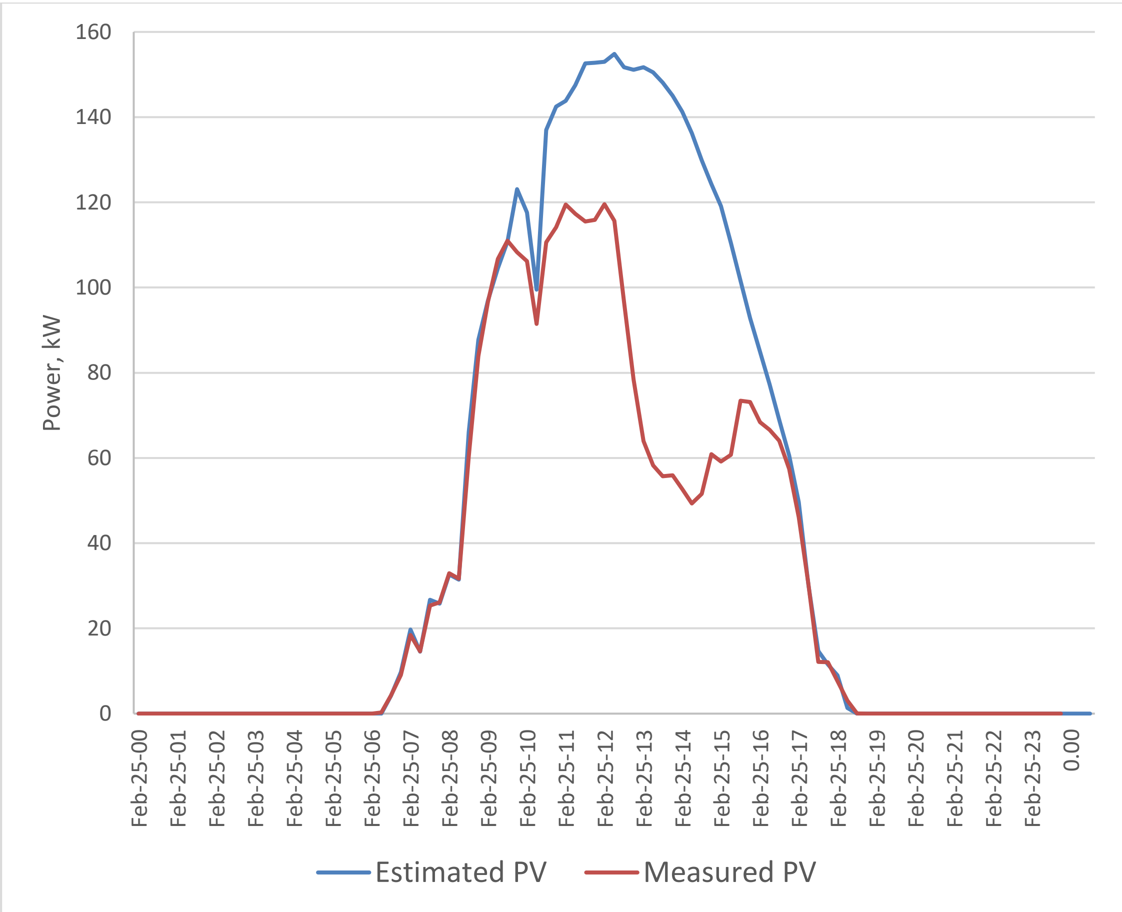

Figure 2.2: Example showing how PV curtailment is identified.

The 21 days identified showing no significant PV curtailment are used to solve for the value of $\mathrm{PV_{derate}}$ which minimizes the normalized root mean square error $NRMSE$ between the calculated and measured PV energy production.

$$NRMSE = \sqrt{\frac{\sum_{i=1}^{N}\left(E_{\mathrm{PVth(i)}} - \mathrm{E}_{\mathrm{PV\ (i)}}\right)^2}{N}} \qquad \textit{Equation 2.24}$$

where N is the number of measurement intervals over the 21 days identified and $E_{\mathrm{PVth(i)}}$ is the theoretical PV energy as per Equation 2.3. PV_{derate} is minimized using Excels built-in generalized reduced gradient nonlinear optimization solver tool. The minimized value of $NRMSE = 1.05$ gives $PV_{\mathrm{derate}} = 0.83$. The uncertainty in PV_{derate} has been estimated by solving for PV_{derate} for each of the 21 days identified: a maximum value of 0.86 and a minimum value of 0.81 were obtained. Therefore, the estimated uncertainty is ±3 %.

2.1.17. Nominal PV Energy

The method for calculating the theoretical PV energy, as per Section 2.1.16, is also used to calculate the nominal PV energy E_{PVnom}. The nominal PV energy is calculated assuming zero power losses excluding the impact of temperature on PV module performance. Therefore, when calculating E_{PVnom}, $PV_{\mathrm{derate}} = 1$, and $\eta_{\mathrm{Inv}} = 100\ \%$.

2.1.1. PV Curtailment

PV curtailment $E_{\mathrm{PVCurt(i)}}$ is calculated as the difference between $E_{\mathrm{PVth(i)}}$ and $\mathrm{E}_{PV(i)}$:

$$E_{\mathrm{PVCurt(i)}} = E_{\mathrm{PVth(i)}} - \mathrm{E}_{PV(i)}\ kWh \qquad \textit{Equation 2.25}$$

2.1.2. Averages, Sums, and Annual Projections

A generic equation is used to calculate the average value of variable V_{A} over the entire measurement period:

$$V_{\mathrm{A}} = \frac{\sum_{\mathrm{i}=1}^{n} V_{\mathrm{i}}}{n} \qquad \textit{Equation 2.26}$$

where V_{i} is the variable in question, and n is the number of non-zero measurements of V_{i} over the measurement period. Similarly, a generic equation is used to calculate sums of any variable V_{S} over the measurement period:

$$V_{\mathrm{S}} = \sum_{\mathrm{i}=1}^{n} V_{\mathrm{i}} \qquad \textit{Equation 2.27}$$

For example, V_{i} would assume the value of $P_{\mathrm{DG(i)}}$ when calculating the average DG power, or $E_{\mathrm{DG(i)}}$ when calculating the sum of DG energy over the measurement period. To project annual values from the dataset, sums over each variable are divided by the number of measurement days (183) and multiplied by 365. Therefore, it has been assumed that the measured values are representative of the entire year. In most locations, seasonal variations could be expected to skew the projection of these annual values. However, due to the proximity of the site to the equator, seasonal variations over the 6-month period are assumed to be negligible.

Projecting the annual PV energy using this method can be improved by considering the historical monthly global horizontal irradiation (GHI) and the timeframe of the measurement period. The timeframe is October 12th, 2017 to April 12th, 2018, spanning 183 days, which is approximately half of the year straddling the winter solstice. Examining the GHI from five different meteorological databases shows the total GHI is nearly equal for opposite halves of the year, see Table 2.2. On average (considering all 5 databases), there is 3.4 % more GHI during October to March than during April to September. Therefore, a scaling factor reducing the projected PV energy generated from April to September by -3.4 % is applied to account for the difference when projecting the annual PV energy yield.

Table 2.2: Comparison of GHI from 5 different meteorological databases showing half year totals and percentage differences for each database.

Month	**GHI** *kWh/m²/month*				
	SolarGIS	**Meteonorm**	**PVGIS-CMSAF**	**NASA**	**PVGIS-Helioclim**
January	197.9	160.7	157.8	199.0	168.0
February	190.3	140.0	150.4	189.8	158.5
March	202.5	156.3	177.9	205.8	174.5
April	185.7	151.2	150.0	186.0	158.4
May	182.6	161.3	148.5	186.9	157.5
June	172.2	147.4	140.4	175.8	148.8
July	180.1	156.1	146.0	180.1	155.9
August	193.3	156.9	152.5	194.1	165.2
September	195.4	156.1	155.7	195.9	166.8
October	194.6	157.3	163.1	192.5	168.0
November	178.5	151.4	147.0	179.7	157.8
December	193.2	151.9	151.3	194.7	165.5
Total Oct-Mar	**1157.0**	**917.6**	**947.4**	**1161.6**	**992.4**
Total Apr-Sept	**1109.3**	**929.0**	**893.1**	**1118.8**	**952.6**
Percent Difference Apr-Sept vs. Oct-Mar	4.3 %	-1.2 %	6.1 %	3.8 %	4.2%

3 System Analysis: Derivations of Proposed Interventions

In this section, interventions are formulated though analysis of the results generated with the Excel tool described in Section 2. All proposed interventions target reduced emissions and costs. The outcomes from each intervention are speculated through analytical and logical reasoning. As described earlier, simulation analysis is required to quantify changes in system performance. The iHOGA simulation models used to accomplish this are described in Section 4, and simulation results are then presented in Section 5.

Before deriving interventions, general results from Section 2 are discussed to familiarize the reader with the characteristic behavior of the K-HES.

The average allocation of power to the load/battery from each energy source is depicted in Figure 3.1. The areas shaded below the neon green line represent power delivered to the load, while areas above the neon green line represent charging power to the batteries. The black line is the average PV power. Several observations are made characterizing the typical behavior of the K-HES:

- The average load ranges between 38 kW overnight, to 61 kW during the day.
- PV power supplies the entire load for approximately 8 hours daily, from 8:00-16:00. The peak load is covered during this period and excess PV power is used to charge the batteries.
- Battery discharge occurs primarily between 2:00-8:00 and 16:00-18:00, which is consistent with the control measures.
- In the morning from 8:00-11:00, if PV power cannot supply the load, the remaining load is primarily covered by the grid or DG. Therefore, the battery is typically depleted by 8:00.
- After 11:00, if PV cannot supply the load, the remaining load is covered primarily by the battery system; thus, the battery is generally charged enough by this time to support the load when required.
- The average PV charging power during midday is 50 kW-60 kW.
- When the DG or the grid operates there is a small consistent charge on the battery bank; represented by the thin blue and green areas above the black load line.
- From 12:00-16:00, the shape of the average PV profile becomes concave: suggesting the PV output is typically curtailed in the afternoon.
- The primary sources of energy to the load between 18:00 and 2:00 are the DG and the grid.

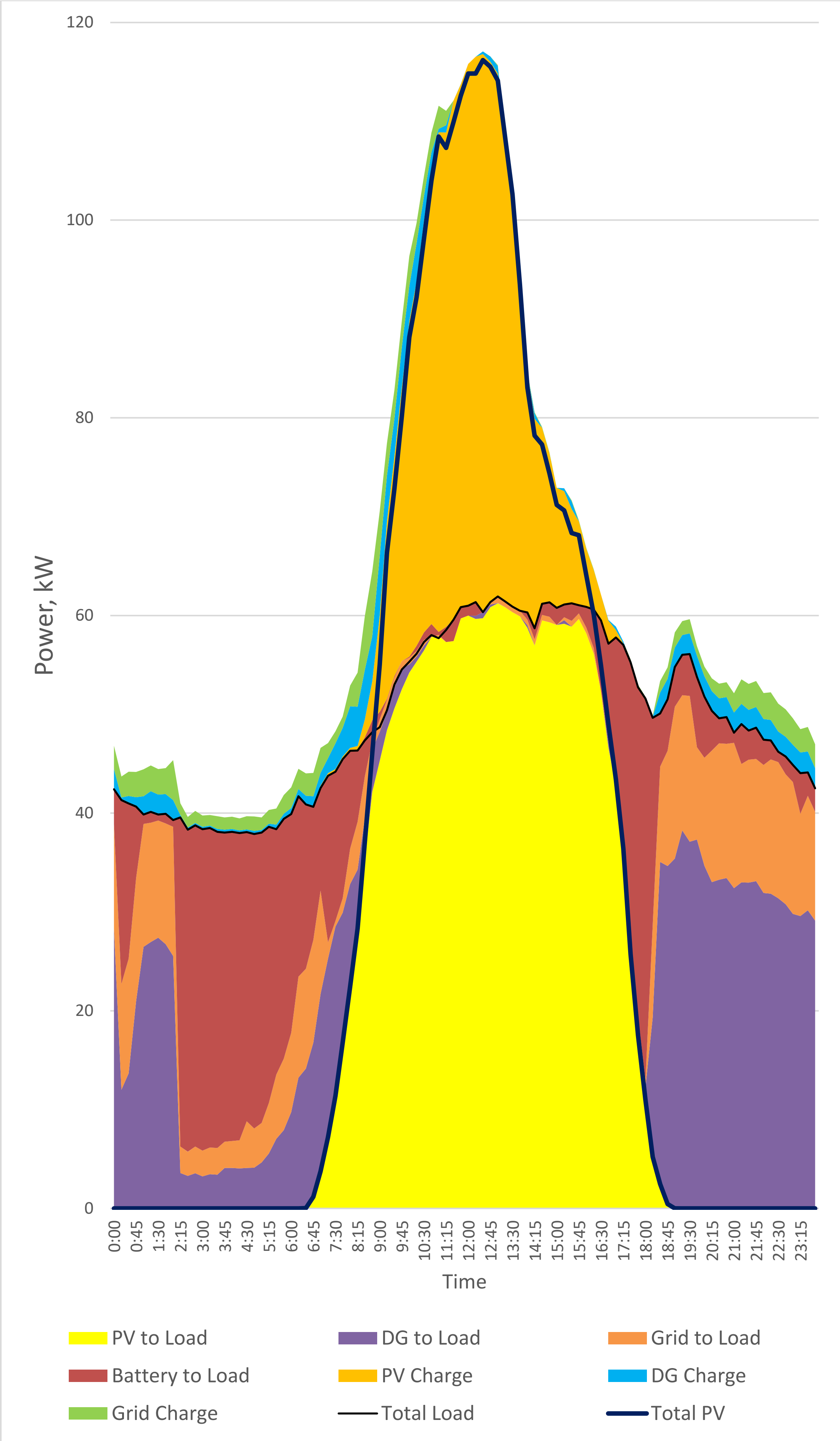

Figure 3.1: Average daily allocation of power to the load and batteries defined by source over the 6-month measurement period.

Shifting perspective from power to energy, Figure 3.2 shows the percentage of electricity produced by each generator in the system. The amount of electricity generated by the grid and DG is nearly equivalent, while the largest portion is generated by the PV system.

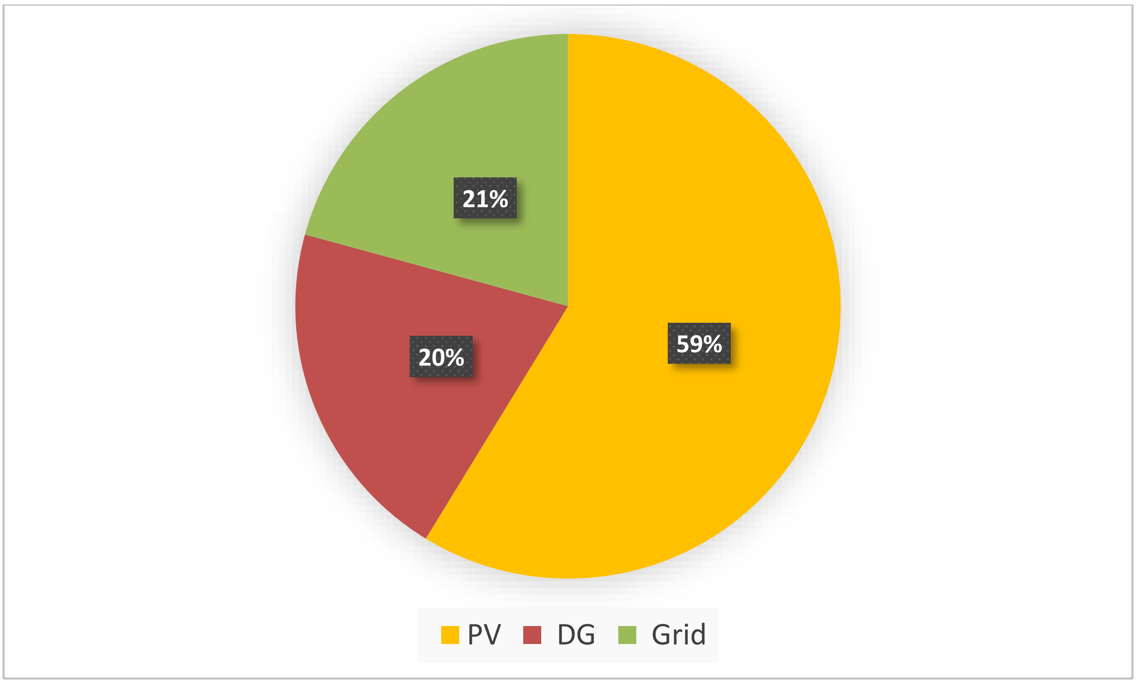

Figure 3.2: Total energy produced by percentage from each energy generator in the K-HES over the 6-month measurement period.

The battery must be considered when determining the source of electricity to the load. Figure 3.3 shows the battery provides 19 % of the load, which is nearly equal to the percentage from the grid 18 %, and the DG 19 %. Conversely, 44 % of the load is supplied directly by the PV system.

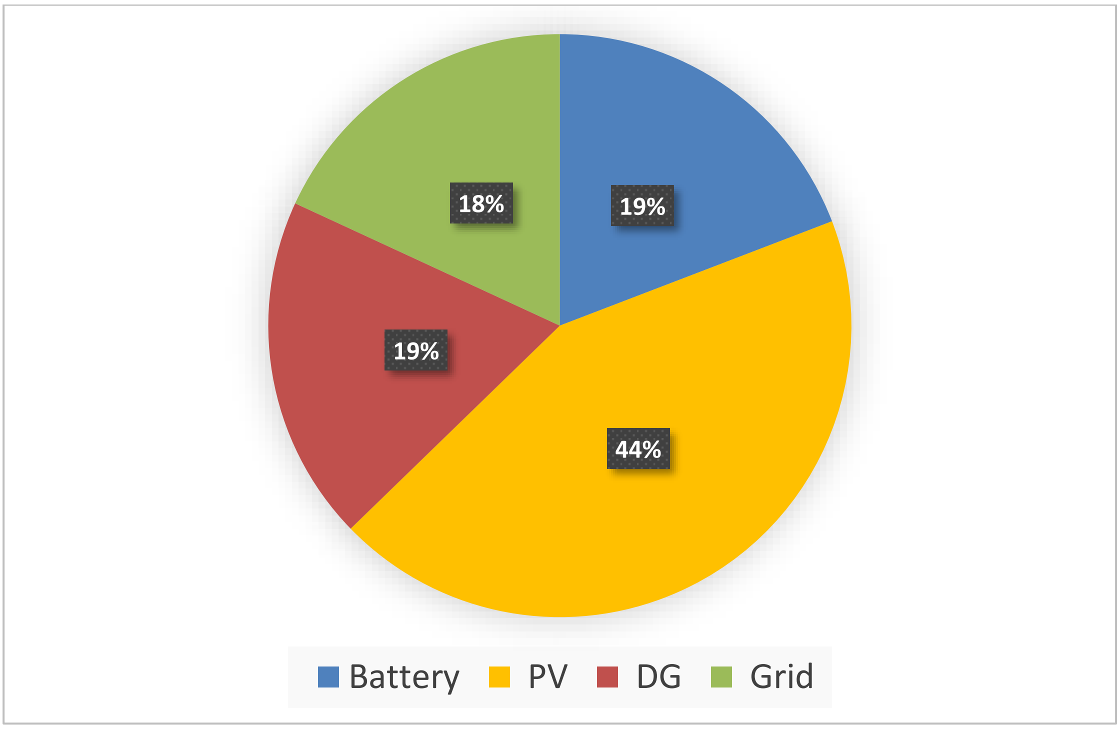

Figure 3.3: Percentage of energy to supply the load from each energy source in the K-HES over the 6 month measurement period.

Significant propagation of uncertainty prohibits an accurate calculation of electrical losses in the system. However, as a rough estimation, an energy balance on the system reveals

13200±4700 kWh more electrical energy is generated than consumed by the load (note, this value does not include PV curtailment losses). Assuming losses to be 13200 kWh, the battery system's round-trip efficiency is 76 %.

The charge electrical energy generated by each power generator is shown in Figure 3.4. Despite the control strategy, which aims to maximize PV charging and eliminate grid and DG charging, a cumulative 25 % of charge energy is attributed to the grid and DG.

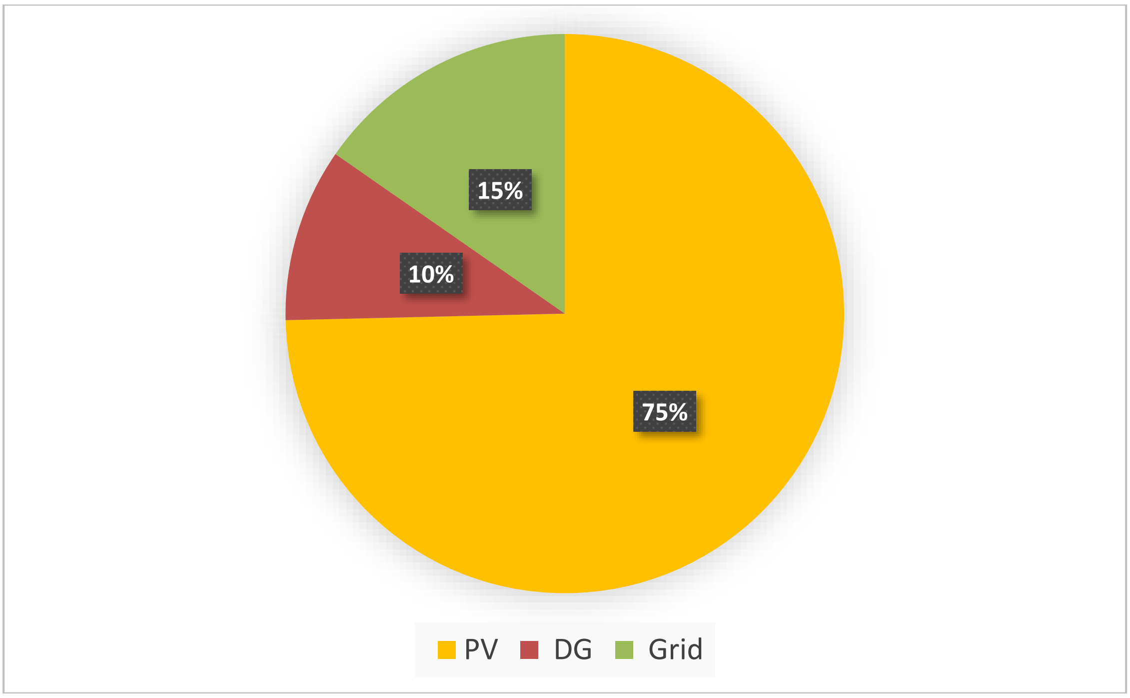

Figure 3.4: Percentage of energy charging the battery from each energy source in the K-HES.

The grid and DG charge energy is a result of an average charge of 3.6 kW on the battery bank when the grid or DG operates, as shown in Figure 3.5. The average charge of 3.6 kW is the minimum charging current permitted by the bi-Inv's.

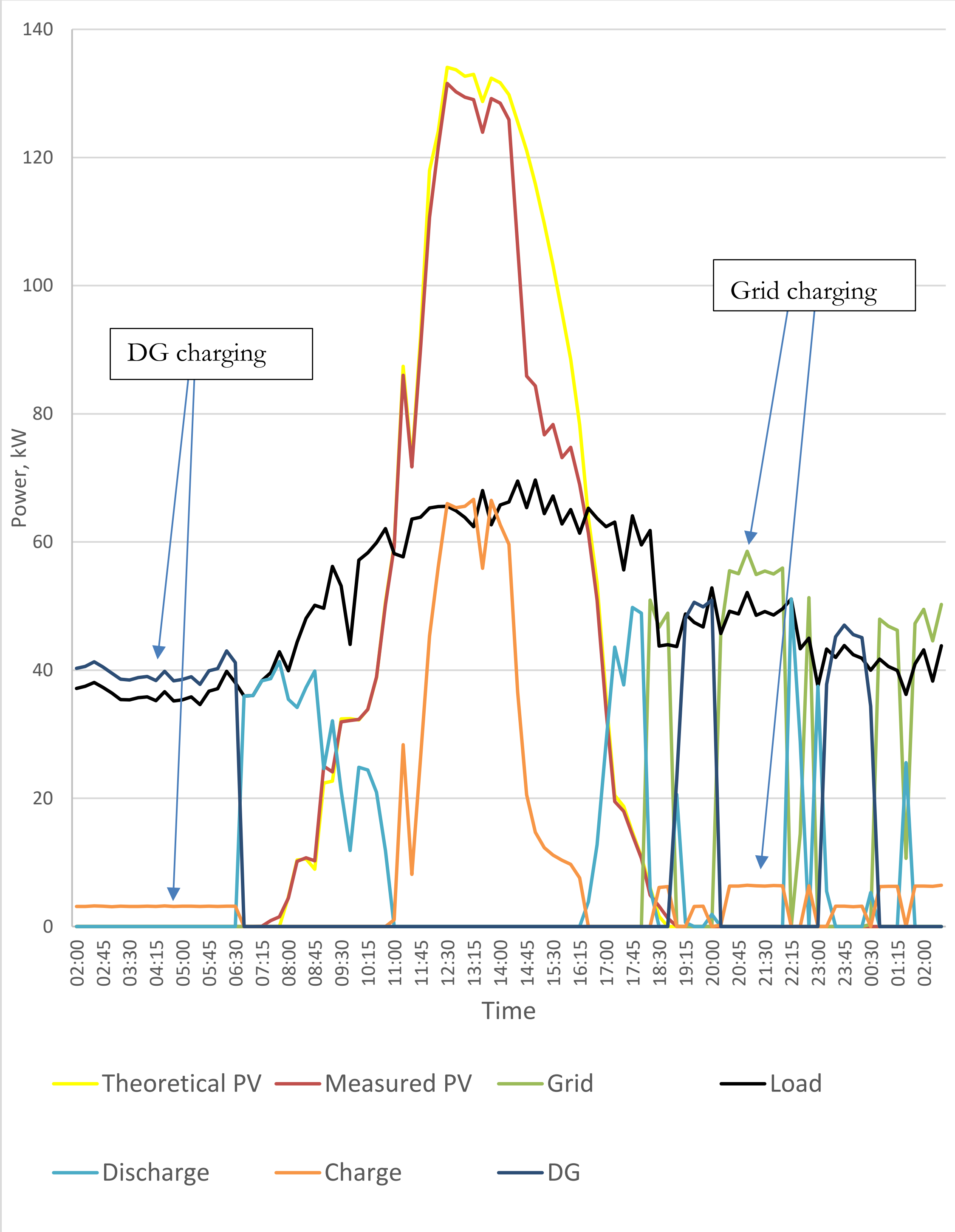

Figure 3.5: Example from January 2nd, 2018 showing grid and DG charging.

Table 3.1 contains the calculated parameters and their associated uncertainties for the quantities depicted in Figure 3.2-Figure 3.4. The annual projections for each quantity, calculated as per Section 2.1.2, are also found in this table.

Table 3.1: Summary of calculated and measured parameters. Sums are taken over the entire measurement period (Oct 12th, 2017-April 12th, 2017).

Parameter	***Symbol***	***Measurement Period***	***Uncertainty (±%)***	***Annual Projection***
PV Energy, kWh	E_{PV}	141400	1	277300
DG Energy, kWh	E_{DG}	49400	1	98500
Grid Energy, kWh	E_{Gr}	50000	1	100000
Load Energy, kWh	E_{L}	228000	1	453900
Total Energy Production, kWh	E_{T}	244000	1	480000
Battery Discharge, kWh	E_{DChrg}	44000	2	87000
Battery Charge, kWh	E_{Chrg}	57000	7	114000
PV Charge Energy, kWh	E_{PVChrg}	42000	5	84000
DG Charge Energy, kWh	E_{DGChrg}	6000	16	12000
Grid Charge Energy, kWh	E_{GrChrg}	9000	12	18000
PV Energy to Load, kWh	E_{PV-L}	90000	3	197000
DG Energy to Load, kWh	E_{DG-L}	44000	3	87000
Grid Energy to Load, kWh	E_{Gr-L}	41000	3	82000
DG Runtime, h	t_{G}	1000	8	2000
Nominal PV Energy, kWh	E_{PVnom}	204000	9	407000
Theoretical PV Energy, kWh	E_{PVth}	166000	13	331000
Estimated PV Curtailment, kWh	E_{PVCurt}	25000	95	50000

Throughout the following subsections, interventions are proposed to improve the performance of the K-HES. Each intervention is introduced with reference to the results presented above as well as additional supporting results.

3.1 Intervention #1: Reduce DG size

The average load on the DG is found to be 49.0±0.5 kW, which is less than the average load on the entire system of 53.2±0.5 kW. This is a result of the daily peak load typically being covered by the PV system, which is evident from Figure 3.1. The DG is rated for 220 kW; therefore, on average, it operates at 22 % of its rated output power. The widely accepted minimum recommended load for a DG is 30 % of its rated output power, and the optimal load is between 50-80 % [32]. Therefore, the DG is oversized, meaning it suffers from the effects of operating at low power, which results in poor fuel efficiency and premature ageing [17]. Likely as a direct consequence of this, the DG is in poor condition and no longer capable of delivering its rated output power. In fact, the DG struggles to output 100 kW. Measured fuel consumption data is not available; however, given the DG's poor condition, its fuel consumption may be higher than the manufacturer's reported values. Nonetheless, the manufacturer's reported values are assumed to be valid for the simulation work in this study. However, this is considered the most optimistic scenario given the condition of the DG. Referring to Figure 3.6, the difference in fuel efficiency when operating at 80 % (176 kW) versus 22 % (49 kW) of the rated output power is 12 %. If the same amount of energy were generated when operating at 176 kW, the 12 % change in fuel efficiency would correspond to a 50 % reduction in fuel consumption. In other words, an appropriately sized DG (with equivalent fuel efficiency) could reduce fuel consumption by up to 50 %. Therefore, the first proposed intervention to improve the performance of the K-HESs is to reduce the size of the DG. The simulation work in Section 5.1 aims to quantify fuel, emission, and cost savings which could be achieved by reducing the size of the DG.

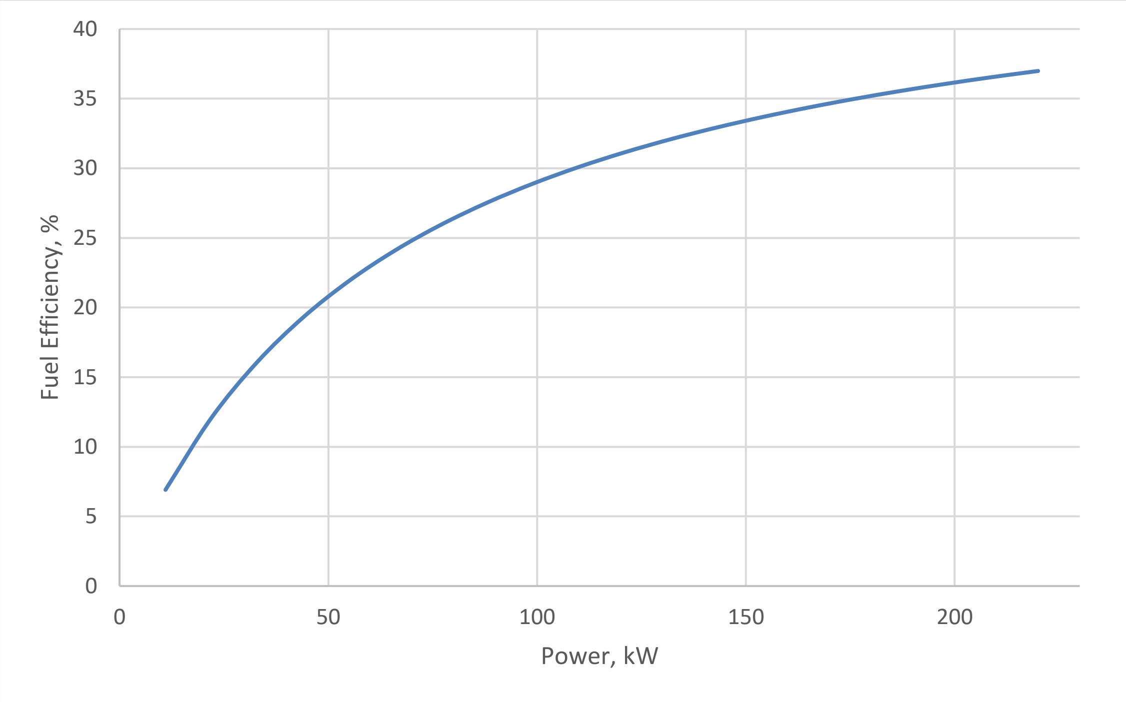

Figure 3.6: Primary DG rated 275 kVA/220 kW fuel efficiency curve under standby operating conditions derived from fuel consumption values in Table 1.4.

3.2 Intervention #2: Install A/C in the Battery Room

The battery system is housed in a heavily constructed concrete building with a large thermal mass. The building holds the indoor temperature relatively constant to the ambient temperature, see Figure 3.7. The battery temperature is typically 0.5-3°C above the battery room temperature, which is due to heat generation during charging and discharging. Table 3.2 contains the minimum, maximum, and average temperatures from Oct 12th 2017-April 12th 2018.

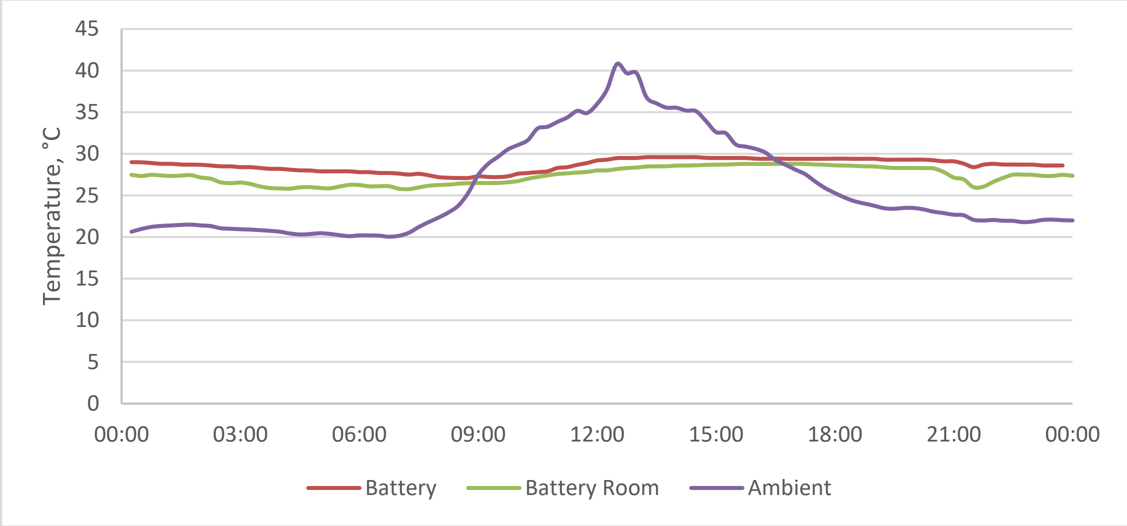

Figure 3.7: Typical daily temperature profile of battery, battery room, and ambient temperatures. Example from December 3rd, 2017.

Table 3.2: Average, maximum, and minimum temperatures from Oct 12th 2017 to April 5th 2018.

	Battery Temperature, °C	*Battery Room Temperature, °C*
Average	29	28
Maximum	34	33
Minimum	23	18

The lifetime of the batteries is directly related to the temperature in the battery room, which of course governs the average temperature of the batteries. Hoppecke, the manufacturer of the 16OPzS batteries states: "the battery service life will be halved in case the temperature rises by 10 K" [33], referring to a 10˚C rise above 20˚C. The average temperature of the battery room is 29˚C. Based on Hoppecke's statement, the service life of the battery is currently being reduced by more than 40 % when compared to a battery room temperature of 20˚C. The strong dependence of battery lifetime on temperature provides motivation for installing Air Conditioning (A/C) to cool the battery room and extend battery lifetime.

The tradeoff to installing A/C is the increased electrical load and the cost of purchasing and installing the A/C unit. A thermodynamic analysis of the battery room (conducted by P3r Engineers, see Appendix A) concluded that a peak electrical cooling load of 4 kW is required to keep the temperature of the battery room below 22˚C. The report did not provide the estimated energy consumption of the A/C unit. However, if it is assumed that 6 hours of cooling per day at peak capacity is enough to achieve an average battery room temperature of 20°C or below, then it can be shown the additional electrical load could be supplied primarily with PV power. With appropriate control measures it is foreseeable that all cooling could be achieved only with PV power. From Figure 3.1 it is clear the average PV power is more than 4 kW greater than the average load between 9:00-16:00. Figure 3.8 shows the relative quantities of energy to supply the load during this period: 95 % is directly from the PV system, and 3 % is from the battery (recall, the battery was previously shown to be charged with 75 % PV power). It is expected that adding a 4 kW load during this period will not significantly alter the relative quantities of energy which supply the load. Therefore, the majority of power used to run the A/C is expected to be generated by the PV system. Furthermore, if 6 hours of cooling at 4 kW is not enough to lower the battery room temperature to 20°C or below, a larger cooling capacity could be installed and run over the same timeframe. For example, using the same argument, an 8 kW cooling system would still be expected to utilize a high percentage of PV power.

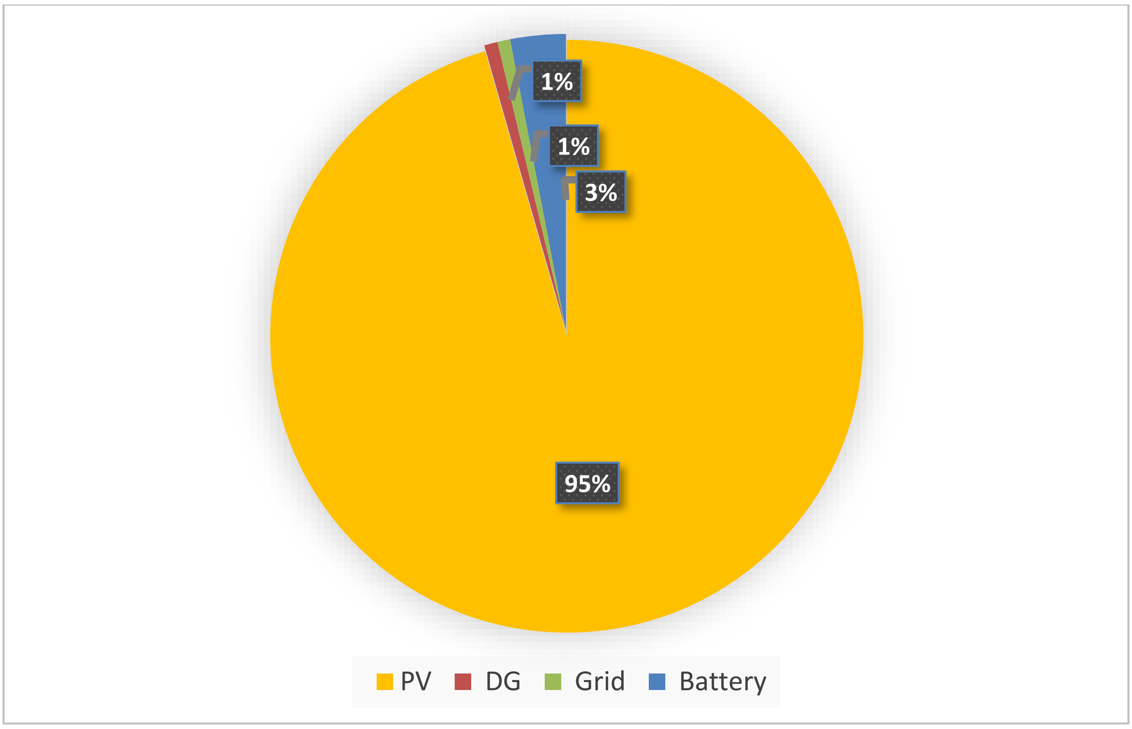

Figure 3.8: Relative quantities of energy supplying the load from 9:00-16:00.

With potential consequence, the additional A/C load could reduce the amount of PV energy available to charge the batteries. On the other hand, there could also be a significant reduction in PV curtailment. PV curtailment typically occurs between 10:00-17:00 when the batteries are approaching 100 % SoC, and more severely after the batteries have reached 100 % SoC, see Figure 3.9.

A comparison of the projected annual PV energy to the theoretical PV energy, from Table 3.1, shows that PV curtailment occurs. However, uncertainty inhibits the accurate calculation of the annual PV curtailment, which is found to be 25000 kWh ±95 %. The large uncertainty is primarily due to the propagation of error from the measured PV power, irradiance, and calculation of PV_{derate}. Nonetheless, from Figure 3.10, it is concluded with certainty, on average, PV curtailment occurs between 13:00-16:00. Moreover, when only considering PV curtailment occurring between 13:00-16:00, the measurement accuracy can be marginally reduced to 15000 kWh ±61 %. Therefore, there is at least 6000 kWh of PV energy curtailed annually and installing A/C is expected to utilize a portion of this energy.

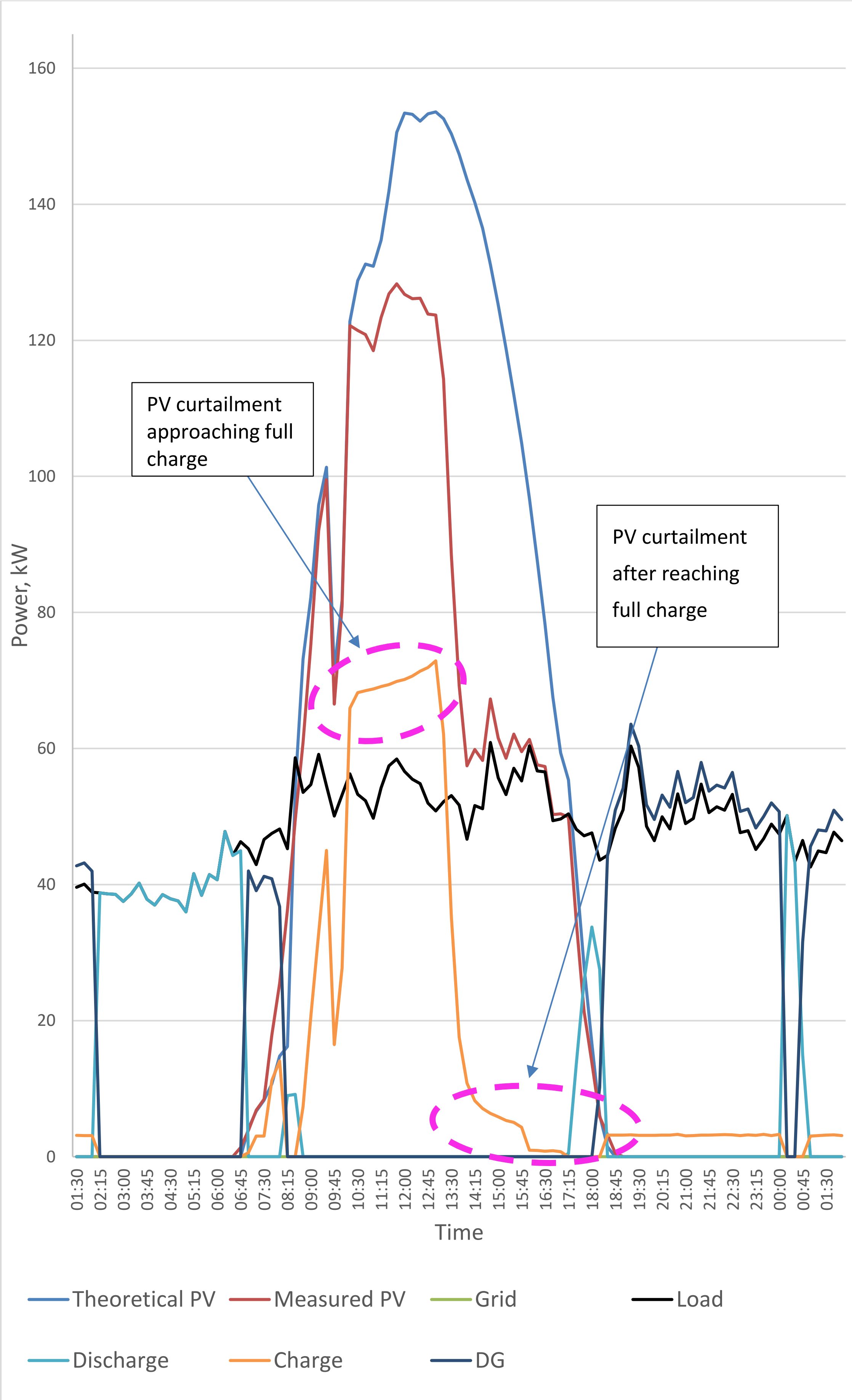

Figure 3.9: Typical PV curtailment example from Dec 1st, 2017.

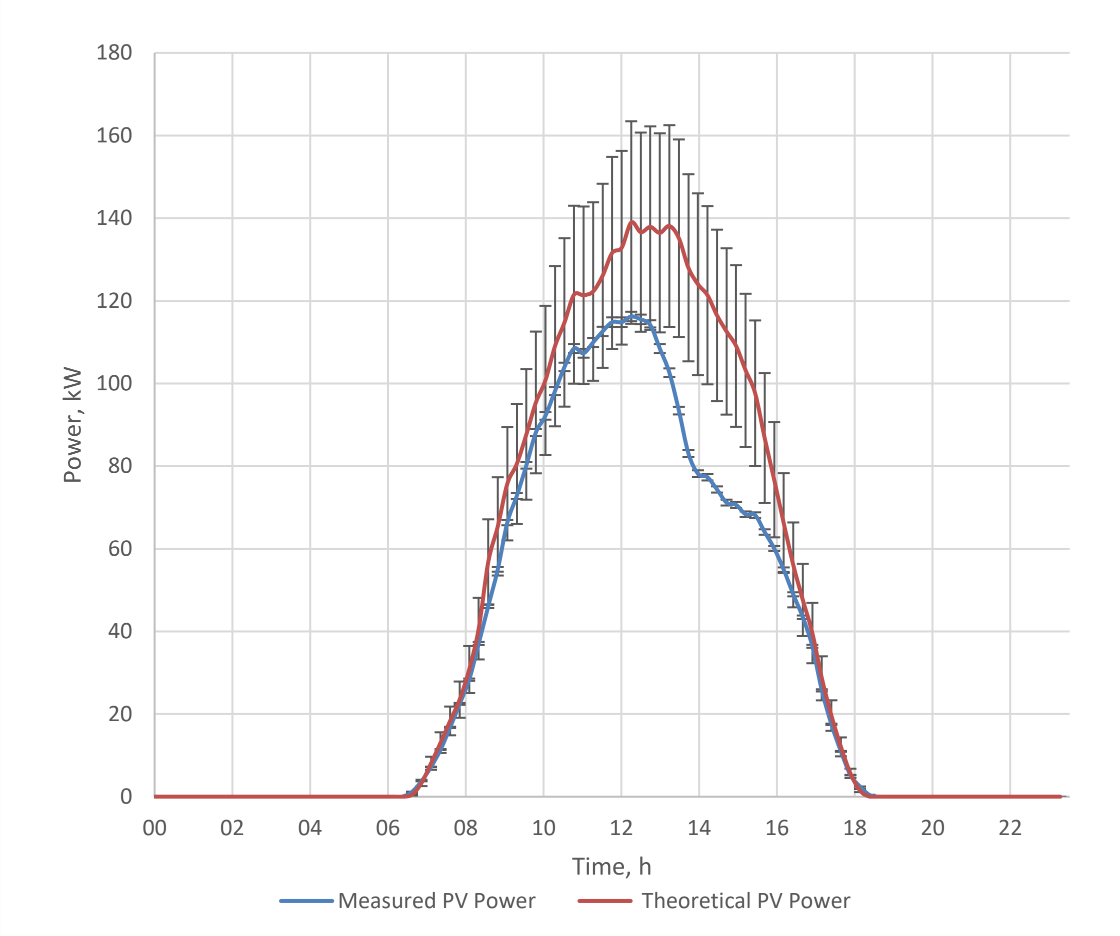

Figure 3.10: Estimated average PV power compared to average measured PV power.

The second proposed intervention is to install A/C in the battery room. The simulation work in Section 5.2 aims to investigate the impacts of increased energy consumption and the additional cost of A/C in comparison to the benefits of extending battery lifetime.

3.3 Intervention #3: Adjusting Power Source Priority Order

The power source priority order, as described in Section 1.3.4, is defined by two operating modes:

- Mode #1 from 2:00-17:59, the priority order is PV, battery, grid, DG
- Mode #2 from 18:00-1:59, the priority order is PV, grid, DG, battery

PV power, on average, is much less than the load at 18:00. Thus, when switching from mode #1 to mode #2 the grid/DG is called. If the grid is not available, the DG is started. While in mode #2, the initial response to grid outages is to call the DG. As the DG warms up, the battery system provides power until it is connected to the load. If the grid becomes available, the DG enters a cool down cycle but remains connected until the minimum runtime of 15-minutes expires. The battery supports the load until the minimum runtime expires. Figure 3.11 depicts a typical example of the DG response to a short duration outage. In this example, a 15-minute grid outage results in 19 minutes of battery discharge time, and only 7 minutes of the DG supporting the load. The total time disconnected from grid is 26 minutes.

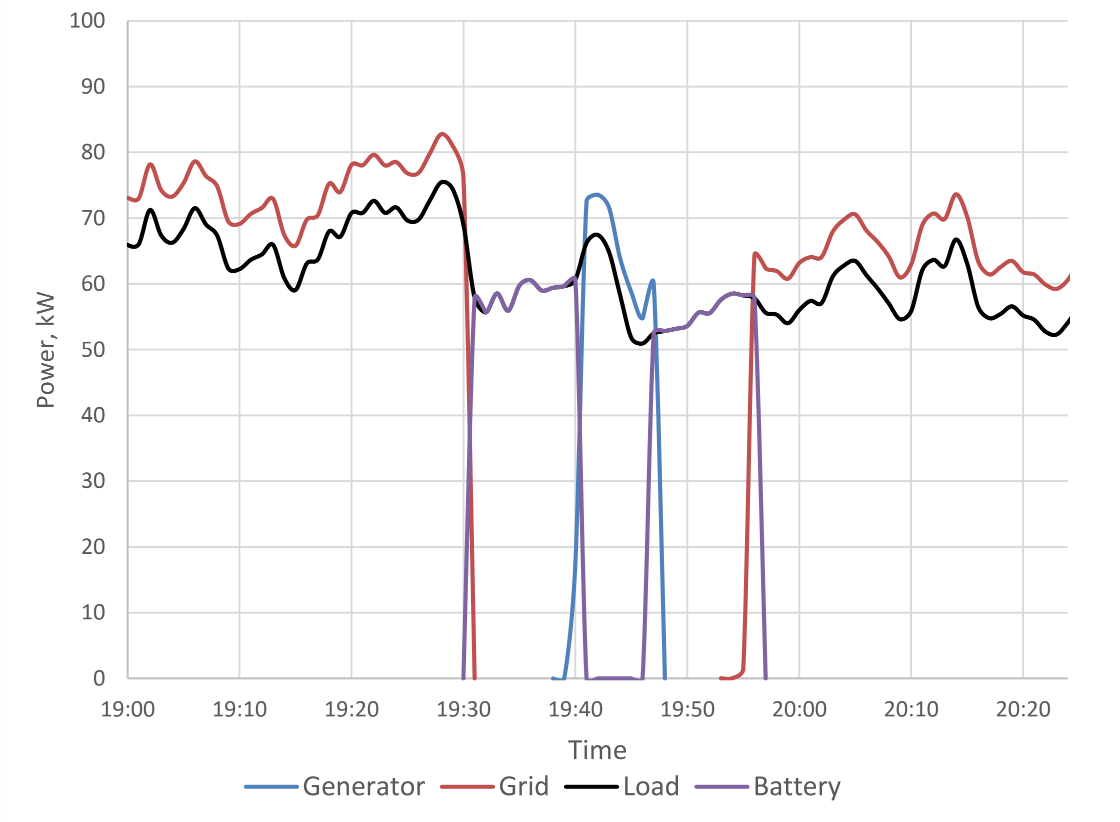

Figure 3.11: Typical DG response to grid outage, example from November 15th.

In general, the DG response to a short grid outage results in disconnection from the grid for 20-30 minutes, meaning even short duration grid outages (1-10 minutes) result in 20-30 minutes of grid outage time. Of course, each grid outage is accompanied by wear and tear on the DG and fuel consumption during start up, runtime, and cool down. It can be concluded the DG system, although well-suited for long duration grid outages, has a poor response to short duration grid outages. Therefore, the number of DG responses to short duration grid outages should be minimized. There were 353 grid outages between October 12th, 2017 and April 12th, 2018; of which 36 % had a duration of less than 15 minutes, 50 % were less than 30 minutes, and 88 % were less than 240 minutes, as shown in Figure 3.12.

As an alternative to relying on the DG for responding to numerous short duration grid outages, the battery can be given priority ahead of the DG. The priority order in mode #2 would then be: PV, grid, battery, DG. Effectively the DG would become the energy systems last resort in both operating modes. It would only be called if the battery SoC reached its minimum value while the grid was unavailable. In contrast to the DG system, the battery system is well suited for short duration grid outages. Upon grid failure the battery behaves as a UPS. When the grid becomes available again, the battery seamlessly transitions back to grid power. Therefore, the battery can effectively maximize the amount of time connected to the grid, while also reducing DG runtime, fuel consumption, and emissions.

Consider an example in an extreme case when a grid outage has a one-minute duration. Given the current control modes, the outage results in at least 15 minutes of battery discharge time and 5 minutes when the DG is connected to the load. With a total disconnection time from the grid of 20 minutes. Whereas if the battery were given priority ahead of the DG, the total disconnection time from the grid would be one minute and the battery would be discharged for one minute.

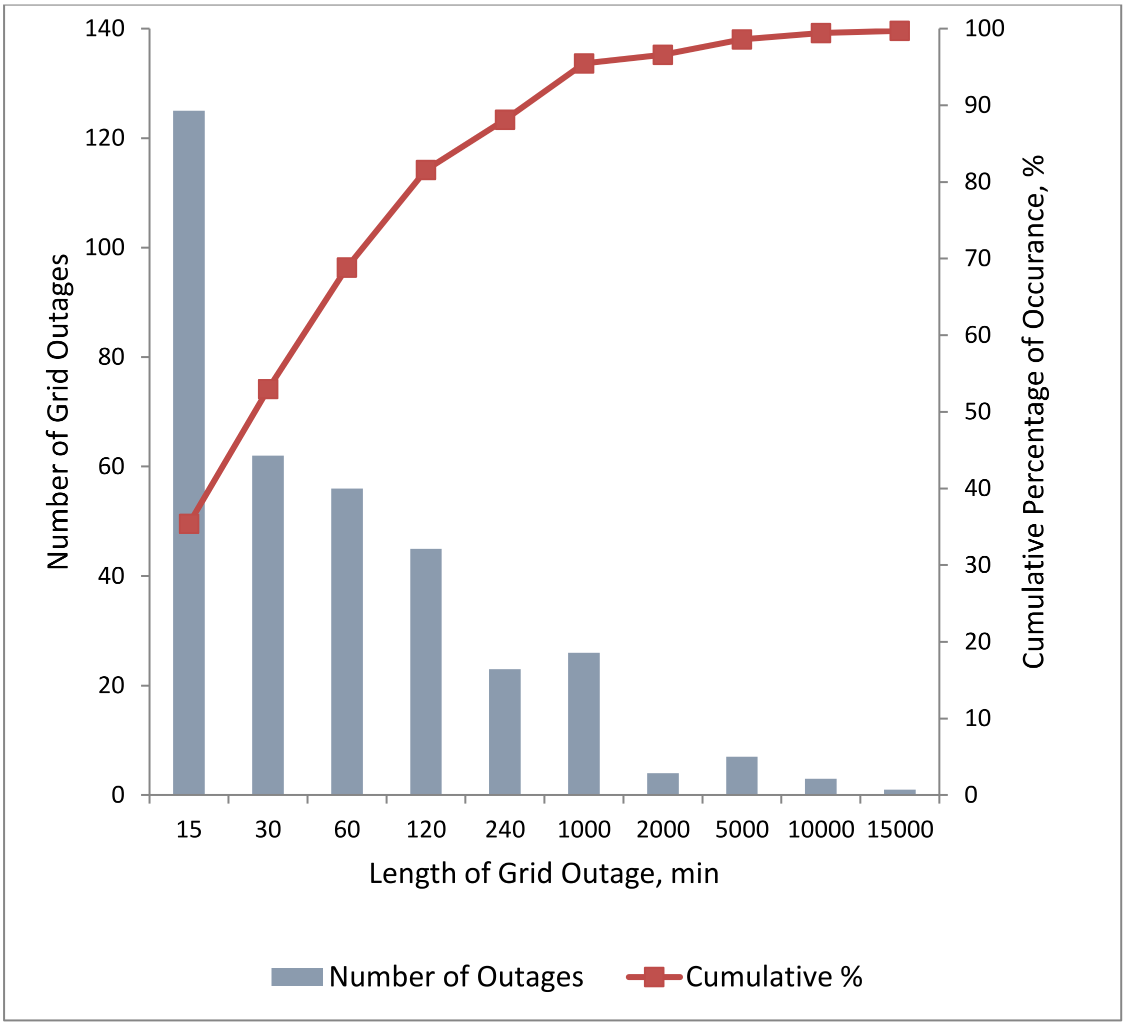

Figure 3.12: Histogram showing grid outages between 12/10/2017 and 05/04/2018 sorted by duration.

The change of priority in mode #2 is expected to be achieved at a low cost. A simple change of settings on the master bi-Inv to cancel the forced grid/DG call from 18:00 to 1:59 would result in the system always operating in mode #1. To give the grid priority ahead of the battery in mode #2, a bypass switch in the external control loop needs to be activated from 18:00 to 1:59. This can be achieved using hardware and software which already exists in the system.

A potential consequence to adjusting the priority order is increasing the amount of time the battery spends at a low SoC, which would occur after long duration grid outages. For example, if a long grid outage were to occur at 18:00, the battery could be expected to fully discharge and remain at a low SoC until PV charging the following day. From Figure 3.12, long duration outages are much less common than short outages; therefore, benefits are expected to outweigh the costs. The simulation work in Section 5.3 aims to determine if and how much the battery lifetime is impacted, and furthermore, to quantify the resulting changes to total system costs and emissions.

3.4 Intervention #4: Introduce Grid Charging

Grid charging has the potential to reduce the K-HES's dependence on DG energy and mediate the risk of battery damage. From Figure 3.1, it is clear most of the DG runtime is between 18:00 and 2:00. Figure 3.13 substantiates this fact, showing that during any given hour between 18:00 and 2:00, the DG runs approximately 50 % of the time. The relative quantities of energy supplying the load from each energy source are shown in Figure 3.3. The grid has the smallest contribution followed closely by the battery and the DG. PV provides the largest share of the load. If it is assumed the PV system's share of the load cannot be increased and the load cannot be decreased, then only option for decreasing DG energy production is to increase the contribution of energy from the grid. This can be achieved by charging the batteries with the grid. Grid charging is expected to increase the availability of the battery system to respond to grid outages, thereby reducing the probability of calling the DG. The grid power limit is based on the 200 kW transformer, which far exceeds the requirements for charging the batteries and simultaneously meeting the load. The sum of the peak load (104 kW) and the maximum charging power (1200 A, approximately 60 kW) is well under the 200 kW power limit. Therefore, even during peak loading the battery can be charged with the maximum charging power. Implementing grid charging would require the control priority order to be defined as per Section 3.3; therefore, this intervention requires that intervention #3 is simultaneously implemented.

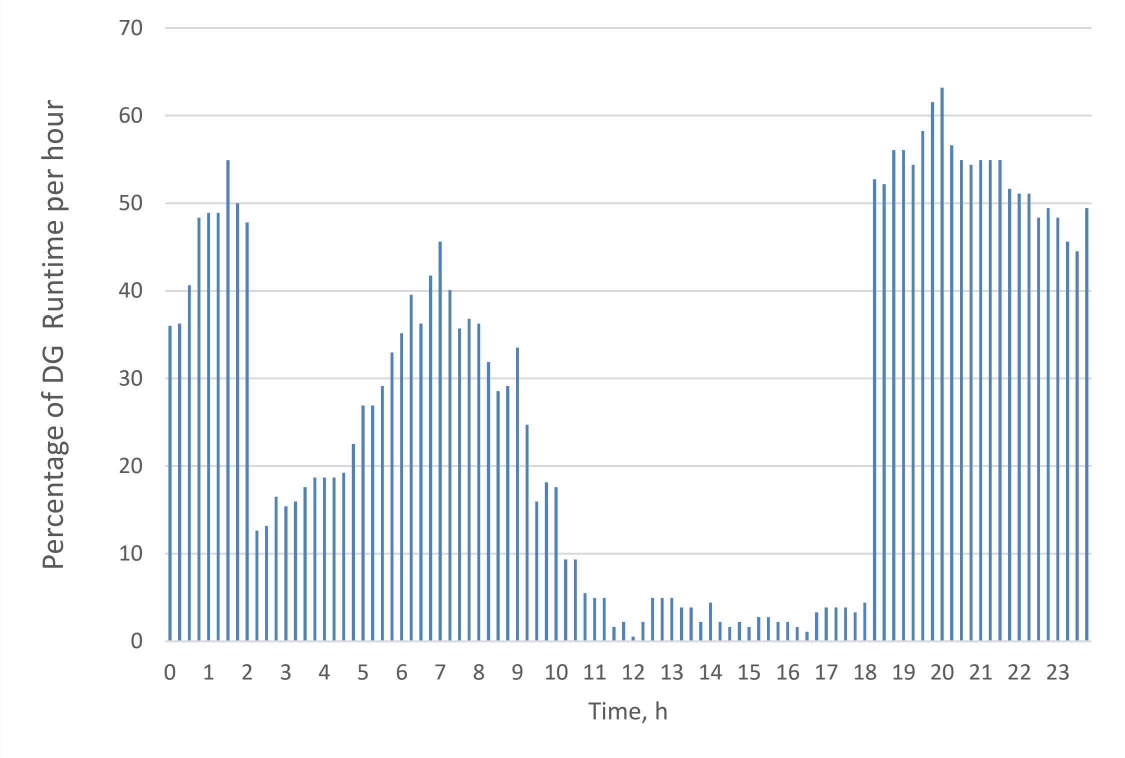

Figure 3.13: The percentage of DG runtime for each hour of the day from October 12th, 2017 to April 5th, 2018.

An additional benefit to grid charging would be a reduced risk of severe battery damage from prolonged periods at a low SoC. The existing control system only allows PV power to deliver significant charging power; therefore, periodically the batteries remain at a low SoC when climatic conditions prohibit substantial PV charging. Moreover, it was identified the previous intervention could increase the risk of prolonged time at a low SoC. Grid charging would provide a means of reducing this risk.

The tradeoff is expected to be a significant increase of throughput energy on the battery system, and thus a reduction of battery lifetime. Reduced battery lifetime could increase lifecycle emissions and costs. However, the benefits of lower DG costs and emissions could outweigh the impacts of reduced battery lifetime. The simulation analysis in Section 5.4 aims to investigate the balance of this tradeoff.

3.5 Intervention #5: Improved Hierarchical Control Philosophy and Optimized Setpoints

It has become clear from the measurement data and evaluation results the existing control philosophy is not optimal. Therefore, a new control philosophy is considered. The control philosophy is adapted from a modern predictive control philosophy based on optimized setpoints. The control philosophy has been described and demonstrated by the authors of [17]. An interpretation of their method has been adapted to the K-HES and is presented in this section.

The control setpoints to be optimized are:

- Threshold Battery Power, $P_{ThrBatt}$: the cost of energy from the batteries increases with increasing power. Therefore, as the load increases, there is a point when the batteries are no longer the cheapest option to supply the load. $P_{ThrBatt}$ defines the point when it becomes more economical to utilize the DG than the battery system.
- Minimum DG power, P_{DGMin}: assigning an optimized minimum DG power can be advantageous to increase fuel efficiency and reduce O&M costs. In practice, the minimum DG power must be maintained by charging the batteries or, if the batteries are fully charged, by supplying a dump load.
- Critical DG power, $P_{DGcritical}$: Below the critical power, the DG will operate at its rated power, and surplus power is used to charge the batteries. For loads above the critical power point, the DG follows the load.
- Stop DG SoC, SoC_{stp}: Once the DG is running, it runs until the battery SoC reaches SoC_{stp}.
- Minimum SoC, SoC_{min}: The optimal value of the minimum SoC is dependent on the relative cost of supplying power with the battery system compared to the other system components. The cost of supplying power with the battery system generally increases when SOC_{min} is reduced due to a reduction in battery lifetime. However; if the cost of producing power with other components is high enough, a lower SOC_{min} can be justified.

In Section 5.5 the control setpoints described above are optimized using iHOGA's built in genetic algorithms to solve for values which minimize system emissions and costs. In [17], the authors optimize the control setpoints daily, which yields improved results when compared to yearly optimization. In this study, the optimization is conducted over the entire lifetime of the system.

The costs to implement this control method are difficult to estimate. An attempt is not made in this study. Alternatively, the simulation analysis will assume zero added costs and the predicted change in NPC will be used to guide a discussion regarding the feasibility.

3.6 Intervention #6: Minimize PV Ramp-up Charge

In Section 3.4, it is assumed the share of PV energy in the system could not be increased. The assumption would be valid if:

1. PV energy is always given priority to the load.
2. The batteries are fully discharged each morning before PV charging begins - leaving maximum battery capacity available for PV charging.

For all intents and purposes, the first assumption is found to be accurate. However, an investigation into the second assumption reveals the maximum battery capacity is not always available for PV charging.

As depicted in Figure 3.1 , a share of battery charging from the grid and DG often occurs between 6:00-11:00. This behavior has been found to reduce the available battery capacity available for PV charging. The cause is evident when observing the behavior of the system during PV ramp up. Referring to Figure 3.14, the battery begins discharging at 2:00 in the morning and is flat by 5:30 - 7:00. On average, the PV system ramps up to surpass the load by 9:17. This creates a window, of approximately 2.5 hours in duration, when the grid/DG overlaps with PV ramp up each morning (see caption on Figure 3.14). During this period the cumulative sum of energy produced is greater than the load, and therefore the battery is charged. All charging is attributed to the grid and DG because the PV power is less than the load. Preferably, the battery would not be charged before PV power exceeds the load. In this example, the battery receives 36 kWh of charging energy during PV ramp up. Again, referring to Figure 3.14, it is clear the charging rate is heavily curtailed in the afternoon. Therefore, it can be assumed that an additional 36 kWh of PV energy could have been utilized, and 36 kWh of DG energy generation could have been eliminated, if charging during PV ramp up had not occurred.

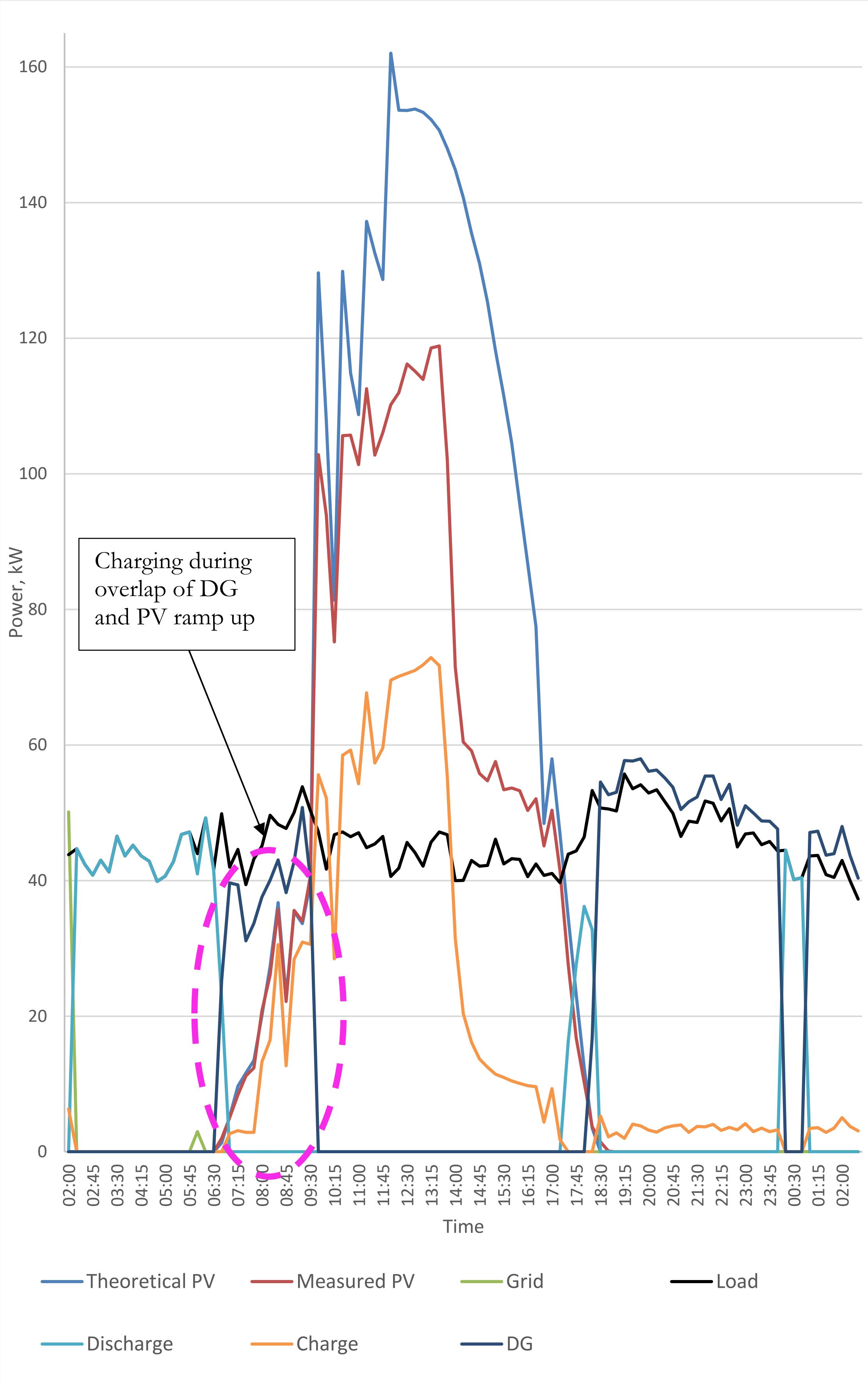

Figure 3.14: Measurement data from Nov 26th, 2017, showing battery charging during PV ramp up.

Furthermore, overlaying the theoretical average hourly PV power on top of Figure 3.1 shows that DG and grid charging could be causing PV curtailment in the morning. No conclusion is drawn as the measured PV power is within the calculation error of the theoretical PV power, as illustrated in Figure 3.15.

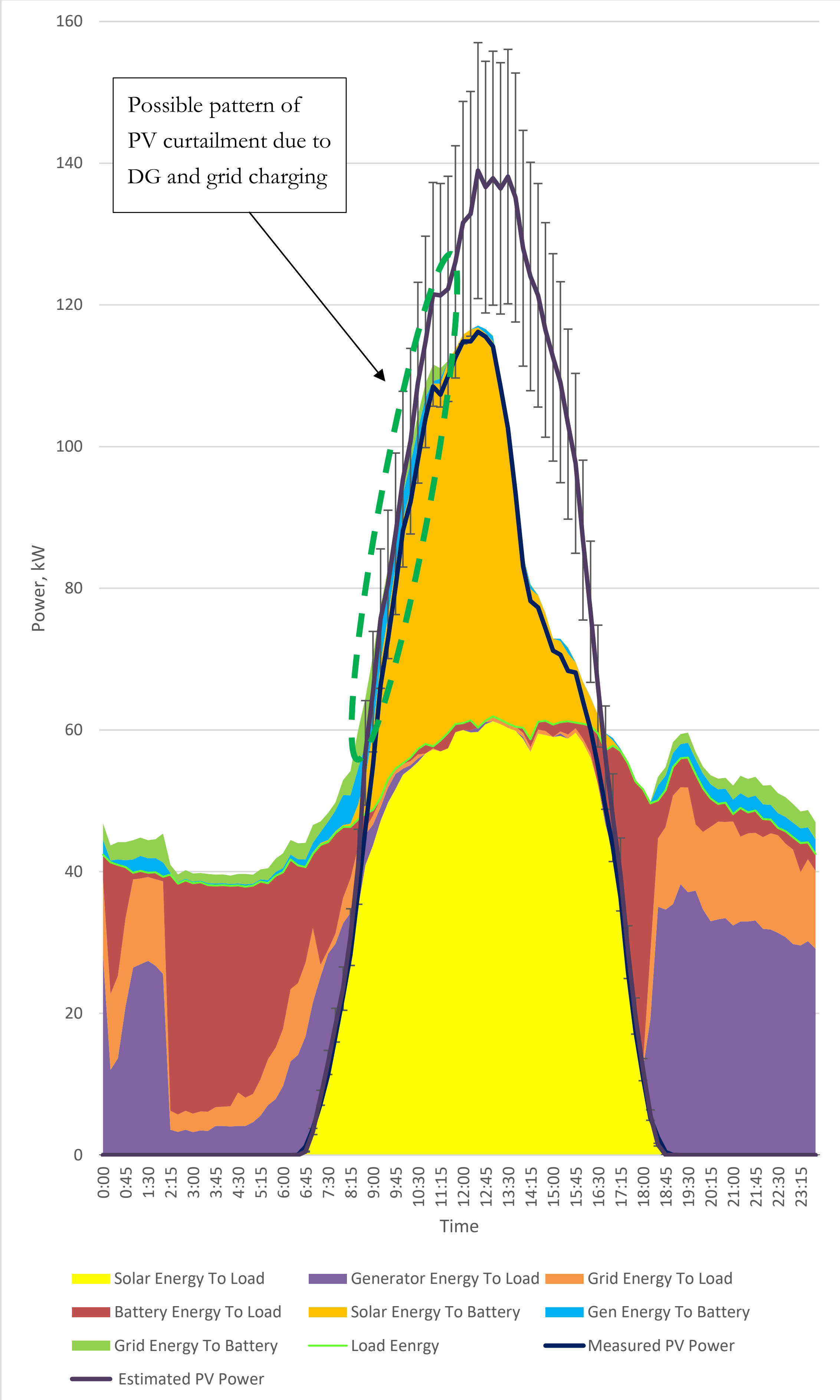

Figure 3.15: The estimated hourly PV power curve is overlaid on Figure 3.1 to identify a possible source of PV curtailment.

Grid and DG charging during PV ramp up could be eliminated by ensuring the battery is always discharged during PV ramp up. This can be achieved by implementing control which:

1. Ensures the battery begins discharging when PV ramp up begins.
2. Ensures the battery has sufficient SoC for the duration of the PV ramp up period.

An attempt to implement this control is already in place by allowing the SoC_{min} to drop from 60 % to 55 % starting at 7:00. However, the method has not been effective because the SoC is commonly reduced to 55 % before 7:00. Allowing the SoC_{min} to drop an additional 10 % or 15 % at 7:00 could prove more effective. Ideally, a more sophisticated control algorithm would calculate the battery discharge in response to PV ramp up each morning. This intervention is proposed based on analysis of the measurement data, but it is not simulated with iHOGA due to limitations of the software.

4 iHOGA Simulation Models

iHOGA is the simulation tool selected for this study. It is used to construct a base-case simulation model of the K-HES, and simulation models of each of the proposed interventions. The iHOGA models and the relevant input data are presented in this section. Assumptions and differences between the iHOGA models and the real system components are identified and discussed.

4.1 Component Models

4.1.1. Load Profile

The load profile in iHOGA is defined by an annual series with a temporal resolution of one hour. The 15-minute load measurement data is averaged over each hour to match this format. The result is a smoothing of the measurement data as shown in Figure 4.1.

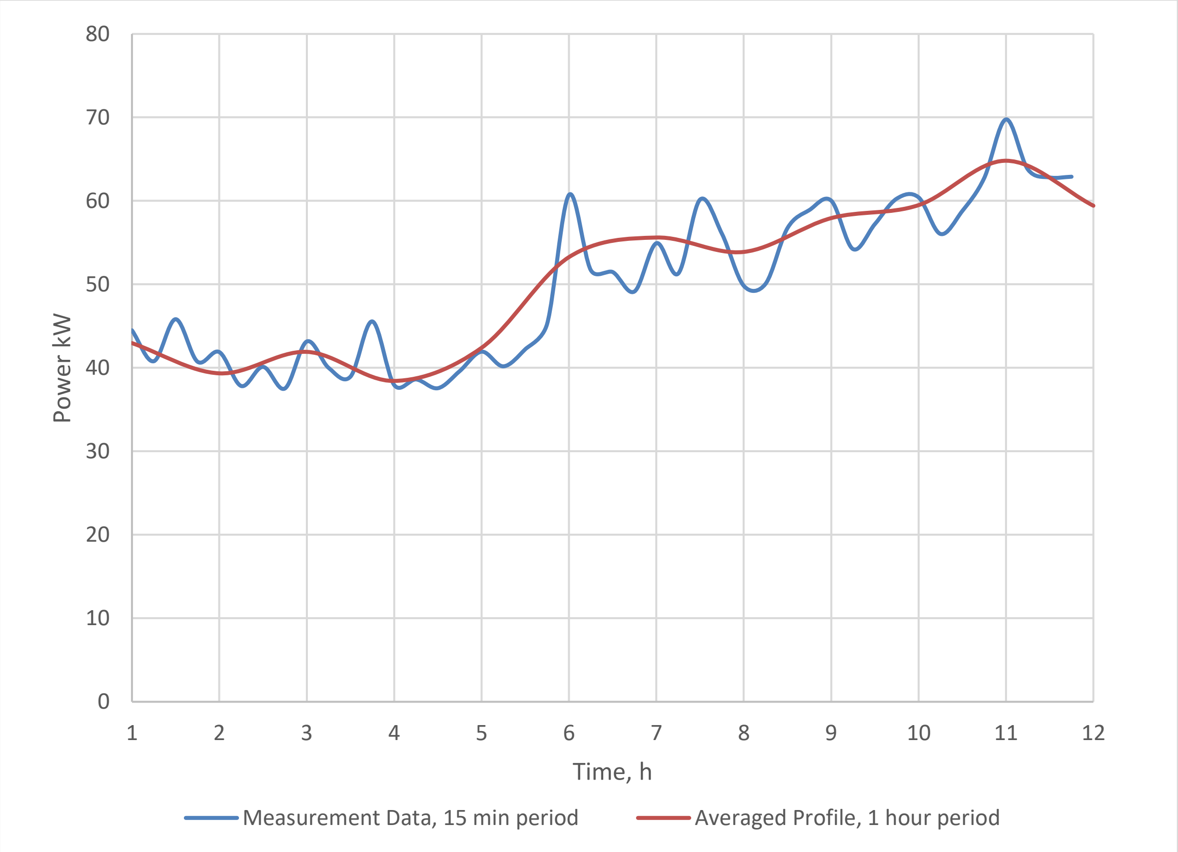

Figure 4.1: Example of smoothing effect on load measurement data averaged from 15-minute to 1-hour intervals.

The duration of the measurement period spans six months. The measurement series is repeated on itself to obtain a full year load profile whereby, the profile over the first and second half of the year is identical, see Figure 4.2. This load profile is used for all simulation models in this study, except when considering the additional load of the A/C unit in Section 5.2.

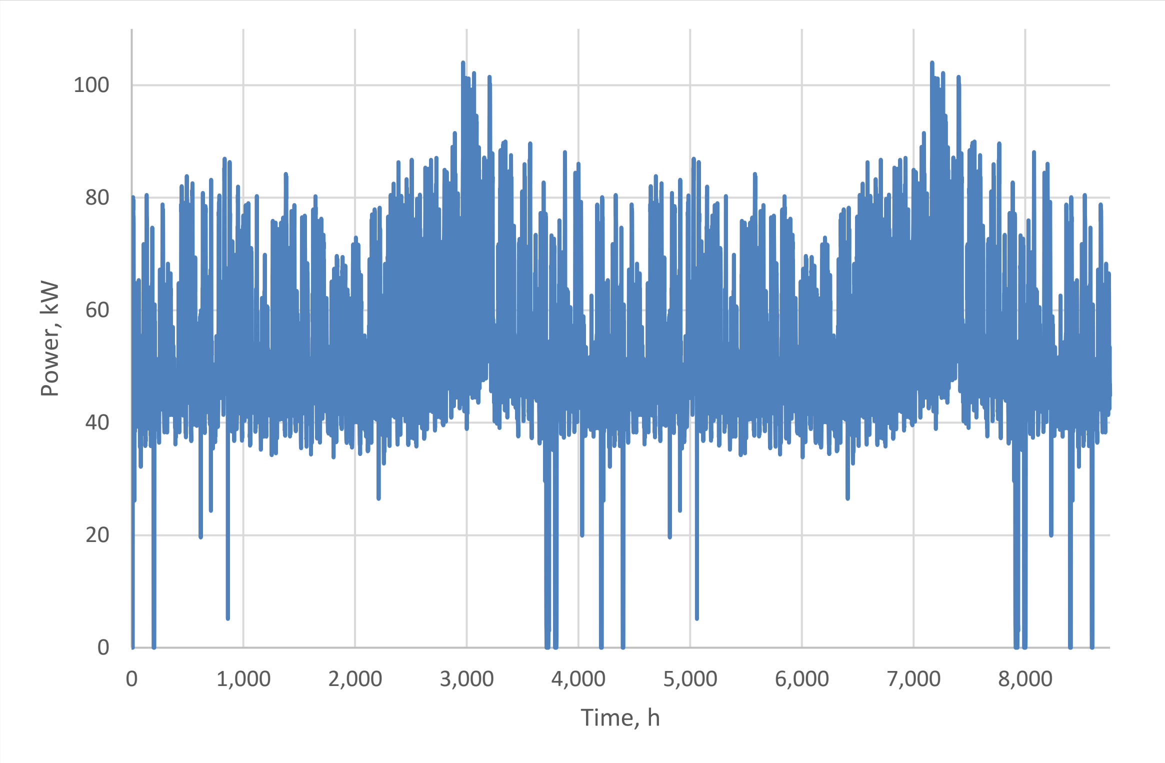

Figure 4.2: One-year hourly load profile for iHOGA base-case simulation model.

For all simulations it is assumed:

1. The measurement period is representative of the annual load.
2. The load is not expected to change over the 25-year lifetime of the K-HES.

4.1.2. PV Model

The PV array power $P_{PV}(t)$ available at the AC bus is calculated according to:

$$P_{PV}(t) = P_{nom} * \frac{G_{PoA}(t)}{G_{STC}} * PV_{derate} * \eta_{Inv} * \left(1 + PV_{Pcoeff} * (PV_T(t) - PV_{TSTC})\right)$$

Equation 4.1

where P_{nom} is the total rated DC power of the array, and $G_{PoA}(t)$ is the irradiance in the plane of array according to Graham's hourly irradiation model (described in [34]) during timestep t. G_{STC}, PV_{derate}, η_{Inv}, PV_{Pcoeff}, and PV_{TSTC} are as defined in Section 2.1.16. $PV_T(t)$ is the calculated PV cell temperature in timestep t:

$$PV_T(t) = T_{amb}(t) + \frac{T_{NOCT} - 20}{0.8} * \left(\frac{G_{PoA}(t)}{G_{STC}}\right)$$

Equation 4.2

where T_{NOTC} is the nominal operating cell temperature of the PV module when the ambient temperature is 20˚C, and $T_{amb}(t)$ is the average ambient temperature during timestep t estimated by the distribution of monthly average temperatures according to Erbs model [35]. The solar resource data used in the simulation model is from the SolarGIS database, see Table 4.1.

Table 4.1: Solar resource data from SolarGIS database.

Month	***GHI, kWh/m²/month***	***Average Temperature, ˚C***
January	197.9	21.5
February	190.3	22.3
March	202.5	22.4
April	185.7	21.6
May	182.6	20.9
June	172.2	20.3
July	180.1	20.1
August	193.3	20.6
September	195.4	21.8
October	194.6	22.1
November	178.5	21.5
December	193.2	21.4

iHOGA cannot model multiple PV orientations simultaneously. Therefore, power generation from the NE/SW facing arrays must be approximated by a single PV array with a single orientation. It is important the array's equivalent annual energy yield and daily PV generation curve are estimated as closely as possible. The annual energy yield can be easily adjusted by changing the value of PV_{derate} in Equation 4.1. To determine which orientation best represents the daily generation profile PVsyst simulations were conducted. Figure 4.3 and Figure 4.4 show the simulated PV generation profiles of the K-HES in 3 different orientations (NE/SW, south, and flat) on the winter solstice (December 21st) and the spring equinox (March 21st), respectively. Each system has been calibrated to have the same annual energy yield. On December 21st the south orientation has significantly reduced output power in contrast to the SW/NE and the flat arrays, which have very similar generation profiles. On March 21st all the arrays have nearly equivalent output power. The flat array is clearly a closer approximation to the generation profile of the SW/NE array than the south facing array. The largest discrepancy between the output power of the flat array and SW/NE arrays is during the early morning and late afternoon. It is assumed the small difference is consistent throughout the year and has a negligible impact on the modeled behavior of the system.

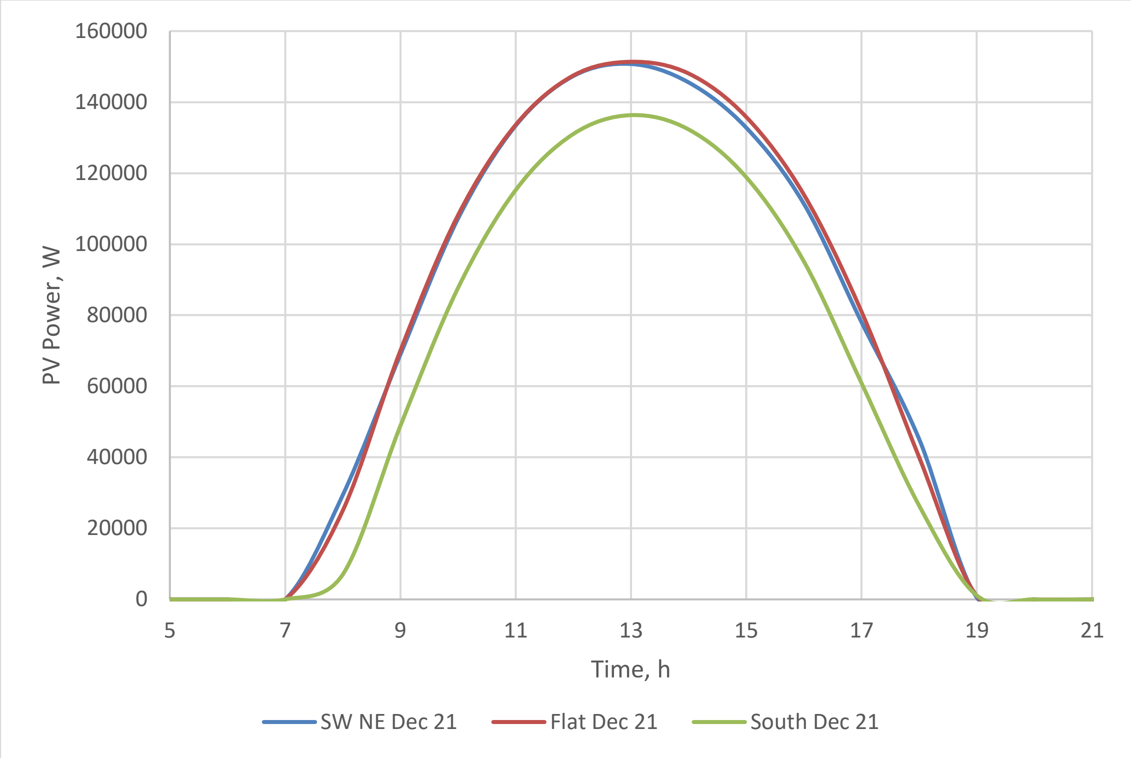

Figure 4.3: PVsyst simulation results of the K-HES PV output power for 3 different orientations on December 21st.

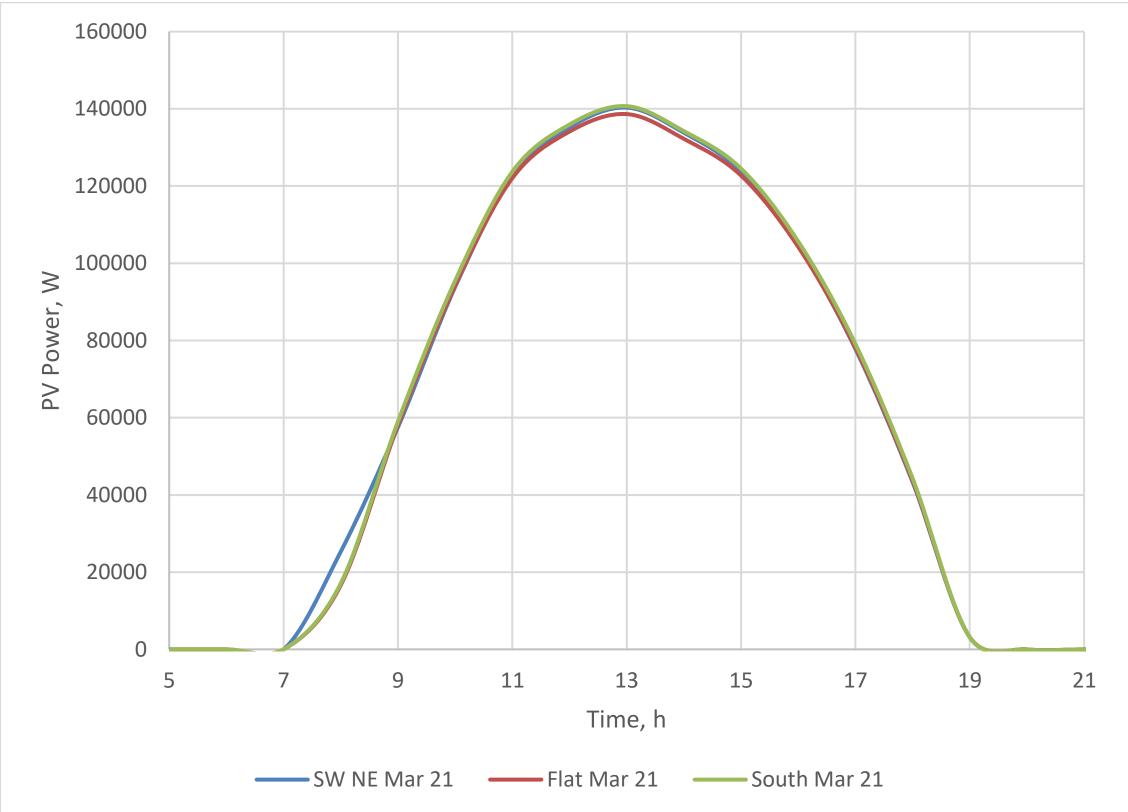

Figure 4.4: PVsyst simulation results of the K-HES PV output power for 3 different orientations on March 21st.

Input data for the PV model is summarized in Table 4.2. Table 4.3 describes the PV inverter efficiency.

Table 4.2: PV model input data.

Technical Input Data	
Module Name	Can Sol CS6K-275M
Nominal Power, W	275
Expected Lifetime, years	25
T_{NOTC}, °C	20
Temperature Coefficient Pmpp, %/°C	-0.41
Manufacturing Emissions, kgCO_2/kW	2,200
Number of Modules per string	20
Total Number of Modules	740
PV_{derate}	0.71

Table 4.3: PV inverter efficiency data.

Percent of Nominal Inverter Output Power	***Inverter Efficiency, %***
0	0
2	94
3	95
4	96
5	97
10	97.5
20	98.2
30	98.25
40	98.3
50	98.25
60	98.2
70	98.15
80	98.1
90	98.05
100	98

The PV model does not account for:

1. Fluctuations of PV derating factors barring the impact of temperature on PV output.
2. Temperature impact of inverter efficiency.
3. PV degradation over the lifetime of the system.

4.1.3. DG Model

The DG model in iHOGA is responsible for calculating fuel consumption C_{Fuel}. The calculation is conducted using the widely accepted model described in [17], which assumes a linear fuel consumption curve:

$$C_{Fuel} = B \cdot P_{DG-rated} + A \cdot P_{DG}(t) \qquad \textit{Equation 4.3}$$

where $A = 0.212$ L/kWh and $B = 0.063$ L/kWh are fuel consumption curve coefficients generated from the fuel consumption information on the DG datasheet, see Table 1.4. $P_{\mathrm{DG-rated}}$ is the rated power of the DG, and $P_{\mathrm{DG}}(t)$ is the operating power of the DG during the simulation at time t. Based on the work of [32], the iHOGA DG model accounts for higher O&M costs when the DG is operated outside of its optimal operating range of 50 %-80 % of its rated power. O&M costs are scaled linearly by the extra aging factors in Table 4.4.

Table 4.4: DG model input data.

Parameter	Value
Rated power, kW	220
Rated power, kVA	275
Expected lifespan, h	25000
Minimum Power, %	0
Fuel type	Diesel
Fuel cost, $/L	1.00
Fuel inflation rate*	3.5
Fuel emissions, kgCO2/L	3.5
Manufacturing Emissions, kgCO2/kW- rated power	270
Fuel Curve Coefficients, L/kWh	A =0.212
	B =0.063
Extra ageing factor, at 30 % of rated Power	1.25
Extra ageing factor, at 100 % of rated Power	1.5

*based on fuel prices in Kenya between 1991-2016, cost data from the world bank [36].

4.1.4. Grid Connection Model

The grid model simply behaves as an infinite energy source with a defined availability and power limit. Grid availability is defined annually with a temporal resolution of one hour. The availability profile is identical for each year of the simulation. The hourly profile is calculated as described in Section 2.1.14. The profile is repeated on itself to obtain a full year profile. By using measurement data to construct the grid availability profile, the behavior of the interaction between the grid and the K-HES are incorporated in the model. The grid power limit is 200 kW based on the size of the transformer at the connection point to the grid. The emissions associated with power generated from the grid are based the percentage of power generated from various sources in Kenya: 52.1 % from hydro, 32.5 % from fossil fuels, 13.2 % from geothermal, 1.8 % from biogas cogeneration and 0.4 % from wind from [37]. The model parameters concerning the grid connection are summarized in Table 4.5.

Table 4.5: Simulation inputs for grid connection model in iHOGA.

Parameter	***Value***
Max import power, kW	200
Allow energy import	YES
Allow energy export	NO
Emissions, $kgCO_2/kWh$	0.4
Priority to AC grid over DG	YES

4.1.5. Energy Storage Model

iHOGA allows the user to model several battery chemistries with different modeling options. For this study, the flooded lead acid Hoppecke 16-OPzS solar.power 2900 batteries are modeled with the Schiffer battery model. The Schiffer battery model is designed for lead acid batteries operating under similar conditions to those found in hybrid renewable energy systems. The model is discussed in [38], and described by Schieffer et al:

> The model is a weighted Ah throughput approach based on the assumption that operating conditions are typically more severe than those used in standard tests of cycling and float lifetime. The wear depends on the depth-of-discharge, the current rate, the existing acid stratification, and

the time since the last full charging. The actual Ah throughput is continuously multiplied by a weight factor that represents the actual operating conditions [28].

In iHOGA, the Schieffer model is configured as a multiyear battery model which runs until the battery reaches its end of life. End of life occurs when the nominal capacity of the battery falls below 70 % of its original capacity. Particularly important to this study is the consideration of how temperature impacts battery lifetime and performance, specifically for determining the impact of the proposed intervention to install A/C in the battery room. Unlike standard battery models employed by most HES software, the Schieffer model estimates how important battery characteristics change under various operating conditions. The model calculates battery efficiency, and the reduction in battery capacity during each timestep. Therefore, the performance consequences of battery degradation are considered during each timestep in the simulation.

The emissions associated with manufacturing and installing the battery system are also accounted for in the battery model. The term "manufacturing emissions" in Table 4.8 accounts for the emissions produced while transporting materials to and from the site, during installation, and for recycling materials at the end of their life.

Input data for the Schiffer battery model is summarized in Table 4.6, Table 4.7, and Table 4.8.

Table 4.6: Detailed Schiffer battery model input data. Default values for OPZS batteries from [28] are used except for parameters from the battery manufacturer marked with (*).

Parameter	***Value***
Battery Type	Flooded Lead Acid OPZS
Open Circuit Voltage at Full Charge, V *	2.08
Gradient of change in OCV with SOC, V	0.1
Initial effective internal resistance (charge), ohm-Ah	0.43
Initial effective internal resistance (discharge), ohm-Ah	0.38
Resistance representing charge-transfer process based on SOC	0.36
Resistance representing charge-transfer process based on SOC	0.29
Normalized capacity of the battery, Charge	1.001
Normalized capacity of the battery, Discharge	1.642
Normalized reference current for current factor, A/Ah	-0.1
Height of battery, cm *	81.5
Corrosion Voltage of fully charged battery without current flow, V	1.75
Nominal voltage for gassing, V	2.23
Normalized gassing current, mA/100A	20
SoC considered full charge, % *	95
Min SoC for bad charges, % *	99
SoC to reset number of bad recharges, 95 *	95
End of battery lifetime, % of nominal capacity *	70

Table 4.7: Curve of corrosion speed vs. potential of positive electrode (Hg/Hg2SO$_4$), required for Schieffer battery model. Default values for OPZS battery cells are used. Reference electrode voltage is 0.616V.

Potential of positive electrode vs. the reference electrode, V	**Corrosion speed, mg/cm^2/h**
0.6	4
0.8	4.5
0.95	5
1	6
1.1	8.5
1.12	5
1.14	2.5
1.18	2
1.25	2.5
1.35	7
1.4	15

Table 4.8: iHOGA battery model parameters.

Parameter	**Value**
Nominal battery capacity, Ah @ C10	2150
Parallel batteries	6
Nominal Voltage, V	48
Average Battery Temperature, °C (basecase)	29
Minimum SoC, %	55
Self-Discharge, %/month	5
Maximum Current, A	200
Round trip efficiency	Schiffer model
Cycles to Failure	Schiffer model
Float Life@ 20 °C	10
Manufacturing Emissions, kgCO$_2$/kWh	300*
SoC at start of simulation, %	55

*assuming freight to and from Germany with an average of 0.1 kgCO$_2$/tonne-km and 3 kgCO$_2$ per kg of battery manufactured.

4.1.6. Bi-directional Inverter Model

The bi-Inv model accounts for inverter (DC to AC) and rectifier (AC to DC) power conversion losses. The rectifier efficiency is 94 %, and the inverter efficiency is modeled by linear interpolation between the points on the efficiency curve described in Table 4.10. In the real K-HES system, the bi-Inv is responsible for implementing control measures; however, in the simulation, control is modeled with a separate component (control parameters are discussed in Section 4.1.8). Input data for the bi-Inv is summarized in Table 4.9. The model allows the inverter output power to surge above its nominal power according to Table 4.11, which is representative of the surge power levels summarized in the Sunny Island bi-Inv datasheet [22].

Table 4.9: Summary of bi-directional inverter model data.

Parameter	Value
Bi-directional Inverter Model	SMA Sunny Island 8.0
Nominal Inverter Power, kW	108
Lifespan, years	10
Maximum DC Charge Current, A	1200
Rectifier Efficiency, %	94
Low Voltage Cutoff, V DC	41
High Voltage Cutoff, V DC	63

Table 4.10: Bi-directional inverter efficiency as a function the percentage of nominal output power.

Percent of Nominal Inverter Power, %	Inverter Efficiency, %
0	10
2	30
3	50
4	70
5	85
10	94
20	96
30	96
40	95.5
50	95
60	94.5
70	94
80	93
90	92.5
100	92

Table 4.11: Inverter surge power and duration.

Power Surge Duration, min	Output Power above Nominal, %
30	27
15	30
10	35
5	47

4.1.7. Cost Model

iHOGA employs standard methods for calculating lifecycle costs. The model considers the nominal interest rate, annual inflation rate, system lifetime, and salvage values when calculating NPC. Costs are considered for each component: capital costs, replacement costs (based on annual inflation of component capital costs), and O&M costs. Of course, perpetual cost such as the cost of energy from the grid and the cost of diesel fuel are also considered. The detailed calculations performed by the cost model are described in the iHOGA user manual [38]. All cost input data for the model is summarized in Table 4.12 and Table 4.13, the values have been guided by data provided by Solarcentury Ltd.

Table 4.12: Financial data.

Financial Data	
Nominal Interest Rate, %	7
Annual Inflation rate, %	3
Study Period/Project Lifetime	25
Consider Salvage Values	Yes
Currency	USD

Table 4.13: Cost data for system components.

Parameter	***Lifetime, years***	***Capital cost, $***	***Inflation of Capital Cost, %***	***O&M***	***Cost Type***
PV System	25	250000	N/A	6000 $/year*	Fixed
DG (Diesel Fuel)	Variable	99000 (1.0 $/L)	0 (3.5)	1.4 $/h,	Variable
Batteries	Variable	80000	-5	N/A*	Variable
bi-Inv	10	61000	0	N/A*	Fixed
Grid Power	N/A	0.2 $/kWh	3	N/A	Variable
Other**	25	175000	N/A	N/A	Fixed

*The PV system O&M costs are inclusive of all system O&M costs aside from those for the DG which are accounted for separately
**refers to costs such as design, management, freight, and installation.

4.1.8. Microgrid controller Model

All simulation variables influencing control have been grouped together for ease of explanation under the umbrella term "microgrid controller". The microgrid controller decides which power source is selected and how much power is delivered by each source. Unfortunately, all the K-HES control measures discussed in Section 1.3.4 cannot be replicated in the simulation environment. The discrepancies between the real control measures and the simulated control measures are presented in this section.

As per the observed behavior of the K-HES, the microgrid controller model does not allow the grid and the DG to operate in parallel; however, the PV system can operate in parallel with the DG, grid, or the battery. The model calculates the output power of the non-PV source as the difference between the load and the PV power. The battery system output power is observed to match this behavior; however, the observed parallel operation of the DG and the grid with the PV system is more complex. From observation, the grid and the DG power are not flexibly adjusted as the PV power ramps up. Instead, as PV power increases, their output power is only marginally curtailed. Resulting in a net sum of power generation greater than the load. This behavior is discussed in Section 3.6 and depicted in Figure 3.14, where it is shown that the excess power production charges the batteries. In consequence, the model underestimates DG power, grid power, and the charge/discharge power during PV ramp up. It is found the simulation control variable for the minimum DG power $P_{\mathrm{DG\_min}}$ can be employed to replicate the observed behavior. The appropriate value of $P_{\mathrm{DG\_min}}$ is determined in Section 4.2. Unfortunately, a minimum grid power variable does not exist in the simulation environment and thus cannot be defined to achieve a similar result during grid operation. An alternative solution presents itself later in this section.

Recall from Section 1.3.4 the priority order for each control mode is:

- Mode #1: from 2:00-17:59, the priority order is PV, battery, grid, DG
- Mode #2: from 18:00-1:59, the priority order is PV, grid, DG, battery

There are only two priority orders to choose from in iHOGA: PV, grid, battery, DG or PV, battery, DG, grid. Additionally, on top of the defined priority orders, iHOGA also allows the user to schedule times when the battery has priority next to PV, and when the grid has priority next to PV. Selecting the priority order PV, battery, DG, grid, and the appropriate scheduled priority for the battery and the grid gives the closest possible replication of the real control modes:

- Simulation mode #1: from 2:00-17:59, the priority order is PV, battery, DG, grid
- Simulation mode #2: from 18:00-1:59, the priority order is PV, grid, battery, DG

The last two terms of both simulation control modes are reversed when compared to mode #1 and mode #2; however, the behavior of the system can be approximated to a reasonable degree of accuracy, which will be shown in Section 4.2.

Only one value of SoC_{min} and SoC_{stp} can be defined in iHOGA. The lowest value of SoC_{min} $(55\ \%)$ and the highest value of SoC_{stp} $(65\ \%)$ are adopted in the model. The simulation treats SoC_{min} as an absolute minimum value, thus the SoC never drops below 55 % during the simulation. Ideally, the real system would behave in the same way. However, the measurement data shows several instances when the SoC drops below 55 %. Such instances typically occur due to faults occurring when connecting the DG or the grid to the load. The battery remains connected to the load until these faults are cleared causing the SoC to drop below SoC_{min}. Such faults are not accounted for in the simulation; the simulated battery lifetime would be expected to be reduced if they were.

There is a difference in how SoC_{stp} is enforced in the simulation as compared to the real system. In the real system, the DG and the grid remain connected to the load until SoC_{stp} is reached. This is to ensure the battery has enough reserve capacity to support the load in the event the PV power is suddenly reduced. However, in the simulation, if the PV power surpasses the load, the DG and the grid are disconnected. Furthermore, the real system provides additional reserve capacity by connecting the grid or DG when the load exceeds 95 kW. Reserve capacity cannot be simulated in iHOGA. In Section 4.2 results are used to show the limited impact on the model.

In Section 3 it is found that the grid and DG charge the batteries with an average power of 3.6 kW when connected to the load. A parameter exists in iHOGA to model this behavior for the DG, but not for the grid. For the DG, a limit is set for the maximum DC charging current: $I_{DG\_chrgmax} = \frac{3600W}{48V} = 75\ A$. The result is a constant charge of 3.6 kW anytime the DG is running. For the grid model, there are only two simulation variables impacting the charging power: the maximum grid power (defined in the grid model), and the maximum charging current (defined in the energy storage model). The software selects the lowest of the two values when calculating the charging power. Without adjusting these variables, the simulation model overestimates the grid energy and the charge/discharge energy. Therefore, one of the two variables must be artificially lowered to enforce a pseudo charging limit during the simulation. The maximum charging current cannot be modified as the PV charging rate is dependent on this variable. However, adjusting the maximum grid power is found to be a suitable solution. The appropriate value of the maximum grid power and the impacts of modifying this value in the simulation are discussed in Section 4.2.

Lastly, the bi-Inv charge control settings from Table 1.6 are accounted for in the model with a similar set of parameters, see Table 4.14. The model settings and the real settings are nearly identical. Charge control in the simulation and the real system have been assumed to be equivalent.

Table 4.14: Simulation charge control settings.

Parameter	**Value**
Float Charging Voltage, V	2.23
Boost Charging Voltage, V	2.4
Boost Charge Duration, h	3
Boost Activated if SoC is less than, %	80
Equalization Charging Voltage,	2.5
Equalization Duration, h	8
Equalization Activated if SoC less than, %	40
Equalization Cycle time, days	180

4.2 Model Adjustments and Accuracy

The iHOGA simulation models are used to derive technical, economic, and environmental KPIs, namely: generator runtime, NPC, fuel consumption, CO_2 emissions, battery lifetime, and renewable fraction (RF). The proposed method for evaluating the interventions is to compare KPIs generated with the base-case model, to KPIs generated with the intervention models. Obviously, the base-case model must accurately represent the performance of the K-HES to obtain meaningful results. The accuracy is assessed by comparing the simulation results against the measurement results. Without any adjustments to the model, as described in Section 4.1, it is found the base-case model does not match the measurement results (see Table 4.15). The excess energy parameter has not been defined: it is simply the sum of the PV, grid, and DG energy minus the load. Furthermore, the simulation values of discharge energy and charge energy are reported prior to inverter losses, whereas the analysis results are inclusive of inverter losses. For ease of comparison, the simulated values have been scaled assuming rectifier and inverter efficiencies of 94 %.

Table 4.15: Pre-adjustment base-case simulation results compared against analysis results.

Parameter	**Analysis Results**	**Simulation Results**	**Difference, %**
Theoretical PV Energy, kWh	325000	325000	-0.2
DG Energy, kWh	98500	65000	-34
Grid Energy, kWh	99000	117000	18
Load Energy, kWh	454000	454000	0.0
Excess Energy, kWh	70000	53000	-24
Battery Discharge, kWh	87000	101000	16
Battery Charge, kWh	114000	129000	14
DG Runtime, h/year	2000	1500	-21

There are many substantial differences between the analysis results and the simulation results. The DG runtime and energy production are underestimated, and the grid energy is overestimated along with battery charge/discharge energy. As expected, the skewed results are consistent with the microgrid controller not imposing a grid charging limit. Therefore, a pseudo charging limit is imposed on the grid by limiting the maximum grid power. By trial and error, the grid transformer power is reduced until the grid energy is found to equal the expected value. At this point, the transformer power is found to be 65 kW. A drastic improvement to the simulation results is observed. The grid energy and charge/discharge energy are shifted to within ±1 % of the expected values, and the DG runtime is within ±4 %. However, the DG energy remains slightly lower than expected. As previously discussed, P_{DG_min} can be implemented to improve the behavior of the model. The value of P_{DG_min} is increased until the measured DG energy production is achieved in the simulation. $P_{DG\_min} = 23\ \%$ is found to increase the DG energy to the expected value. The simulation results (with $P_{DG\_min} = 23\ \%$, and the maximum grid power equal to 65 kW) are shown in

Table 4.16. All simulated values are within ±4 % of the measured results. The largest discrepancy is a 4 % underestimation of the DG runtime. This is likely because the simulation model does not account for the reserve capacity provided by the real system, or due to the control discrepancies discussed regarding simulation mode #2. From the measurement data, it can be shown that approximately 60 hours per year of DG runtime attributed to reserve capacity. Adding 60 hours to the simulation results would correct the estimated DG runtime to within 1 %.

Table 4.16: Base-case simulation results compared to analysis results.

Parameter	***Analysis Results***	***Simulation Results***	***Difference, %***
Estimated PV Production, kWh	325000	325000	-0.2
DG Energy, kWh	98500	98500	0
Grid Energy, kWh	99000	98700	-1
Load Energy, kWh	454000	454000	0
Excess Energy, kWh	70000	68000	-2
Battery Discharge, kWh	87000	87000	0
Battery Charge, kWh	114000	112000	-1
DG Runtime, h/year	2000	1900	-4

The limits introduced for grid power and P_{DG_min} are not based on real properties of the K-HES. Therefore, they could introduce unintended impacts when they are passed on to the intervention simulation models. Potential impacts are now considered.

Limiting the grid power periodically inhibits the grid model from supplying the load when it exceeds 65 kW. When this occurs, the DG or the battery must support the load instead. Comparing the grid availability profile against the load profile shows that the load only exceeds 65 kW while the grid is available for 57 hours each year. At least 3700 kWh of energy is generated during this period, or closer to 4500 kWh if it is assumed the load surpasses 65 kW by a significant margin during these 57 hours. Therefore, by imposing the grid limit of 65 kW, the simulated values of grid energy, DG energy, and battery charge/discharge energy are assumed to have an uncertainty of ±4500 kWh. The impact of this uncertainty cannot easily be propagated to the calculated KPIs; therefore, it is neglected in this study. Furthermore, instead of achieving a constant 3.6 kW charging power as observed, the simulated charging power is dependent on the load, and is found to fluctuate between 0 kW-25 kW, which could impact battery lifetime. It has been assumed this does not significantly impact the calculated KPIs given that the total charging energy and discharging energy are within 1 % of the analysis results, and the maximum charging power of 25 kW is not expected to have a significant impact on battery lifetime.

The effects of employing P_{DG_min} in the model are also considered. It was earlier found the real average DG power is 22 % of its rated power. However, in the simulation the average DG power is found to be 23 % of its rated power. The higher average power means the modelled DG efficiency is slightly increased, and thus the fuel consumption reduced. This has the effect of making the DG appear slightly more efficient during the simulation. The small difference in average DG power is assumed to be negligible.

Given the complexity of the iHOGA simulation environment, calculating uncertainty for the absolute value of the KPIs generated by iHOGA is a tedious process. To do so comprehensively, would require compounding input data error with error introduced by the model structure itself. However, the absolute value of the KPIs are not the primary concern of this study. Instead, the primary concern is the difference in the KPIs resulting from modifications to the model. It is assumed that if the input data remains constant, then the uncertainty of the difference between KPIs is primarily generated from error introduced by the model structure. Therefore, the model structure of iHOGA needs to be evaluated to estimate the uncertainty of the results. Such a task is beyond the scope of this study. The accuracy of the simulation results is thus contingent on the assumption that the iHOGA model structure does not generate compromising uncertainty.

The parameters and KPIs generated by the base-case simulation model are summarized in Table 4.17. The previously known parameter "DG runtime" is replaced with "Equivalent DG runtime", which differs from the DG runtime by including the extra DG ageing hours added by the DG model discussed in Section 4.1.3. Additionally, the renewable fraction RF parameter is introduced in the last row of Table 4.17, it is calculated as:

$$RF = \frac{E_L - E_{Gr} - E_{DG}}{E_L} \qquad \textit{Equation 4.4}$$

For the base-case, the simulated value of RF is 57 %, and from

Table 4.16, the analysis results can be shown to give RF=56 %.

Table 4.17: Base-case simulation KPIs.

Parameters	***Base-Case Simulation***
PV Production, kWh	325000
DG Energy, kWh	98500
Grid Energy, kWh	98700
Load Energy, kWh	454000
Excess Energy, kWh	68000
Battery Discharge, kWh	87000
Battery Charge, kWh	112000
KPIs	
Equivalent DG Runtime, h/year	2500
NPC, $	2419000
Fuel Consumption, L/year	47300
Emissions, kgCO_2/year	276000
Battery Life, years	3.9
Renewable Fraction, %	57

In summary, the base-case model is found to replicate the measured behavior of the K-HES within an accuracy of ±4 % or less. It has been assumed this accuracy is high enough for iHOGA to generate KPIs that accurately describe the K-HES. A comprehensive analysis of the uncertainty introduced by iHOGA has not been conducted. Therefore, the simulation results presented in this study are conditional on the accuracy of the iHOGA model structure.

5 Simulation Studies

The base-case simulation model described in Section 4 is the building block for each of the intervention simulation models presented in this section. Aside from the modifications presented for each intervention model, all models can be assumed to be identical to the base-case model. The KPIs generated for each intervention are compared to the KPIs from the base-case simulation in each subsection and the results are discussed.

5.1 Intervention #1: Reduce DG Size

Five DGs of reduced capacities are simulated in place of the existing 220 kW DG. The efficiency of all DGs is assumed to be equal to the efficiency of the 220 kW DG. In other words, the fuel curve coefficients (A and B from Table 4.4) are equal in each model. In reality, the fuel curve coefficients are different for DGs of varying sizes; however, reducing size does not always correspond to improved efficiency or vice versa. Therefore, for comparison's sake, the coefficients are kept equal. The only changes to the base-case model are the rated capacities of the DGs ($P_{DG-rated}$, from Equation 4.2), and the capital cost of the DGs. The changed values are summarized in Table 5.1.

Table 5.1: DG capacities and capital costs.

DG Capacity, kW	***Capital Cost, $***
220	99000
160	72000
120	54000
104	46800
80	36000
64	28800

The simulation results are presented in Table 5.2. NPC, fuel consumption, and emissions are the most heavily impacted KPIs. NPC is reduced by $473000 (or 22 %) when the 220 kW DG is replaced by a 64 kW DG. Furthermore, emissions and fuel consumption are reduced by 29 % and 47 %, respectively. Note, fuel consumption and systems emissions are not reduced proportionally as emissions are not exclusively dependent on fuel consumption. The 47 % reduction in fuel consumption is equivalent to annual savings of $22000. If it is assumed the 64 kW DG would replace the existing 220 kW DG (having no salvage value) the expected simple payback period would be 1.3 years. For comparison, the 104 kW DG is expected to reduce fuel consumption by 37 % and have a payback period of 2.8 years; the 80 kW DG is expected to reduce fuel consumption by 42 %, and have a payback period of 1.8 years.

The predicted NPC and fuel consumption for each of the simulated DG sizes is illustrated in Figure 5.1. As anticipated, there is a clear trend showing that reducing DG size results in a reduction of NPC and fuel consumption. The largest savings are observed over the first 100 kW of reduced DG capacity. However, significant savings continue as capacity is reduced further.

Table 5.2: Simulation results predicting the K-HES performance after reducing the DG size.

Parameters	***DG Capacity***					
	220 kW (base-case)	**160 kW**	**120 kW**	**104 kW**	**80 kW**	**64 kW**
PV Production, kWh	325000	325000	325000	325000	325000	325000
DG Energy, kWh	98500	87600	84700	83700	82700	81900
Grid Energy, kWh	98700	98900	98900	98900	98900	98900
Load Energy, kWh	454000	454000	454000	454000	454000	454000
Excess Energy, kWh	68000	57500	54500	53700	52600	51900
Battery Discharge, kWh	87000	92000	92000	92000	92000	92000
Battery Charge, kWh	112000	105000	105000	105000	105000	105000
KPIs						
Equivalent DG Runtime, h/year	2500	2500	2300	2200	2000	2,100
NPC, $	2419000	2202000	2064000	2010000	1933000	1887000
Fuel Consumption, L/year	47300	38000	32600	30500	27300	25100
Emissions, $kgCO_2$/year	276000	243000	223000	215000	203000	196000
Battery Life, years	3.9	4.0	4.0	4.0	4.0	4.0
Renewable Fraction, %	57	59	60	60	60	60

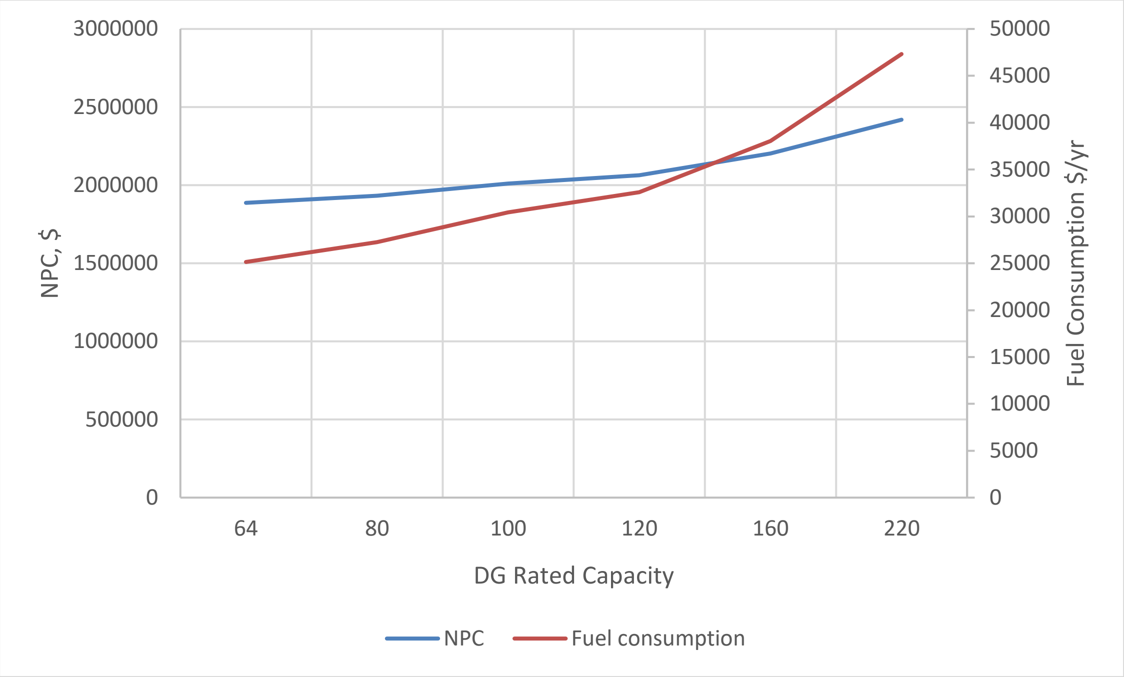

Figure 5.1: Simulation results, NPC and fuel consumption for various DG sizes.

There are potential design issues when implementing a smaller DG. The most obvious being a capacity shortage. For example, the average load on the DG is 48 kW, but there are periods during peak loading when the capacity of a smaller DG could be exceeded. Fortunately, the peak load is typically covered or at least subsidized by the PV system. Nonetheless, a small DG would be impractical for emergency scenarios when the PV/battery/grid are not available. To avoid this problem, the secondary DG in the K-HES could be sized for the peak load. A simple solution would be to retain the original 220 kW, or secondary 200 kW DG for this purpose. Control measures would need to be implemented to ensure a smooth transition between the two generators when the smaller DG becomes overloaded. The bi-Invs have control settings and a relay which could be used to facilitate this control. It would also be possible to run two smaller DGs in parallel; however, the additional complexity and costs could be impractical for a system of this size.

From the measurement data, it can be shown the DG load can be expected to exceed 64 kW for 48 hours annually, 80 kW for 9 hours, and 104 kW for 4 hours. The simulations show that the battery system supports the DGs in all instances, resulting in no unmet load. Regardless, it can still be argued that a larger DG will reduce the risk of unmet load. The 80 kW DG provides a reasonable compromise with NPC savings of 20 %, emission reductions of 26 %, and potential expansion of the load in the future.

5.2 Intervention #2: Install A/C in the Battery Room

Installing A/C in the battery room is modelled in iHOGA by reducing the average battery temperature from 29°C to 20°C and adding 4 kW to the load profile from 10:00-16:00 daily. The cost of installing A/C has been added to the model as a onetime capital cost of $21000. The simulation results are summarized in comparison to the base-case in Table 5.3.

There are several interesting results to discuss. As expected, battery lifetime is significantly improved. The model predicts a lifetime extension of 36 %. This result is consistent with the information provided by the battery manufacturer. The outcome is two fewer battery replacements over the 25-year lifetime of the K-HES. However, it has been speculated that the additional load from the A/C could impact the available PV charging energy. Indeed, the simulation predicts a 1 % reduction in charging energy. Yet, the battery efficiency is improved, which is evident from the battery charge reduction of 1 % while battery discharge is increased by 2 %. Therefore, battery efficiency is increased by 2 %.

Furthermore, excess energy is reduced by 13 %, which is assumed to be a result of the A/C system consuming PV power that otherwise would have been curtailed. Although the total load is increased by 2 %, the DG energy, and fuel consumption are reduced by 1 %, which is attributed to the improved battery efficiency. Total emissions are reduced by 5 %, in part due to the slight reduction in fuel consumption, but largely due to fewer battery replacements over the system lifetime. Whereby the emissions associated with shipping materials to and from the site are saved, as well as the manufacturing emissions of the battery materials. The emission reductions are accompanied by a 1 % reduction in NPC. It is concluded that installing A/C in the battery room is expected to be both economically and environmentally beneficial.

Table 5.3: Simulation results predicting the performance of the K-HES after installing A/C in the battery room.

Parameters	***Base-Case***	***A/C installed***	***Difference, %***
PV Production, kWh	325000	325000	0
DG Energy, kWh	98500	97500	-1
Grid Energy, kWh	98700	99700	+1
Load Energy, kWh	454000	463000	+2
Excess Energy, kWh	68000	60000	-13
Battery Discharge, kWh	87000	94000	+2
Battery Charge, kWh	112000	105000	-1
KPIs			
Equivalent DG Runtime, h/year	2500	2500	0
NPC, $	2419000	2365000	-3
Fuel Consumption, L	47300	46900	-1
Emissions, kgCO_2/year	276000	262000	-5
Battery Life, years	3.9	5.3	+36
Renewable Fraction, %	57	57	0

5.3 Intervention #3: Adjusting Power Source Priority Order

Recall from Section 3.3 the proposed adjustment to the power source priority order is to change the control modes from:

- Mode #1: from 2:00-17:59, the priority order is PV, battery, grid, DG
- Mode #2: from 18:00-1:59, the priority order is PV, grid, DG, battery

to:

- Mode #1: from 2:00-17:59, the priority order is PV, battery, grid, DG
- Mode #2: from 18:00-1:59, the priority order is PV, grid, battery, DG

The priority order in the simulation model is adjusted exactly to the proposed modes using the microgrid controller settings discussed in Section 4.1.8. The cost of changing the priority order is assumed to be zero. The simulation results are presented in Table 5.4. The large increase in grid energy, 19 % or 19,000 kWh/year, confirms that giving the battery priority ahead of the DG provides improved access to grid power. Furthermore, due to a 27 % reduction in DG energy, fuel consumption, and DG runtime, the total system emissions are reduced by 14 %, and the NPC is reduced by 10 %. The results show that adjusting the power source priority order would significantly improve the overall performance of the K-HES in terms of costs and emissions.

Table 5.4: Simulation results predicting the performance of the K-HES after adjusting the source priority order in mode #2.

Parameters	***Base-Case***	***Adjusted Priority***	***Difference, %***
Estimated PV Production, kWh	325000	325000	0
DG Energy, kWh	98500	72000	-27
Grid Energy, kWh	98700	117800	+19
Load Energy, kWh	454000	454000	0
Excess Energy, kWh	68000	61000	-11
Battery Discharge, kWh	87000	94000	+1
Battery Charge, kWh	112000	106000	+1
KPIs			
Equivalent DG Runtime, h/year	2500	1900	-27
NPC, $	2419000	2176000	-10
Fuel Consumption, L/year	47300	34600	-27
Emissions, kgCO_2/year	276000	238000	-14
Battery Life, years	3.9	4.0	+3
Renewable Fraction, %	57	58	+3

5.4 Intervention #4: Introduce Grid Charging

Grid charging is modeled by removing the grid charging limit of 65 kW, and restoring the transformer limit of 200 kW. As discussed earlier, grid charging requires the priority order to be adjusted as per the previous section. Thus, the changes made to the model in the previous section have been carried over to this section. No additional costs have been added to the model. The simulation results are presented in Table 5.5.

Table 5.5: Simulation results predicting the performance of the K-HES after introducing grid charging.

Parameters	***Base-Case***	***Grid Charging***	***Difference, %***
Estimated PV Production, kWh	325000	325000	0
DG Energy, kWh	98500	65300	-34
Grid Energy, kWh	98700	127000	+29
Load Energy, kWh	454000	454000	0
Excess Energy, kWh	68000	63000	-7
Battery Discharge, kWh	87000	109000	+17
Battery Charge, kWh	112000	123000	+17
KPIs			
Equivalent DG Runtime, h/year	2500	1700	-11
NPC, $	2419000	2225000	-6
Fuel Consumption, L/year	47300	31000	-34
Emissions, kgCO_2/year	276000	245000	-11
Battery Life, years	3.9	3.2	-18
Renewable Fraction, %	57	58	2

As expected, the model predicts a significant reduction of DG energy and fuel consumption, which is found to be 34 %. When compared to the results in the previous section, fuel consumption is decreased by an additional 7 %, and grid energy is increased by 10 %. Clearly, the increased consumption of grid energy is displacing DG energy. The negative impact is the reduction in battery lifetime of 18 %, which can safely be assumed to be a result of the 17 % increase in battery charging and discharging energy. Nonetheless, the NPC of the system is reduced by 6 % due to the reduction in fuel consumption. Therefore, the fuel savings outweigh the costs associated with a reduction in battery lifetime. In the previous section an additional 2 % reduction of NPC was predicted. However, as discussed in Section 3.4, grid charging could prevent severe damage to the batteries. The 2 % increase in NPC could prevent a much costlier expenditure. The shorter battery lifetime also increases emissions when compared to the previous section. However, once again, it can be reasoned that if the battery had to be replaced due to severe damage, there would be much larger emission consequences. Moreover, there is still an 11 % reduction in emissions from the base-case. In conclusion, a significant economic and environmental advantage could be achieved by implementing grid charging. The choice to do so is dependent on the priorities of the system owner/operator and the importance of additional fuel savings.

5.5 Intervention #5: Improved Hierarchical Control Philosophy and Optimized Setpoints

The control setpoints described in Section 3.5 are optimized for lowest NPC and emissions over all possible values by iHOGA's built-in genetic algorithm optimization functions. The values of $P_{critical}$ and SOC_{DG-stp} are used to maximize the fuel efficiency of the DG, but to do so, the DG must be able to charge the batteries. Therefore, $I_{DG\_chrgmax}$ is increased to reflect the 1200 A charging limit of the batteries. The simulation results are presented in

Table 5.6, and the optimized control setpoints are found to be:

- Threshold Battery Power, $P_{ThrBatt} = Infinite$
- Minimum DG power, $P_{DGMin} = 21\ \%\ \ (46\ kW)$
- Critical DG power, $P_{DGcriticl} = 86\ kW$
- Minimum SOC, $SOC_{min} = 40\ \%$
- Stop DG SoC, $SOC_{DG-stp} = 51\ \%$

$P_{ThrBatt} = Infinite$ reflects that it is always cheaper to supply power with the batteries than with the DG. In other words, whenever possible, the battery should always have priority ahead of the DG. This result agrees with the conclusion made from intervention #3 which also showed that giving the battery priority improved system performance. $P_{DGMin} = 21\ \%\ (46\ kW)$ reflects that the NPC of the system would be increased due to high O&M costs if the DG were run below 46 kW. $P_{DGcritical} = 86\ kW$ shows that the DG should run to charge the batteries (at maximum charging power) anytime the load is below 86 kW, and the DG should follow the load when it is above 86 kW. During the simulation, the load on the DG does not exceed 86 kW. In effect, the batteries are always charged by the DG when it operates and P_{DGMin} is only enforced when the batteries are fully charged while the DG is running. $SOC_{min} = 40\ \%$ is a surprising result. Typically, lead acid batteries are not discharged to such low levels due to steep reductions in battery lifetime. However, in this case, the model predicts that reducing SOC_{min} does not reduce the lifetime of the batteries. To the contrary, the battery lifetime is expected to increase from 3.9 to 4.2 years even though SOC_{min} has gone from 55 % to 40 %. It is hypothesized that reducing the time the battery spends at a low SoC during the simulation, with constant charging from the DG, outweighs the impact of the lower SOC_{min}. Furthermore, it is observed that the battery SoC is not reduced to SOC_{min} daily. SOC_{min} is only reached during extended grid outages. Otherwise, the combination of grid, DG, and PV charging keep the battery SoC above SOC_{min}. During normal operation with intermittent grid outages, the battery is only reduced to 50-60 % SoC. $SOC_{DG-stp} = 51\ \%$, which means that the DG only runs until the battery SOC is raised by 11 %. This is consistent with the theme of minimizing DG energy in the system.

The improved control setpoints are expected to have a large impact on the performance of the system: reducing fuel consumption by 44 %, emissions by 28 %, and NPC by 21 %.

Table 5.6: Simulation results predicting the performance of the K-HES after introducing grid charging.

Parameters	***Base-Case***	***Optimized Control***	***Difference, %***
Estimated PV Production, kWh	325000	325000	0
DG Energy, kWh	98500	60000	-30
Grid Energy, kWh	98700	101500	+3
Load Energy, kWh	454000	454000	0
Excess Energy, kWh	68000	41000	-39
Battery Discharge, kWh	87000	132000	+43
Battery Charge, kWh	112000	140000	+33
KPIs			
Equivalent DG Runtime, h/year	2500	1000	-61
NPC, $	2419000	1908000	-21
Fuel Consumption, L/year	47300	26300	-44
Emissions, kgCO_2/year	276000	199000	-28
Battery Life, years	3.9	4.2	+8
Renewable Fraction, %	57	62	+10

Implementing the optimized values of SOC_{min} and SOC_{DG-stp} would only require a simple change of the settings on the Sunny Island inverters, and no action is required to implement $P_{ThrBatt}$ =Infinite. P_{DGMin} and $P_{DGcritical}$ are more complex to implement. P_{DGMin} requires installing a controlled dump load that ensures when the batteries are fully charged, and the load is below P_{DGMin}, the DG remains loaded at P_{DGMin}. Implementing $P_{critical}$ could be achieved with an external control system designed to update the settings on the Sunny Island inverters. Adding these control measures clearly adds complexity to the system; however, the performance improvements are shown to be substantial. The costs for implementing this control strategy are not included in the results. However, it has been assumed the costs would be far less than the expected reduction in NPC of $511000. Moreover, a 2-year payback period or less could be achieved if the cost was found to be below $42000.

5.6 All interventions

It is interesting to consider a case where all proposed interventions are implemented simultaneously. This is expected to return the best overall performance results, and a useful reference point for ranking each of the interventions in terms of their effectiveness.

For the simulation, the DG size is chosen to be 80 kW. The control setpoints need to be optimized again considering the additional changes to the model. To mediate the risk of deep discharge and ensure reserve battery capacity is available during emergency scenarios, a minimum limit for SOC_{min} is set to 40 %. Otherwise, all changes to the model are as described in Sections 5.1-5.5. The simulation results are presented in Table 5.7, and the optimized control setpoints are found to be:

- Threshold Battery Power, $P_{ThrBatt} = Infinite$
- Minimum DG power, $P_{DGMin} = 80\ \%\ (64\ kW)$
- Critical DG power, $P_{DGcritical} = 1\ kW$
- Minimum SOC, $SOC_{min} = 40\ \%$
- Stop DG SoC, $SOC_{DG-stp} = 45\ \%$

Once again, $P_{ThrBatt} = Infinite$ means that discharging the battery system is always cheaper and results in fewer emissions than using the DG. $P_{DGMin} = 64\ kW$ reflects that anytime the load is less than 64 kW, it is advantageous to charge the batteries with the difference between 64 kW and the load. High fuel efficiency and minimized O&M costs are assumed to be the primary factors behind this result. P_{DGMin} is greater than $P_{DGcritical}$, which means the DG should never be forced to run at its rated capacity. $SOC_{min} = 40\ \%$ and $SOC_{DG-stp} = 45\ \%$, are once again consistent with minimizing the runtime of the DG. Implementing the control measures can be achieved as described in Section 5.5, but in this case the process is simplified by the low value of $P_{DGcritical}$, which does not require any intervention to achieve.

The combined interventions are expected to reduce DG runtime and fuel consumption by 70 %, and total system emissions by 42 %. The NPC is expected to be reduced by 30 %, or $738000.

Table 5.7: Simulation results predicting the performance of the K-HES after all interventions are implemented simultaneously.

Parameters	***Base-Case***	***All Interventions***	***Difference, %***
Estimated PV Production, kWh	325000	325000	0
DG Energy, kWh	98500	48500	-51
Grid Energy, kWh	98700	132600	+34
Load Energy, kWh	454000	463000	+2
Excess Energy, kWh	68000	43000	-37
Battery Discharge, kWh	87000	131000	+42
Battery Charge, kWh	112000	143000	+36
KPIs			
Equivalent DG Runtime, h/year	2500	800	-70
NPC, $	2419000	1681000	-30
Fuel Consumption, L/year	47300	14100	-70
Emissions, kgCO_2/year	276000	162000	-42
Battery Life, years	3.9	5.0	+28
Renewable Fraction, %	57	61	+8

A side-by-side comparison of NPC, fuel consumption, emissions, and battery lifetime for each intervention is shown in Table 5.8. A color scheme shows the rated effectiveness of individual interventions. The simulation of all interventions has the best results for all KPIs except for battery lifetime. Installing A/C is the top ranked intervention for extending battery lifetime.

Intervention #1 and #5 stand out as the most effective for achieving reductions in NPC, fuel consumption, and emissions. Yet, in both cases, significant additional improvements would be expected if all interventions were implemented simultaneously. Intervention #2 is expected to extend battery lifetime the most. However, the expected reductions in NPC and emissions are the least significant.

Table 5.8: Comparative analysis and rating of individual interventions against all interventions simultaneously.

Intervention	***1***	***2***	***3***	***4***	***5***	***ALL***
80 kW DG	TRUE	FALSE	FALSE	FALSE	FALSE	**TRUE**
A/C Installed	FALSE	TRUE	FALSE	FALSE	FALSE	**TRUE**
Adjusted Priority	FALSE	FALSE	TRUE	TRUE	FALSE	**TRUE**
Grid Charging	FALSE	FALSE	FALSE	TRUE	FALSE	**TRUE**
Optimized Control	FALSE	FALSE	FALSE	FALSE	TRUE	**TRUE**
NPC	-20 %	-3 %	-10 %	-8 %	-21 %	**-30 %**
Fuel Consumption	-42 %	-1 %	-27 %	-34 %	-44 %	**-70 %**
Emissions	-27 %	-5 %	-14 %	-11 %	-28 %	**-42 %**
Battery Life	+3 %	+36 %	+3 %	-18 %	+8 %	**+28 %**
Color scheme	1st	2nd	3rd	4th	5th	

6 Global Impact

PV-HESs are becoming a common solution for replacing backup DG systems. In the future, a large percentage of backup power systems could indeed be PV-HESs. In this case, the predicted savings in costs and emissions outlined in Section 5 could have far reaching impacts. The International Energy Agency (IEA) estimated in 2012 that 16 TWh of energy was generated with backup DGs in sub-Saharan Africa alone. Contributing to the likes of 90 kilo-barrels per day (kb/d) of diesel fuel consumption [1]. Hypothetically, if all backup DG systems in sub-Saharan Africa were replaced with PV-HESs, which achieved 50 % reductions in fuel consumption, then only 45 kb/d of diesel would be consumed each year. If the method presented in this study was then applied to each system, achieving additional fuel reductions of 70 %, an additional 32kb/d of fuel consumption could be mitigated. With the continuous development and improvement of PV-HES design and operation, such large improvements may not be possible for future systems. However, if only 25 % of the fuel savings found in this study were achieved, an additional 8 kb/d of fuel consumption could still be mitigated. In this case, on an annual basis, the savings would be nearly 3000 kb or 500 million liters of diesel, corresponding to 1.7 billion kgCO_2 emissions. Moreover, if the cost of diesel is assumed to be 1.0 $/L, up to $ 500 million of fuel savings could be generated. If the cost of fuel continues to increase in the future, greater savings could be expected. Furthermore, if it is assumed there are regions outside of sub-Saharan Africa, which will also adopt PV-HESs in the future, even greater cost and emission savings could be achieved globally.

7 Discussion

A method to critically analyze the performance of PV-HESs, identify interventions, and predict performance changes is demonstrated for the case-study of the K-HES. The procedure began with gathering data from several measurement instruments over a period of six months. For the most part, it is found the measurement instruments provided data with adequate accuracy to effectively evaluate the performance of the K-HES. To improve the results of the performance evaluation and reduce calculation uncertainties, an additional energy meter in front of and behind the bi-directional inverter would have proven effective. This would have allowed the calculation of inverter and rectifier efficiencies, transmission losses, as well as a more accurate calculation of battery charge and discharge energy. In future applications of the proposed method these additional meters are recommended.

Furthermore, measurement data quantifying fuel consumption and generator runtime would have been beneficial as additional points of comparison to the simulation results. The accuracy of the performance evaluation and the simulation analysis would be expected to improve with these additional measurement parameters. The uncertainty in the irradiation measurements proved to be restrictive when attempting to accurately calculate theoretical PV power and PV curtailment. Improved irradiation measurement accuracy would have been beneficial to this study; however, it was concluded by [24] that high accuracy irradiation measurements require extremely careful work. It is not realistic for all commercial scale systems like the K-HES to carry out such careful work on a continuous basis – the cost would likely be prohibitive. Therefore, it is important that the method is tolerant towards irradiation uncertainty. High accuracy irradiation measurements that do not require extremely careful work would be valuable and this could be an interesting topic for future studies.

Despite uncertainty hurdles, the evaluation method is shown to conclusively identify several opportunities to improve the performance of the K-HES. Furthermore, the method is simplistic, and it is expected that it could be easily adapted and applied to other PV-HESs. The measurement instruments utilized in this study are standard for commercial and industrial applications. Therefore, the quality of data utilized in this study is believed to be representative of the industry, and similar quality results are expected in comparable applications. Of course, this study only concerns a single PV-HES, and additional case studies are needed to test the adaptability of the proposed method.

The product of the evaluation procedure is several interventions with the potential to improve the performance of the K-HES. The interventions target reductions in fuel consumption and the extension of battery lifetime. Both of which are anticipated to reduce NPC and system emissions. The interventions identified are:

1. Intervention #1: Reduce DG Size of the Primary DG
2. Intervention #2: Install A/C in the Battery Room to keep Battery Temperature of 20°C
3. Intervention #3: Adjusting Power Source Priority Order
4. Intervention #4: Introduce Grid Charging
5. Intervention #5: Improved Hierarchical Control Philosophy and Optimized Setpoints
6. Intervention #6: Minimize PV Ramp-up Charge

The K-HES, and each of the proposed interventions, barring intervention #6, are modelled in the iHOGA simulation environment to generate techno-economic KPIs. It must be stressed that the simulation results are contingent on the accuracy of the iHOGA model structure. This assumption is necessary because a comprehensive model verification and validation has not been conducted for iHOGA in this study, nor in any other study discovered following a thorough search of the literature. The component models utilized by iHOGA have been independently verified, as per the references given in this study. However, iHOGA's treatment of the interaction between those models does not appear to have been thoroughly investigated. Importantly, it is shown in this study there are discrepancies between how iHOGA treats the interaction between systems components and the observed interaction of those components in the K-HES. The discrepancies are found to primarily arise during simulated parallel operation of the DG or the grid with the PV system. The lack of model verification and validation is not unique to iHOGA. To the authors knowledge, there has not been a comprehensive model verification and validation of HOMER, which is by far the most widely used HES simulation software. Yet, there are thousands of institutions around the globe using HOMER to develop HESs. It has become evident from this study that both iHOGA and HOMER have not been evaluated to a level where quantifiable uncertainties of their results can be estimated. This realization is of concern to this study and to other applications of these software packages. Both software packages provide an option to perform sensitivity analyses on select variables, which is useful for determining how input data uncertainty impacts the simulation results. However, uncertainty arising from input data should not be confused with uncertainty arising from the model structure itself. The uncertainty introduced by HES software packages, to the likes of iHOGA and HOMER, needs to be thoroughly investigated in future works.

With the discussion above in mind, the simulation results predict substantial reductions in NPC and system emissions. Intervention #1 and #5 have been identified as the most effective interventions. Each of which are predicted to reduce fuel consumption by over 40 %, NPC by over 20 %, and emissions by over 25 %. For intervention #1, it is estimated the cost of replacing the 220 kW DG with an 80 kW DG would be recovered in 1.8 years. The cost of intervention #5 is not determined. However, if it were found to be less than $42000, a payback period of less than 2 years would be expected when considering the fuel savings alone. The predicted reduction in emissions of 25 % is equal to 69000 kgCO_2/year. Intervention #2 is shown to extend battery lifetime by 36 % and results in two fewer battery replacements over the lifetime of the system. Slight reductions in NPC, fuel consumption, and emissions are also predicted. Intervention number #3 was shown to reduce fuel consumption by 27 %, which led to cost and emission reductions of 10 % and 14 %, respectively. Intervention #4 is an extension of intervention #3, which is shown to result in additional fuel savings of 7 %. However, high stress on the battery system is predicted to reduce battery lifetime by 18 %. The reduction in battery lifetime is shown to be a result of increased energy throughput. However, the risk of severe battery damage due to deep discharge is also reduced. The value of risk mediation is not included in the economic analyses in this study; nonetheless, the NPC is shown to be reduced by 8 %. When all interventions are implemented simultaneously the model predicts staggering reductions in fuel consumption by 70 % or 33200 L/year, NPC by 30 % or $738000, and emissions by 42 % or 114000 kgCO_2/year.

Considering PV-HESs are becoming a common solution for replacing backup DG systems, the predicted savings in costs and emissions could have far reaching impacts. The International Energy Agency (IEA) estimated in 2012 that 16 TWh of energy was generated with backup DGs in sub-Saharan Africa alone. Causing 90 kilo-barrels (kb/d) of diesel fuel consumption per day [1]. Hypothetically, if all backup DG systems were replaced with PV-HESs, which achieved 50 % reductions in fuel consumption, then 45 kb/d of diesel would be consumed each year. If the method presented in this study was then applied to each system, and just 25 % of the fuel savings outlined for the K-HES were achieved in each case, an additional 8 kb/d of fuel consumption could be mitigated. On an annual basis, the savings would be nearly 3000 kb or 500 million liters of diesel, corresponding to 1.7 billion kgCO_2 emissions. Moreover, if the cost of diesel is assumed to be 1.0 $/L, up to $500 million of fuel savings could be generated.

The K-HES has been in operation for approximately one year. If it is assumed the K-HES is representative of typical PV-HESs, it has been shown there are economic and environmental motivators to ensure PV-HESs are optimized following their construction. In [6], it is shown that technical and social boundary conditions of HESs change over time. Therefore, it is expected that optimizations later in the system's life would also be beneficial. The proposed method could also be applied to PV-HESs later in their lifetime. In fact, it is expected that the results could be improved with several years of measurement data as opposed to only 6 months. To determine how often this method could practically be applied, a balance will need to be reached between the cost of optimizing the system, and the resulting cost and emissions savings. Therefore, a cost analysis for implementing the proposed method is an important topic for future work.

The evaluation and optimization process in this report has been conducted manually. However, the process can foreseeably be completely automated. Methods utilizing artificial intelligence, forecasting algorithms, and simulation analysis could be implemented to provide frequent or continuous evaluation and optimization of PV-HESs. It is shown by [17], optimizing control parameters daily instead of yearly leads to significantly improved system performance. Similarly, if the method proposed in this study were to be conducted more frequently, it is expected further gains could be achieved. Moreover, an automated process could be used to expand the number of possible interventions considered. For example, interventions such as the following could be considered:

- Time-optimized energy dispatch
- Optimized control setpoints, for example those optimized in intervention #5, but conceivably additional parameters such as maximum and minimum charging rates for each component
- Requests for maintenance (array cleaning, DG servicing, equalization charging)
- Recommendations for optimized component capacities and/or types. For example, increasing PV array capacity or switching to lithium-ion battery technology
- Demand side management (optimal load scheduling based on forecasted resource availability)

Automation of the proposed method is expected to drive the cost of its implementation down, and allow it to be implemented more frequently, or even continuously. This could facilitate greater cost and emission savings. Automation of the proposed method is an interesting topic for future works.

The value of the optimization process increases when boundary conditions have larger variations. Climate change and load growth are examples of technical boundary conditions which could introduce high levels of uncertainty. There could also be changes in economic boundary conditions. A simple example would be the cost of fuel. Temporary or permanent high fuel costs could warrant heavier usage of the battery system. A more complex example of changing economic boundary conditions would be additional revenue streams. In sub-Saharan Africa, poor quality power supply reduces utility revenues by non-payment [1]. In the future (certainly within the 25-year lifetime of the K-HES), PV-HESs could have a role in improving the quality of grid power supply. The configuration of the K-HES is well suited for providing frequency and voltage support to the national grid. Indeed, when connected to the grid, the system already provides this service by operating under droop control, which effectively reduces the grids reactive power load. The behavior of the system currently benefits the operator of the K-HES, but the system could be tuned to the utility's request to generate revenue. Of course, this would result in additional uncertainty in the demand profile of the K-HES. In this case, system optimization would be expected to become even more important.

8 Conclusion

PV-HESs have the potential to alleviate overwhelming costs and emissions incurred from the use of traditional backup DG systems. However, from project conception to end of lifetime, they can be expected to face significant barriers to achieving optimal performance. To maximize cost and emission reductions, PV-HESs must be adapted to their measured boundary conditions. In this study, system performance and boundary conditions of the K-HES are measured to create an opportunity for system optimization and improvement. The method demonstrated is a techno-economic and environmental approach to optimizing and improving PV-HES performance.

An analytical tool has been developed in Excel to interpret measurement data from the K-HES and evaluate system performance in terms of energy and power consumption/generation. It is expected that this tool could be easily adapted to similar PV-HESs. The tool is shown to be effective for identifying corrective interventions which could improve the performance of the K-HES. Six interventions are identified:

1. Intervention #1: Reduce DG Size of the Primary DG
2. Intervention #2: Install A/C in the Battery Room to keep Battery temperature of 20˚C
3. Intervention #3: Adjusting Power Source Priority Order
4. Introduce Intervention #4: Introduce Grid Charging
5. Intervention #5: Improved Hierarchical Control Philosophy and Optimized Setpoints
6. Intervention #6: Minimize PV Ramp-up Charge

To predict the impact of each proposed intervention in terms of technical, economic and environmental considerations, the K-HES and interventions 1-5 are modeled in the iHOGA simulation environment. The sixth intervention could not be modelled due to restrictions of the software. In addition to simulating the interventions individually, they have also been modelled simultaneously. In this case, predictions show that large reductions in NPC (-30 %), fuel consumption (-70 %), and emissions (-42 %) could be achieved. When considering the potential impact of these reductions beyond the K-HES, in sub-Saharan Africa and globally, there is potential for wide spread reductions in the costs and emissions associated with PV-HESs. In sub-Saharan Africa, these reductions could result in annual savings of $500 million USD and 1.7 billion kgCO_2 emissions.

Many uncertainties have been identified throughout this study. When possible, analytical calculations have been utilized to quantify these uncertainties. However, it has been found that there are uncertainties which could not be quantified within the scope of this study. These uncertainties are found to be primarily introduced by the iHOGA simulation software. A greater understanding of these uncertainties must be achieved to report the accuracy of the simulation results presented.

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Appendix

A P3 Engineers Battery Room Thermal Report

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Mbita Battery room

Thermal and ventilation analysis

It is proposed to install PV batteries and associated inverters inside a concrete building formerly used to house a generator set. The batteries have a maximum operation temperature of 22C, and since they produce hydrogen when charging, it is necessary to provide adequate ventilation to control concentrations to a safe level.

A dynamic thermal and natural ventilation simulation has been undertaken to assess the temperature profile throughout the year, and the amount of natural ventilation available. This has shown that the natural ventilation is sufficient to control hydrogen levels, but not to maintain the space temperature of 22C.

It is recommended that a mixed mode system is adopted, using natural ventilation when possible, but mechanical ventilation with cooling at other times. The existing vents should be fitted with motorised dampers which would close when mechanical control is required, for which a commercial 'multi-split' fan coil system and fresh air supply fan are suggested.

Assumptions

Building structure: walls and roof constructed of 230mm reinforced concrete.

Weather data: Kisumu (Kenya) annual weather file (although the building is located in Mbita, this is the closest weather file available)

Thermal model: IESVE dynamic simulation software.

Design criteria

Battery Room

- temperature limit	22°C max
- heat emission	5.7kW
- hours	10am – 5pm, 365 days per annum
- hydrogen limit	4%

Inverter Room

- temperature limits	-25°C to + 60°C
- heat emission	4.4 kW
- hours	10am – 5pm, 365 days per annum

Store/void

- temperature limits	None
- heat emission	0 kW
- hours	N/A

Hydrogen level control

Based on BS EN 50272, we have calculated that the battery room will require a minimum of 1.2 air changes per hour (110 litres/sec, 400 m3/hr) for hydrogen levels to be kept within the acceptable limits (see Appendix 1).

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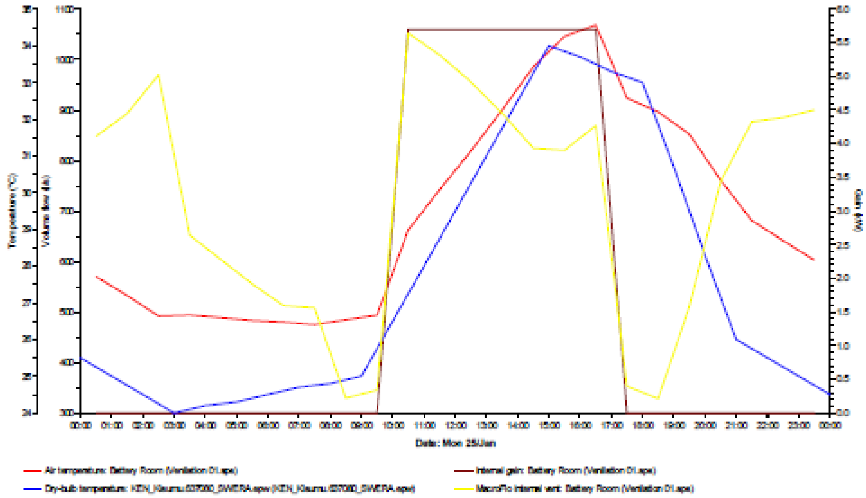
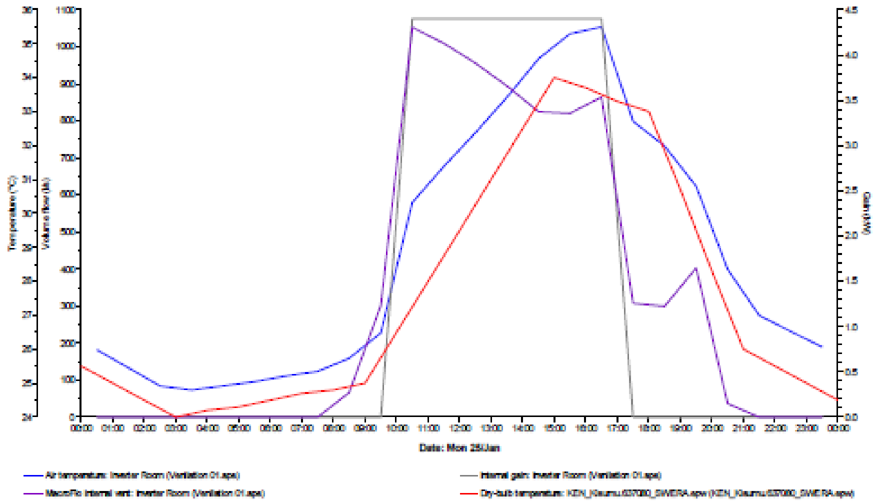
Thermal performance through natural ventilation

The ventilation analysis was based on 100% free area for the internal vents which do not have grilles, and 60% free area for the external vents which do. The vents were assumed to be open day and night, to exploit the cooling effect of the thermal mass of the concrete.
The following graphs indicate the temperature/cooling load profile for the peak day, 25th January;



Graph 1 – Battery Room



Graph 2 – Inverter Room

The results indicate peak space temperatures of 34.5°C in the Battery room and 35.5°C in the Inverter room. This occurs on 25th January when the external temperature is 35°C.

It is clear that natural ventilation alone will not be able to maintain the battery room temperature below 22°C. Mechanical cooling will be necessary. However, the ventilation rate varies between a maximum of 2,800l/s and a minimum of 120l/s during the year, above the limit required to control hydrogen levels below the Lower Explosion threshold.

The results further indicate that natural ventilation in the inverter room will be sufficient for space temperature control within design limits

Battery room mechanical cooling analysis

A further thermal analysis was carried out to calculate the cooling capacity to maintain the space temperature below 22°C.

The peak cooling load with all vents open is 20 kW. However, this is not energy efficient due to the uncontrolled heat gains through varying rates of infiltration of outside air.

To optimise the cooling load and ensure that sufficient fresh air is supplied into the space for hydrogen dispersal, it is recommended that when the room temperature rises above 22°C, the natural vents are closed, and fresh air is mechanically supplied to the space in conjunction with a cooling system. In this way the cooling system capacity reduces to 16kW, and is only required when the outdoor conditions are unsuitable for natural ventilation to maintain the correct indoor environment.

This can be achieved by installing motorised dampers to all the external windows to void/store space and the internal windows between the battery room and inverter room, the windows should be open when outside air temperature is below 20°C and space temperature is below 21°C, when either of these temperatures are exceeded the controls should close all the windows and start the cooling and ventilation plant.

Conclusion

For the Inverter Room, the natural ventilation strategy as proposed is sufficient to maintain the space temperature within the inverters' operating temperature range.

For the Battery Room, the existing natural ventilation openings can provide the minimum ventilation rates for hydrogen dispersal but cannot maintain space temperature under 22C. It is therefore recommended that to control temperature, a simple multi-split air conditioning system is provided, 16kW total capacity. Hydrogen dispersal would be provided by a fresh air supply fan delivering 110l/s of outside air at low level when the batteries are charging/discharging. Air exhaust would discharge at high level through a natural vent.

The space temperature shall be maintained by natural ventilation when conditions allow, and the cooling system would only come on when the space temperature exceeds 21C.

Ventilation requirements calculator

Source: EN 50272-2:2001

Author: Abdul Mohammed
Date: 14/07/2016

Project	ICIPE-MBITA
Manufacturer	Hoppecke
Model/Product	16OPzS 2900

Ventilation requirement

Hydrogen Generation	k	0.00042	
Number cells	n	144	
Charge current	Igas	20	
Safety factor	s	5	
Necessary hydrogen dilution	v	24	
Rated Capacity	Crt	2150	
Ventilation air flow	**Q**	**312.0768**	**m3/hr**

Room volume

Length	10.294	m
Width	6.488	m
Height	5.1	m
Room volume	340.6161	m3

Ventilation rate

Rate	0.086688	m3/s
Fan rating	0.10836	m3/s

Air changes

Fan rating	0.11	m3/s
Room volume	340.62	m3
Air changes	1.15	per hour

Naural ventilation area

Free area (A)	8738.2	cm2
	0.874	m2

Technology inputs

Parameter	Symbol	Unit	Chosen Value	Default Values		
				VLA	VRLA	NiCd
Lower Explosion Limit (maximum hydrogen concentration)	n/a	%	4%	n/a	n/a	n/a
Necessary dilution of hydrogen	v		24	n/a	n/a	n/a
Rate of hydrogen generation	q	m^3/Ah	0.00042	n/a	n/a	n/a
Safety Factor	s		5	n/a	n/a	n/a
Gas Emissions Factor	fg		1	1	0.2	1
Gase emission safety Factor	fs		5	5	5	5
Float current for water dissociation	I_{float}	mA/Ah	1	1	1	1
Boost charge current for dissociation	I_{boost}	mA/Ah	4	4	8	10
Float current	I_{gas}	mA/Ah	5	5	1	5
Boost current	I_{gas}	mA/Ah	20	20	8	50
Float charge voltage	U_{float}	V/cell	2.23	2.23	2.27	1.40
Boost charge voltage	U_{boost}	V/cell	2.40	2.40	2.40	1.55

Extract from BS EN 50272-2,

8.2 Ventilation requirements

The purpose of ventilating a battery location or enclosure is to maintain the hydrogen concentration below the 4 $\%_{vol}$ hydrogen Lower Explosion Limit (LEL) threshold. Battery locations and enclosures are to be considered as safe from explosions, when by natural or forced (artificial) ventilation the concentration of hydrogen is kept below this safe limit.

The minimum air flow rate for ventilation of a battery location or compartment shall be calculated by the following formula:

$$Q = v \cdot q \cdot s \cdot n \cdot I_{gas} \cdot C_{rt} \cdot 10^{-3} \ [m^3/h]$$

where:

Q = ventilation air flow in m^3/h

v = necessary dilution of hydrogen: $\frac{(100\% - 4\%)}{4\%} = 24$

q = $0{,}42 \cdot 10^{-3}\ m^3/Ah$ generated hydrogen

s = 5, general safety factor

n = number of cells

I_{gas} = current producing gas in mA per Ah rated capacity for the float charge current I_{float} or the boost charge current I_{boost}

C_{rt} = capacity C_{10} for lead acid cells (Ah), U_f = 1,80 V/cell at 20 °C
or capacity C_5 for NiCd cells (Ah), U_f = 1,00 V/cell at 20 °C

With $v \cdot q \cdot s$ = $0{,}05\ m^3/Ah$ the ventilation air flow calculation formula is:

$$\mathbf{Q = 0{,}05 \cdot n \cdot I_{gas} \cdot C_{rt} \cdot 10^{-3}\ [m^3/h]}$$

8.3 Natural ventilation

The amount of ventilation air flow shall preferably be ensured by natural ventilation, otherwise by forced (artificial) ventilation.

Battery rooms or enclosures require an air inlet and an air outlet with a minimum free area of opening calculated by the following formula:

$$\mathbf{A = 28 \cdot Q}$$

with Q = ventilation flow rate of fresh air [m^3/h]
A = free area of opening in air inlet and outlet [cm^2]

NOTE For the purpose of this calculation the air velocity is assumed to be 0,1 m/s.

The air inlet and outlet shall be located at the best possible location to create best conditions for exchange of air, i.e.

- openings on opposite walls,
- minimum separation distance of 2 m when openings on the same wall.

8.4 Forced ventilation